机器人学中的基础模型：应用、挑战与未来

Foundation Models in Robotics: Applications, Challenges, and the Future

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核心摘要校订

根据原文摘要单独校订。正文仍为逐页机器翻译初稿；原始图表、公式和英文条目请查看紧邻原文页。

本文综述预训练基础模型在机器人学中的应用。传统机器人深度学习模型常在针对具体任务的小规模数据上训练，跨应用适应性有限；以互联网规模数据预训练的基础模型表现出更强泛化，在部分情况下能够对训练中未出现的问题给出零样本解决方案。基础模型可能增强机器人自主系统中从感知到决策和控制的多个环节，例如 LLM 生成代码与提供常识推理，VLM 支持开放词表视觉识别。但机器人相关训练数据稀缺、安全保证、不确定性量化和实时执行等问题仍然存在。本文考察使用或构建基础模型解决机器人问题的研究，讨论其对感知、决策与控制能力的作用，以及阻碍实际采用的挑战和未来机会。

Foundation Models in Robotics: Applications, Challenges, and the Future

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机器人学中的基础模型：应用、挑战与未来

二
机器人技术中的 基础模型（Foundation Models）：应用、挑战和未来

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***Abstract*—We survey applications of pretrained foundation models in robotics. Traditional deep learning models in robotics are trained on small datasets tailored for specific tasks, which limits their adaptability across diverse applications. In contrast, foundation models pretrained on internet-scale data appear to have superior generalization capabilities, and in some instances display an emergent ability to find zero-shot solutions to problems that are not present in the training data. Foundation models may hold the potential to enhance various components of the robot autonomy stack, from perception to decision-making and control. For example, large language models can generate code or provide common sense reasoning, while vision-language models enable open-vocabulary visual recognition. However, significant open research challenges remain, particularly around the scarcity of robot-relevant training data, safety guarantees and uncertainty quantification, and real-time execution. In this survey, we study recent papers that have used or built foundation models to solve robotics problems. We explore how foundation models contribute to improving robot capabilities in the domains of perception, decision-making, and control. We discuss the challenges hindering the adoption of foundation models in robot autonomy and provide opportunities and potential pathways for future advancements. The GitHub project corresponding to this paper[1] can be found here.**

***Index Terms*—Robotics, Large Language Models (LLMs), Visual-Language Models (VLM), Large Pretrained Models, Foundation Models**

I. Introduction

FOUNDATION models are pretrained on extensive internet-scale data and can be fine-tuned for adaptation to a wide range of downstream tasks. Foundation models have demonstrated significant breakthroughs in vision and language processing; examples include BERT [1], GPT-3 [2], GPT-4 [3], CLIP [4], DALL-E [5], and PaLM-E [6]. Foundation models have the potential to unlock new possibilities in robotics domains such as autonomous driving, household robotics, industrial robotics, assistive robotics, medical robotics, field robotics, and multi-robot systems. Pretrained Large Language Models (LLMs), Large Vision-Language Models (VLMs), Large Audio-Language Models (ALMs), and Large Visual-Navigation Models (VNMs) can be utilized to improve various tasks in robotics settings. The integration of foundation models into robotics is a rapidly evolving area, and the robotics community has very recently started exploring ways to leverage these large models within the robotics domain for perception, prediction, planning, and control.

Prior to the emergence of foundation models, traditional deep learning models for robotics were typically trained on limited datasets gathered for distinct tasks [7]. Conversely, foundation models are pre-trained on extensive and diverse data, which has been proven in other domains (such as natural language processing, computer vision, and healthcare [8]) to significantly expand adaptability, generalization capability, and overall performance. Ultimately, foundation models may hold the potential to yield these same benefits in robotics. Knowledge transfer from foundation models may reduce training time and computational resources compared to task-specific models. Particularly relevant to robotics, multimodal foundation models can fuse and align multimodal heterogeneous data gathered from various sensors into compact homogeneous representations needed for robot understanding and reasoning [9]. These learned representations hold the potential to be used in any part of the autonomy stack including perception, decision-making, and control. Furthermore, foundation models provide zero-shot capabilities, which refer to the ability of an AI system to perform tasks without prior examples or dedicated training data for that specific task. The would enable robots to generalize their learned knowledge to novel cases, enhancing adaptability and flexibility for robots in unstructured settings.

Integrating foundation models into robotic systems may enable context-aware robotic systems by enhancing the robot's ability to perceive and interact with the environment. For example in the perception domain, Large Vision-Language Models (VLMs) have been found to provide cross-modal understanding by learning associations between visual and textual data, aiding tasks such as zero-shot image classification, zero-shot object detection [10], and 3D classification [11]. As another example, language grounding in the 3D world [12] (aligning contextual understanding of VLMs to the 3-dimensional (3D) real world) may enhance a robot's spatial awareness by associating words with specific objects, locations, or actions within the 3D environment.

In the decision-making or planning domain, LLMs and VLMs have been found to assist robots in task specification for high-level planning [13]. Robots can perform more complex tasks by leveraging linguistic cues in manipulation, navigation, and interaction. For example, for robot policy learning techniques like imitation learning [14] and reinforcement learning [15], foundation models seem to offer the possibility to improve data efficiency and enhance contextual understanding. In particular, language-driven rewards can be used to guide RL agents by providing shaped rewards [16]. Also, researchers have employed language models to provide feedback for policy learning techniques [17]. Some works have shown that a VLM model's visual question-answering (VQA) capability can be harnessed in robotics use cases. For example, researchers have used VLMs to answer questions related to visual content to aid robots in accomplishing their tasks [18]. Also, researchers have stated utilizing VLMs to help with data annotation, by generating descriptive labels for visual content [19].

Despite the transformative capabilities of foundation models in vision and language processing, the generalization and fine-tuning of foundation models for real-world robotics tasks remain challenging. These challenges include: 1) **Data Scarcity:** how to obtain internet-scale data for robot manipulation, locomotion, navigation, and other robotics tasks, and how to perform self-supervised training with this data, 2) **High Variability:** how to deal with the large diversity in physical environments, physical robot platforms, and potential robot tasks while still maintaining the generality required for a foundation model, 3) **Uncertainty Quantification:** how to deal with (i) instance-level uncertainty such as language ambiguity or LLM hallucination; (ii) distribution-level uncertainty; and (iii) distribution-shift, especially resulting from closed-loop

[1]Preliminary release. We are committed to further enhancing and updating this work to ensure its quality and relevance

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摘要：我们调查了预训练 基础模型（Foundation Models） 在机器人领域的应用。机器人技术中的传统深度学习模型是在针对特定任务定制的小型数据集上进行训练的，这限制了它们在不同应用程序中的适应性。相比之下，在互联网规模数据上进行预训练的 基础模型（Foundation Models） 似乎具有卓越的泛化能力，并且在某些情况下表现出为训练数据中不存在的问题找到零样本解决方案的能力。 基础模型（Foundation Models） 可能具有增强机器人自主堆栈的各个组件（从感知到决策和控制）的潜力。例如，大型语言模型可以生成代码或提供常识推理，而视觉语言模型可以实现开放词汇视觉识别。然而，重大的开放研究挑战仍然存在，特别是在机器人相关训练数据、安全保证和不确定性量化以及实时执行方面的缺乏。在本篇综述中，我们研究了最近使用或构建 基础模型（Foundation Models） 来解决机器人问题的论文。我们探索 基础模型（Foundation Models） 如何有助于提高机器人在感知、决策和控制领域的能力。我们讨论了阻碍 基础模型（Foundation Models） 在机器人自主中采用的挑战，并为未来的进步提供了机会和潜在途径。本文1对应的 GitHub 项目可以在这里找到。

索引术语-机器人、大型语言模型 (LLM)、视觉语言模型 (VLM)、大型预训练模型、基础模型（Foundation Models）

一、简介

基金会互联网规模的数据可以进行微调以适应模型，并针对广泛的下游任务进行了预训练。 基础模型（Foundation Models）在视觉和语言处理方面表现出重大突破；示例包括 BERT [1]、GPT-3 [2]、GPT-4 [3]、CLIP [4]、DALL-E [5] 和 PaLM-E [6]。 基础模型（Foundation Models） 有潜力在自动驾驶、家用机器人、工业机器人、辅助机器人、医疗机器人、现场机器人和多机器人系统等机器人领域解锁新的可能性。预训练的大型语言模型 (LLM)、大型视觉语言模型 (VLM)、大型音频语言模型 (ALM) 和大型视觉导航模型 (VNM) 可用于改进机器人设置中的各种任务。 基础模型（Foundation Models） 与机器人的集成是一个快速发展的领域，机器人社区最近开始探索如何在机器人领域利用这些大型模型进行感知、预测、规划和控制。

在 基础模型（Foundation Models） 出现之前，传统的机器人深度学习模型通常是在为不同任务收集的有限数据集上进行训练的 [7]。相反，基础模型（Foundation Models） 在广泛且多样化的数据上进行了预训练，这已在其他领域（例如自然语言处理、计算机视觉和医疗保健 [8]）中得到证明，可以显着扩展适应性、泛化能力和整体性能。最终，基础模型（Foundation Models） 可能有潜力在机器人领域带来同样的好处

。与特定任务相比，来自 基础模型（Foundation Models） 的知识转移可以减少训练时间和计算资源

> 1初步发布。我们致力于进一步加强和更新这项工作，以确保其质量和相关性

模型。与机器人技术特别相关的是，多模态 基础模型（Foundation Models） 可以将从各种传感器收集的多模态异构数据融合并对齐为机器人理解和推理所需的紧凑同质表示[9]。这些学习到的表示有可能用于自主堆栈的任何部分，包括感知、决策和控制。此外，基础模型（Foundation Models） 提供零样本能力，这是指人工智能系统无需先前示例或特定任务的专用训练数据即可执行任务的能力。这将使机器人能够将其学到的知识推广到新的案例中，从而增强机器人在非结构化环境中的适应性和灵活性。

将 基础模型（Foundation Models） 集成到机器人系统中可以通过增强机器人感知环境并与环境交互的能力来实现上下文感知机器人系统。例如，在感知领域，大型视觉语言模型 (VLM) 被发现可以通过学习视觉和文本数据之间的关联来提供跨模态理解，从而帮助完成零样本图像分类、零样本目标检测 [10] 和 3D 分

类 [11] 等任务。另一个例子，3D 世界中的语言 落地关联（Grounding） [12]（将 VLM 的上下文理解与 3D 现实世界保持一致）可以通过将单词与 3D 环境中的特定对象、位置或动作相关联来增强机器人的空间意识。

在决策或规划领域，LLM 和 VLM 被发现可以协助机器人进行高级规划的任务规范 [13]。机器人可以利用操纵、导航和交互中的语言线索来执行更复杂的任务。例如，对于 模仿学习（Imitation Learning） [14] 和 强化学习（Reinforcement Learning） [15] 等机器人策略学习技术，基础模型（Foundation Models）似乎提供了提高数据效率和增强上下文理解的可能性。特别是，语言驱动的奖励可以通过提供形状奖励来指导强化学习智能体[16]。此外，研究人员还利用语言模型为策略学习技术提供反馈[17]。一些研究表明，VLM 模型的视觉问答 (VQA) 功能可以在机器人用例中利用。例如，研究人员使用 VLM 来回答与视觉内容相关的问题，以帮助机器人完成任务 [18]。此外，研究人员表示利用 VLM 通过为视觉内容生成描述性标签来帮助数据注释 [19]。

尽管 基础模型（Foundation Models） 在视觉和语言处理方面具有变革性的能力，但 基础模型（Foundation Models） 对于现实世界机器人任务的泛化和微调仍然具有挑战性

。这些挑战包括：1）数据稀缺：如何获取用于机器人操作、运动、导航和其他机器人任务的互联网规模数据，以及如何使用这些数据进行自我监督训练，2）高可变性：如何处理物理环境、物理机器人平台和潜在机器人任务的巨大多样性，同时仍然保持 基础模型（Foundation Model） 所需的通用性，3）不确定性量化：如何处理（i）实例级不确定性，例如语言歧义或法学硕士幻觉； (ii) 分布水平的不确定性； (iii) 分布转移，特别是由闭环引起的

robot deployment, 4) **Safety Evaluation:** How to rigorously test for the safety of a foundation model-based robotic system (i) prior to deployment, (ii) as the model is updated throughout its lifecycle, and (iii) as the robot operates in its target environments. 5) **Real-Time Performance:** how to deal with the high inference time of some foundation models which could hinder their deployment on robots and how to accelerate inference in foundation models to the speed required for online decision-making.

In this survey, we study the existing literature on the use of foundation models in robotics. We study current approaches and applications, present current challenges, suggest directions for future research to address these challenges, and identify potential risks exposed by integrating foundation models into robot autonomy. Another survey on foundation models in robotics appeared simultaneously with ours on arXiv [20]. In comparison with that paper, ours emphasizes future challenges and opportunities, including safety and risk, and ours has a stronger emphasis on comparisons in applications, algorithms, and architectures among the existing papers in this space. In contrast to some existing surveys that focus on a specific in-context instruction, such as prompts [21], vision transformers [22], or decision-making [13], [23], we provide a broader perspective to connect distinct research threads in foundation models organized around their relevance to and application to robotics. Conversely, our scope is much narrower than the paper [24], which explores the broad application of foundation models across many disciplines, of which robotics is one. We hope this paper can provide clarity regarding areas of recent progress and existing deficiencies in the research, and point the way forward to future opportunities and challenges facing this research area. Ultimately, we aim to give a resource for robotics researchers to learn about this exciting new area.

We limit the scope of this survey to papers that fall into one of the following categories:

1) **Background Papers:** Papers that do not explicitly link to robotics, but are nonetheless required for understanding foundation models. These papers are discussed in the background section (section II) of the survey paper.
2) **Robotics Papers:** Papers that integrate a foundation model into a robotic system in a plug-and-play fashion, papers that adapt or fine-tune foundation models for robotic systems, or papers that build new robotic-specific foundation models.
3) **Robotics-Adjacent Papers:** Papers that present methods or techniques applied to areas adjacent to robotics (e.g., computer vision, embodied AI), with a clear path to future application in robotics.

This survey is organized as follows: In Section II, we provide an introduction to foundation models including LLMs, vision transformers, VLMs, embodied multimodal language models, and visual generative models. In addition, in the last part of this section, we discuss different training methods used to train foundation models. In Section III, we present a review of how foundation models are integrated into different tasks for decision-making in robotics. First, we discuss robot policy learning using language-conditioned imitation learning, and language-assisted reinforcement learning. Then, we discuss how to use foundation models to design a language-conditioned value function that can be used for planning purposes. Next, robot task specification and code generation for task planning using foundation models are presented. In Section IV, we study various perception tasks in robotics that have the potential to be enhanced by employing foundation models. These tasks include semantic segmentation, 3D scene representation, zero-shot 3D classification, affordance prediction, and dynamics prediction. In Section V, we present papers about Embodied AI agents, generalist AI agents, as well as simulators and benchmarks developed for embodied AI research. In Section VI, we conclude the survey by discussing different challenges for employing foundation models in robotic systems and proposing potential avenues for future research. Finally, in Section VII we offer the concluding remarks.

II. Foundation Models Background

Foundation models have billions of parameters and are pretrained on massive internet-scale datasets. Training models of such scale and complexity involve substantial costs. Acquiring, processing, and managing data can be costly. The training process demands significant computational resources, requiring specialized hardware such as GPUs or TPUs, as well as software and infrastructure for model training which requires financial resources. Additionally, training a foundation model is time-intensive, which can translate to even higher costs. Hence these models are often used as plug-and-play modules (which refers to the integration of foundation models into various applications without the need for extensive customization). Table I provides details about commonly used foundation models. In the rest of this section, we introduce LLMs, vision transformers, VLMs, embodied multi-modal language models, and visual generative models. In the last part of this section, we introduce different training methods that are used to train foundation models.

A. Terminology and Mathematical Preliminaries

In this section, we first introduce common terminologies in the context of foundation models and describe basic mathematical details and training practices for various types of foundation models.

Tokenization: Given a sequence of characters, tokenization is the process of dividing the sequence into smaller units, called tokens. Depending on the tokenization strategy, tokens can be characters, segments of words, complete words, or portions of sentences. Tokens are represented as 1-hot vectors of dimension equal to the size of the total vocabulary and are mapped to lower-dimensional vectors of real numbers through a learned embedding matrix. An LLM takes a sequence of these embedding vectors as raw input, producing a sequence of embedding vectors as raw output. These output vectors are then mapped back to tokens and hence to text. GPT-3, for example, has a vocabulary of 50,257 different tokens, and an embedding dimension of 12,288.

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机器人部署，4) 安全评估：如何严格测试基于 基础模型（Foundation Model） 的机器人系统的安全性 (i) 在部署之前，(ii) 在模型在其整个生命周期中更新时，以及 (iii) 当机器人在其目标环境中运行时。 5）实时性能：如何处理某些基础模型（Foundation Models）的推理时间过长，这可能会阻碍其在机器人上的部署，以及如何将基础模型（Foundation Models）的推理加速到在线决策所需的速度。

在本篇综述中，我们研究了有关 基础模型（Foundation Models） 在机器人技术中使用的现有文献。我们研究当前的方法和应用，提出当前的挑战，提出解决这些挑战的未来研究方向，并确定将 基础模型（Foundation Models） 集成到机器人自主性中所暴露的潜在风险。另一项关于机器人领域 基础模型（Foundation Models） 的调查与我们在 arXiv 上的调查同时出现[20]。与那篇论文相比，我们的论文强调了未来的挑战和机遇，包括安全性和风险，并且我们的论文更加强调该领域现有论文在应用程序、算法和架构方面的比较。与一些现有的专注于特定上下文指令（例如提示[21]、视觉转换器[22]或决策[13]、[23]）的调查相反，我们提供了更广泛的视角来连接基础模型（Foundation Models）中围绕其与机器人的相关性和应用而组织的不同研究线索。相反，我们的范围比论文[24]要窄得多，该论文探讨了 基础模型（Foundation Models） 在许多学科中的广泛应用，其中机器人学就是其中之一。我们希望本文能够阐明研究领域的最新进展和现有缺陷，并为该研究领域未来面临的机遇和挑战指出前进的方向。最终，我们的目标是为机器人研究人员提供资源，以了解这个令人兴奋的新领域。

我们将本篇综述的范围限制为属于以下类别之一的论文：

- 1) 背景论文：没有明确链接到机器人技术，但仍然需要理解 基础模型（Foundation Models） 的论文。这些论文在调查论文的背景部分（第二部分）中进行了讨论。

- 2) 机器人论文：以即插即用方式将 基础模型（Foundation Model） 集成到机器人系统中的论文、为机器人系统调整或微调 基础模型（Foundation Models） 的论文，或构建新的机器人专用 基础模型（Foundation Models） 的论文。

- 3) 机器人相关论文：提出应用于机器人相关领域的方法或技术的论文（例如计算机视觉，具身智能（Embodied AI）），并为机器人技术的未来应用提供清晰的路径。

本篇综述的组织如下：在第二节中，我们介绍了 基础模型（Foundation Models），包括 LLM、视觉转换器、VLM、具体化多模态语言模型和视觉生成模型。此外，在本节的最后一部分，我们讨论了用于训练 基础模型（Foundation Models） 的不同训练方法。在第三节中，我们回顾了 基础模型（Foundation Models） 如何集成到机器人决策的不同任务中。首先，我们讨论使用语言条件 模仿学习（Imitation Learning） 进行机器人策略学习，

和语言辅助的 强化学习（Reinforcement Learning）。然后，我们讨论如何使用 基础模型（Foundation Models） 设计可用于规划目的的语言条件值函数。接下来，介绍使用 基础模型（Foundation Models） 进行任务规划的机器人任务规范和代码生成。在第四节中，我们研究了机器人技术中的各种感知任务，这些任务有可能通过使用 基础模型（Foundation Models） 得到增强。这些任务包括语义分割、3D 场景表示、零样本 3D 分类、可供性（Affordance） 预测和动态预测。在第五节中，我们介绍了有关 具身智能（Embodied AI） 智能体、通用 AI 智能体以及为 具身智能（Embodied AI） 研究开发的模拟器和基准的论文。在第六节中，我们通过讨论在机器人系统中使用 基础模型（Foundation Models） 的不同挑战并提出未来研究的潜在途径来结束调查。最后，在第七节中我们提出结论性意见。

二. 基础模型（Foundation Models）背景

基础模型（Foundation Models）拥有数十亿个参数，并在大规模互联网规模数据集上进行了预训练。如此规模和复杂性的训练模型需要大量成本。获取、处理和管理数据的成本可能很高。训练过程需要大量的计算资源，需要 GPU 或 TPU 等专用硬件，以及模型训练的软件和基础设施，这需要财政资源。此外，训练 基础模型（Foundation Model）非常耗时，这可能会导致更高的成本。因此，这些模型通常用作即插即用模块（指将 基础模型（Foundation Models）集成到各种应用程序中，而不需要进行大量定制）。表 I 提供了常用 基础模型（Foundation Models）的详细信息。在本节的其余部分中，我们将介绍 LLM、视觉变换器、VLM、具体多模态语言模型和视觉生成模型。在本节的最后一部分，我们介绍用于训练 基础模型（Foundation Models）的不同训练方法。

A. 术语和数学预备知识

在本节中，我们首先介绍 基础模型（Foundation Models）背景下的常用术语，并描述各种类型 基础模型（Foundation Models）的基本数学细节和训练实践。

标记化：给定一个字符序列，标记化是将序列划分为更小的单元（称为标记）的过程。根据标记化策略，标记可以是字符、单词片段、完整单词或句子的一部分。标记被表示为维度等于总词汇量大小的 1-hot 向量，并通过学习的嵌入矩阵映射到实数的低维向量。 LLM 将这些嵌入向量的序列作为原始输入，生成一系列嵌入向量作为原始输出。然后，这些输出向量被映射回标记，从而映射回文本。例如，GPT-3 的词汇表包含 50,257 个不同的标记，嵌入维度为 12,288。

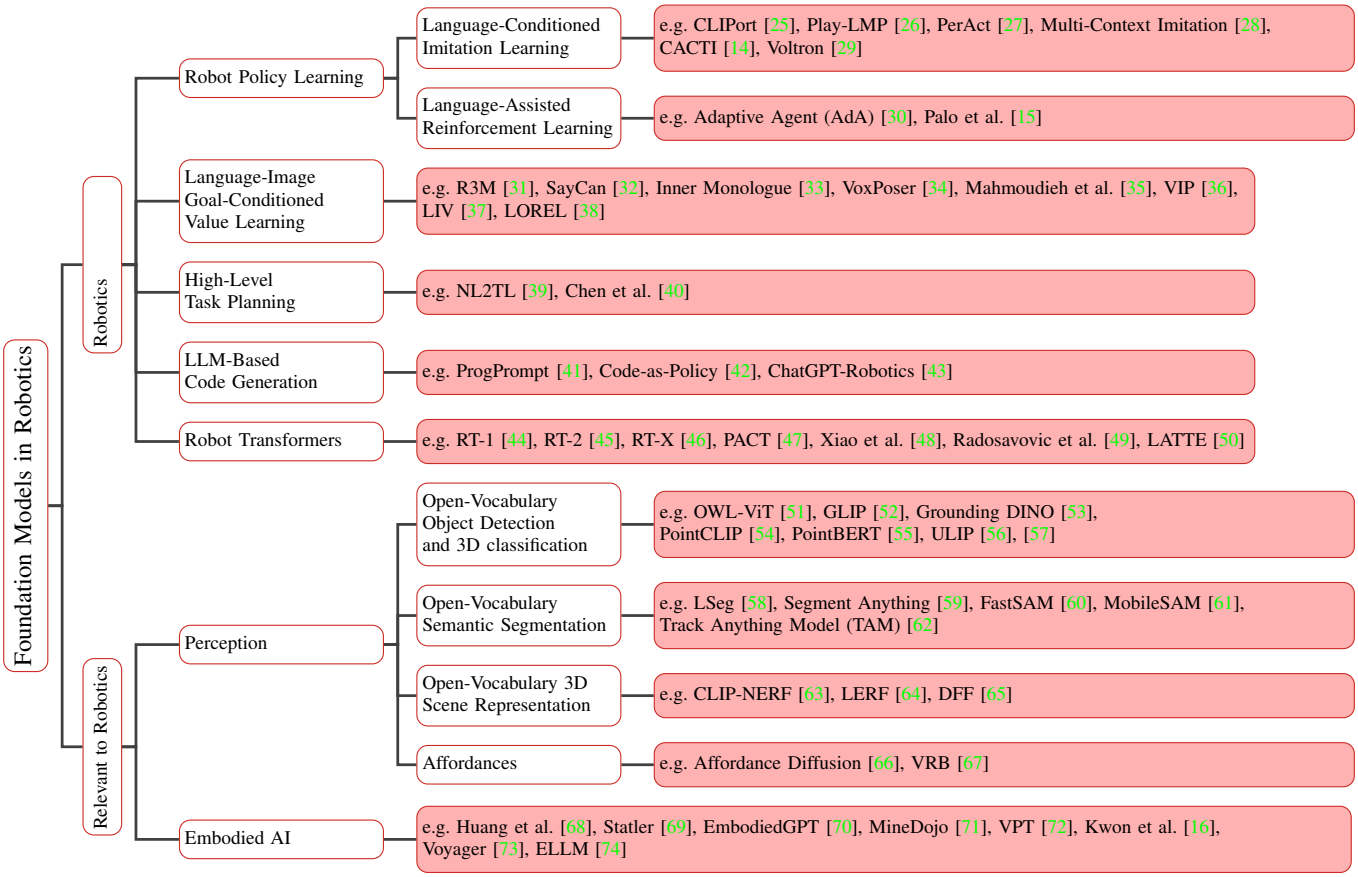


Fig. 1. Overview of Robotics Tasks Leveraging Foundation Models.

The token decoding (from low-dimension real-valued embedding vectors to high-dimension 1-hot vectors) is not deterministic, resulting in a weighting for each possible token in the vocabulary. These weightings are often used by LLMs as probabilities over tokens, to introduce randomness in the text generation process. For example, the temperature parameter in GPT-3 blends between always choosing the top-weighted token (temperature of 0) and drawing the token based on the probability distribution suggested by the weights (temperature of 1). This source of randomness is only in the token decoding process, not in the LLM itself. To the authors' knowledge, this is, in fact, the only source of randomness in the GPT family of models.

One of the most common tokenization schemes, which is used by the GPT family of models, is called byte-pair encoding [75]. Byte-pair encoding starts with a token for each individual symbol (e.g., letter, punctuation), then recursively builds tokens by grouping pairs of symbols that commonly appear together, building up to assign tokens to larger and larger groups (pairs of pairs, etc) that appear frequently together in a text corpus. The tokenization process can extend beyond text data to diverse contexts, encompassing various data modalities like images, videos, and robot actions. In these scenarios, the respective data modalities can be treated as sequential data and tokenized similarly to train generative models. For example, just as language constitutes a sequence of words, an image comprises a sequence of image patches, force sensors yield a sequence of sensory inputs at each time step, and a series of actions represent the sequential nature of tasks for a robot.

Generative Models: A generative model is a model that learns to sample from a probability distribution to create examples of data that seem to be from the same distribution as the training data. For example, a face generation model can produce images of faces that cannot be distinguished from the set of real images used to train the model. These models can be trained to be conditional, meaning they generate samples from a conditional distribution conditioned on a wide range of possible conditioning information. For example, a gender conditional face generator can generate images of female or male faces, where the desired gender is given as a conditioning input to the model.

Discriminative Models: Discriminative models are used for regression or classification tasks. In contrast to generative models, discriminative models are trained to distinguish between different classes or categories. Their emphasis lies in learning the boundaries between classes within the input space. While generative models learn to sample from the distribution over the data, discriminative models learn to evaluate the probability distribution of the output labels given the input features, or (depending on how the model is trained) learn to evaluate some statistic of the probability distribution over the outputs, such as the expected output given an input.

Transformer Architecture: Most foundation models are built on the transformer architecture, which has been instru-

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语言条件，例如CLIPort [25]、Play-LMP [26]、PerAct [27]、多上下文模仿 [28]、
模仿学习（Imitation Learning） CACTI [14]、Voltron [29]
机器人策略学习
语言辅助
例如自适应智能体（AdA）[30]，Palo 等人。 [15]
强化学习（Reinforcement Learning）
语言-图像
目标条件例如R3M [31]、SayCan [32]、内心独白 [33]、VoxPoser [34]、Mahmoudieh 等人。 [35]、VIP [36]、
LIV [37]、LOREL [38]
价值学习
高级
例如NL2TL [39]，Chen 等人。 [40]
任务规划
基于LLM
代码生成ProgPrompt [41]、代码即策略 [42]、ChatGPT-Robotics [43]
机器人变压器，例如RT-1 [44]、RT-2 [45]、RT-X [46]、PACT [47]、Xiao 等人。 [48]，拉多萨沃维奇等人。 [49]、LATTE [50]
开放词汇
例如OWL-ViT [51]、GLIP [52]、落地关联（Grounding） DINO [53]、
目标检测
PointCLIP [54]、PointBERT [55]、ULIP [56]、[57]
和 3D 分类
开放词汇，例如LSeg [58]、Segment Anything [59]、FastSAM [60]、MobileSAM [61]、
语义分割跟踪 Anything 模型 (TAM) [62]
感知
开放词汇 3D
e .g。 CLIP-NERF [63]、LERF [64]、DFF [65]
场景表示
可供性（Affordances） 可供性（Affordance） 扩散 [66]、VRB [67]
具身智能（Embodied AI） 例如黄等人。 [68]、Statler [69]、EmbodiedGPT [70]、MineDojo [71]、VPT [72]、Kwon 等人。 [16],
Voyager[73], ELLM[74]
机器人
FoundationModelsinRobotics
相关机器人

图 1. 利用 基础模型（Foundation Models） 的机器人任务概述。

令牌解码（从低维实值嵌入向量到高维 1-hot 向量）不是确定性的，导致词汇表中每个可能令牌的权重。这些权重通常被法学硕士用作标记的概率，以在文本生成过程中引入随机性。例如，GPT-3 中的温度参数混合了始终选择权重最高的标记（温度为 0）和根据权重建议的概率分布（温度为 1）绘制标记。这种随机性来源仅存在于令牌解码过程中，而不存在于 LLM 本身。据作者所知，这实际上是 GPT 系列模型中唯一的随机性来源。

GPT 系列模型使用的最常见的标记化方案之一称为字节对编码 [75]。字节对编码从每个单独符号（例如字母、标点符号）的标记开始，然后通过对经常一起出现的符号对进行分组来递归地构建标记，然后将标记分配给在文本语料库中经常一起出现的越来越大的组（成对的对等）。标记化过程可以从文本数据扩展到不同的上下文，涵盖图像、视频和机器人动作等各种数据模式。在这些场景中，各个数据模态可以被视为顺序数据，并与训练生成模型类似地进行标记化。例如，正如语言构成单词序列一样，图像包含图像块序列，力传感器产生

每个时间步的感官输入序列以及一系列动作代表了机器人任务的顺序性质。

生成模型：生成模型是一种学习从概率分布中采样以创建似乎与训练数据具有相同分布的数据示例的模型。例如，面部生成模型可以生成无法与用于训练模型的真实图像集区分开来的面部图像。这些模型可以被训练为有条件的，这意味着它们从以各种可能的条件信息为条件的条件分布中生成样本。例如，性别条件面部生成器可以生成女性或男性面部的图像，其中所需的性别作为模型的条件输入给出。

判别模型：判别模型用于回归或分类任务。与生成模型相反，判别模型被训练来区分不同的类或类别。他们的重点在于学习输入空间内类之间的边界。生成模型学习从数据分布中进行采样，而判别模型学习评估给定输入特征的输出标签的概率分布，或者（取决于模型的训练方式）学习评估输出概率分布的某些统计量，例如给定输入的预期输出。

变压器架构：大多数基础模型（Foundation Models）都建立在变压器架构之上，该架构已被证实

mental in the rise of foundation models and large language models. The following discussion was synthesized from [76], as well as online blogs, unpublished reports, and wikipedia [77]–[79]. A transformer acts simultaneously on a collection of embedded token vectors $(x_1, \ldots, x_N)$ known as a context window. The key enabling innovation of the Transformer architecture is the multi-head self-attention mechanism originally proposed in the seminal work [76]. In this architecture, each attention head computes a vector of importance weights that corresponds to how strongly a token in the context window x_i correlates with other tokens in the same window x_j. Each attention head mathematically encodes different notions of similarity, through different projection matrices used in the computation of the importance weights. Each head can be trained (backward pass) and evaluated (forward pass) in parallel across all tokens and across all heads, leading to faster training and inference when compared with previous models based on RNNs or LSTMs.

Mathematically, an attention head maps each token x_i in the context window to a "query" $q_i = W_q^T x_i$, and each other token in the context head x_j to a "key" $k_j = W_k^T x_j$. The similarity between query and key is then measured through a scaled dot product, $q_i^T k_j / \sqrt{d}$, where d is the dimension of the query and key vectors. A softmax is then taken over all j to give weights α_{ij} representing how much x_i "attends to" x_j. The tokens are then mapped to "values" with $v_j = W_v^T x_j$, and the output of the attention for position i is then given as a sum over values weighted by attention weights, $\sum_j \alpha_{ij} v_j$. One of the key reasons for the success of the transformer attention model is that it can be efficiently computed with GPUs and TPUs by parallelizing the preceding steps into matrix computations,

$$\mathrm{attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{1}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are matrices with rows q_i^T, k_i^T, and v_i^T, respectively. Each head in the model produces this computation independently, with different W_q, W_k, W_v matrices to encode different kinds of attention. The outputs from each head are then concatenated, normalized with a skip connection, passed through a fully connected ReLU layer, and normalized again with a skip connection to produce the output of the attention layer. Multiple layers are arranged in various ways to give "encoders" and "decoders," which together make up a transformer.

The size of a transformer model is typically quantified by (i) the size of the context window, (ii) the number of attention heads per layer, (iii) the size of the attention vectors in each head, and (iii) the number of stacked attention layers. For example, GPT-3's context window is 2048 tokens (corresponding to about 1500 words of text), each attention layer has 96 heads, each head has attention vectors of 128 dimensions, and there are 96 stacked attention layers in the model.

The basic multi-head attention mechanism does not impose any inherent sense of sequence or directionality in the data. However, transformers—especially in natural language applications—are often used as sequence predictors by imposing a positional encoding on the input token sequence. They are then applied to a token sequence autoregressively, meaning they predict the next token in the sequence, add that token to their context window, and repeat. This concept is elaborated below.

Autoregressive Models: The concept of autoregression has been applied in many fields as a representation of random processes whose outputs depend causally on the previous outputs. Autoregressive models use a window of past data to predict the next data point in a sequence. The window then slides one position forward, recursively ingesting the predicted data point into the window and expelling the oldest data point from the window. The model again predicts the next data point in the sequence, repeating this process indefinitely. Classical linear autoregressive models such as Auto-Regressive Moving Average (ARMA) and Auto-Regressive Moving Average with eXogenous input (ARMAX) models are standard statistical tools dating back to at least the 1970s [80]. These modeling concepts were adapted to deep learning models first with RNNs, and later LSTMs, which are both types of learnable nonlinear autoregressive models. Transformer models, although they are not inherently autoregressive, are often adapted to an autoregressive framework for text prediction tasks.

For example, the GPT family [81] builds on the original transformer model by using a modification introduced in [82] that removes the transformer encoder blocks entirely, retaining just the transformer decoder blocks. This has the advantage of reducing the number of model parameters by close to half while reducing redundant information that is learned in both the encoder and decoder. During training, the GPT model seeks to produce an output token from the tokenized corpus $\mathcal{X} = (x_1, ..., x_n)$ to minimize the negative log-likelihood within the context window of length N,

$$\mathcal{L}_{\mathrm{LLM}} = -\sum_i \log P\left(x_i \mid x_{i-N}, \ldots, x_{i-1}\right). \tag{2}$$

This results in a large pretrained model that autoregressively predicts the next likely token given the tokens in the context window. Although powerful, the unidirectional autoregressive nature of the GPT family means that these models may lag in performance on bidirectional tasks such as reading comprehension.

Masked Auto-Encoding: To address the unidirectional limitation of the GPT family and allow the model to make bidirectional predictions, works such as BERT [1] use masked autoencoding. This is achieved through an architectural change, namely the addition of a bidirectional encoder, as well as a novel pre-training objective known as masked language modeling (MLM). The MLM task simply masks a percentage of the tokens in the corpus and requires the model to predict these tokens. Through this procedure, the model is encouraged to learn the context that surrounds a word rather than just the next likely word in a sequence.

Contrastive Learning: Visual-language foundation models such as CLIP [4] typically rely on different training methods from the ones used with large language models which encourage explicitly predictive behavior. Visual-language models use contrastive representation learning, where the goal is to learn a

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基础模型（Foundation Models） 和大型语言模型的兴起。以下讨论综合自[76]以及在线博客、未发表的报告和维基百科[77]-[79]。转换器同时作用于称为上下文窗口的嵌入标记向量（x1，...，xN）的集合。Transformer 架构的关键创新是最初在开创性工作 [76] 中提出的多头自注意力机制。在此架构中，每个注意力头计算一个重要性权重向量，该向量对应于上下文窗口 xi 中的标记与同一窗口 xj 中的其他标记的相关程度。每个注意力头通过计算重要性权重时使用的不同投影矩阵，对不同的相似性概念进行数学编码。每个 head 都可以在所有 token 和所有 head 上并行训练（后向传递）和评估（前向传递），与之前基于 RNN 或 LSTM 的模型相比，训练和推理速度更快。

从数学上讲，注意力头将上下文窗口中的每个标记 xi 映射到“查询”qi = WqTxi，并将上下文头 xj 中的每个标记映射到“键”kj = WkTxj。然后通过缩放点积 qiTkj/ √d 来测量查询和键之间的相似性，其中 d 是查询的维度和关键向量。然后对所有 j 进行 softmax 计算，给出权重 αij 来表示 xi “关注”xj 的程度。然后将标记映射到 vj = WvTxj 的“值”，然后将位置 i 的注意力输出作为注意力权重加权值的总和给出，∑ jαijvj。Transformer 注意力模型成功的一个关键原因是，通过将前面的步骤并行化为矩阵计算，可以使用 GPU 和 TPU 高效地进行计算，

attn(Q, K, V) = softmax V, (1)
∑ QK√dkT ∑

其中 Q、K、V 是分别具有行 qiT,k iT 和 v iT, 的矩阵。模型中的每个头独立生成此计算，使用不同的 Wq、Wk、Wv 矩阵来编码不同类型的注意力。然后将每个头的输出连接起来，使用跳跃连接进行归一化，通过完全连接的 ReLU 层，并再次使用跳跃连接进行归一化，以产生注意层的输出。多个层以各种方式排列以提供“编码器”和“解码器”，它们共同构成了一个变压器。

Transformer 模型的大小通常通过（i）上下文窗口的大小，（ii）每层注意力头的数量，（iii）每个头中注意力向量的大小以及（iii）堆叠注意力层的数量来量化。例如，GPT-3的上下文窗口是2048个token（对应于大约1500个文本单词），每个注意力层有96个头，每个头有128维的注意力向量，模型中有96个堆叠的注意力层。

基本的多头注意力机制不会在数据中强加任何固有的顺序感或方向性

。然而，转换器（尤其是在自然语言应用中）通常通过对输入标记序列施加位置编码来用作序列预测器。然后将它们自回归地应用于标记序列，这意味着

他们预测序列中的下一个标记，将该标记添加到上下文窗口中，然后重复。下面详细阐述这个概念。

自回归模型：自回归的概念已应用于许多领域，作为随机过程的表示，其输出因果关系取决于先前的输出。自回归模型使用过去数据的窗口来预测序列中的下一个数据点。然后窗口向前滑动一个位置，递归地将预测的数据点引入窗口并从窗口中排出最旧的数据点。该模型再次预测序列中的下一个数据点，无限期地重复此过程。经典线性自回归模型，例如自回归移动平均 (ARMA) 和具有外源输入的自回归移动平均 (ARMAX) 模型，是标准统计工具，至少可以追溯到 20 世纪 70 年代 [80]。这些建模概念首先适用于 RNN 的深度学习模型，后来适用于 LSTM，这两种模型都是可学习的非线性自回归模型。 Transformer 模型虽然本质上不是自回归的，但通常适用于文本预测任务的自回归框架。

例如，GPT 系列 [81] 通过使用 [82] 中引入的修改建立在原始 Transformer 模型的基础上，该修改完全删除了 Transformer 编码器块，仅保留 Transformer 解码器块。这样做的优点是可以将模型参数的数量减少近一半，同时减少在编码器和解码器中学习到的冗余信息。在训练期间，GPT 模型寻求从标记化语料库 X = (x1, ..., xn) 生成输出标记，以最小化长度 N 的上下文窗口内的负对数似然，

$$L_{LLM} = -\sum_{i} \log P(x_i \mid x_{i-N}, \ldots, x_{i-1}) 。 \quad (2)$$

这会产生一个大型预训练模型，该模型根据上下文窗口中的标记自动回归预测下一个可能的标记。尽管功能强大，但 GPT 系列的单向自回归性质意味着这些模型在阅读理解等双向任务上的性能可能会落后。

屏蔽自动编码：为了解决 GPT 系列的单向限制并允许模型进行双向预测，BERT [1] 等作品使用屏蔽自动编码。这是通过架构更改（即添加双向编码器）以及称为掩码语言建模 (MLM) 的新颖预训练目标来实现的。 MLM 任务只是屏蔽语料库中一定比例的标记，并要求模型预测这些标记。通过此过程，鼓励模型学习单词周围的上下文，而不仅仅是序列中下一个可能的单词。

对比学习：诸如 CLIP [4] 之类的视觉语言 基础模型（Foundation Models） 通常依赖于与鼓励明确预测行为的大型语言模型所使用的训练方法不同的训练方法

。视觉语言模型使用对比表示学习，其目标是学习

joint embedding space between input modalities where similar sample pairs are closer than dissimilar ones. The training objective for many VLMs is some variation of the objective function,

$$\ell_i^{(v\to u)} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{v}_i, \mathbf{u}_i\right)/\tau\right)}{\sum_{k=1}^{N}\exp\left(\text{sim}\left(\mathbf{v}_i, \mathbf{u}_k\right)/\tau\right)}, \tag{3}$$

$$\ell_i^{(u\to v)} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{u}_i, \mathbf{v}_i\right)/\tau\right)}{\sum_{k=1}^{N}\exp\left(\text{sim}\left(\mathbf{u}_i, \mathbf{v}_k\right)/\tau\right)}, \tag{4}$$

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\left(\lambda\ell_i^{(v\to u)} + (1-\lambda)\ell_i^{(u\to v)}\right). \tag{5}$$

This objective function was popularized for multimodal input by ConVIRT [83] and first presented in prior works [84]–[87]. This objective function trains the image and text encoders to preserve mutual information between the true text and image pairs. In these equations, $\mathbf{u}_i$ and $\mathbf{v}_i$ are the ith encoded text and image respectively from $i \in 1, ..., N$ image and text pairs. The sim operation is the cosine similarity between the text and image embeddings, and τ is a temperature term. In CLIP [4] the authors use a symmetric cross-entropy loss, meaning the final loss is an average of the two loss components where each is equally weighted (i.e. $\lambda = 0.5$).

Diffusion Models: Outside of large language models and multi-modal models such as VLMs, diffusion models for image generation (e.g. DALL-E2) [88] are another class of foundation models considered in this survey. Although diffusion models were established in prior work [89], [90] the diffusion probabilistic model presented in [91] popularized the method. The diffusion probabilistic model is a deep generative model that is trained in an iterative forward and reverse process. The forward process adds Gaussian noise to an input x_0 in a Markov chain until x_T when the result is zero mean isotropic noise. This means the forward process produces a trajectory of noise $q(x_{1:T}|x_0)$ as,

$$q\left(\mathbf{x}_{1:T} \mid \mathbf{x}_0\right) := \prod_{t=1}^{T} q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}\right). \tag{6}$$

At each time step $q(x_t|x_{t-1})$ is described by a normal distribution with mean $\sqrt{1-\beta_t}\mathbf{x}_{t-1}$ and covariance $\beta_t\mathbf{I}$ where β_t is scheduled or a fixed hyperparameter.

The reverse process requires the model to learn to the transitions that will de-noise the zero mean Gaussian and produce the input image. This process is also defined as a Markov chain where the transition distribution at time t is $p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right) := \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta\left(\mathbf{x}_t, t\right), \boldsymbol{\Sigma}_\theta\left(\mathbf{x}_t, t\right)\right)$. For completeness, the reverse process Markov chain is given by,

$$p_\theta\left(\mathbf{x}_{0:T}\right) := p\left(\mathbf{x}_T\right)\prod_{t=1}^{T} p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right). \tag{7}$$

Diffusion models are trained using a reduced form of the evidence lower bound loss function that is typical of variational generative models like variational autoencoders (VAEs). The reduced loss function used for training is

$$\begin{aligned}\mathcal{L} = \mathbb{E}_q[&D_{\text{KL}}(q(\mathbf{x}_T \mid \mathbf{x}_0)\|p(\mathbf{x}_T)) \\ &+ \sum_{t>1} D_{\text{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \\ &- \log p_\theta(\mathbf{x}_0 \mid \mathbf{x}_1)],\end{aligned} \tag{8}$$

where $D_{\text{KL}}(q||p)$ denotes Kullback–Leibler divergence, which is a measure of how different a distribution q is from a distribution p.

B. Large Language Model (LLM) Examples and Historical Context

LLMs have billions of parameters and are trained on trillions of tokens. This large scale has allowed models such as GPT-2 [92] and BERT [1] to achieve state-of-the-art performance on the Winograd Schema challenge [93] and the General Language Understanding Evaluation (GLUE) [94] benchmarks, respectively. Their successors include GPT-3 [2], LLaMA [95], and PaLM [96] has grown considerably in the number of parameters (typically now over 100 billion), the size of the context window (typically now over 1000 tokens), and the size of the training data set (typically now 10s of terabytes of text). GPT-3 is trained on the Common Crawl dataset. Common Crawl contains petabytes of publicly available data over 12 years of web crawling and includes raw web page data, metadata, and text extracts. LLMs can also be multilingual. For example, ChatGLM-6B and GLM-130B [97] is a bilingual (English and Chinese) pretrained language model with 130 billion parameters. LLMs can also be fine-tuned, a process by which the model parameters are adjusted with domain-specific data to align the performance of the LLM to a specific use case. For example, GPT-3 and GPT-4 [3] have been fine-tuned using reinforcement learning with human feedback (RLHF).

C. Vision Transformers

A Vision Transformer (ViT) [98]–[100] is a transformer architecture for computer vision tasks including image classification segmentation, and object detection. A ViT treats an image as a sequence of image patches referred to as tokens. In the image tokenization process, an image is divided into patches of fixed size. Then the patches are flattened into a one-dimensional vector which is referred to as linear embedding. To capture the spatial relationships between image patches, positional information is added to each token. This process is referred to as position embedding. The image tokens incorporated with position encoding are fed into the transformer encoder and the self-attention mechanism enables the model to capture long-term dependencies and global patterns in the input data. In this paper, we focus only on those ViT models with a large number of parameters. ViT-G [101] scales up the ViT model and has 2B parameters. Additionally, ViT-e [102] has 4B parameters. ViT-22B [103] is a vision transformer model at 22 billion parameters, which is used in PaLM-E and PaLI-X [104] and helps with robotics tasks.

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输入模态之间的联合嵌入空间，其中相似的样本对比不相似的样本对更接近。许多 VLM 的训练目标是目标函数的一些变体，

exp (sim (vi, ui)/ τ)
ℓ (i v→u) = − log ∑Nk=1 exp (sim (vi, uk)/ τ) , (3)
exp (sim (ui, vi)/ τ)
ℓ (i u→v) = − log ∑Nk=1 exp (sim (ui, vk)/ τ) , (4)

N
L = N 1 ∑ λℓ (i v→u) + (1 − λ)ℓ (i u→v) 。 (5)
i=1

该目标函数由 ConVIRT [83] 推广用于多模态输入，并首次在先前的工作中提出[84]-[87]。该目标函数训练图像和文本编码器以保留真实文本和图像对之间的相互信息。在这些方程中，ui 和 vi 分别是来自 i ∈ 1, ..., N 个图像和文本对的第 i 个编码文本和图像。 sim 运算是文本和图像嵌入之间的余弦相似度，τ 是温度项。在 CLIP [4] 中，作者使用对称交叉熵损失，这意味着最终损失是两个损失分量的平均值，其中每个损失分量的权重相等（即 λ = 0.5 ）。

扩散模型：除了大型语言模型和多模态模型（例如 VLM ）之外，用于图像生成的扩散模型（例如 DALL-E2 ）[88] 是本篇综述中考虑的另一类 基础模型（Foundation Models ）。尽管先前的工作[89]建立了扩散模型，但[90][91]中提出的扩散概率模型推广了该方法。扩散概率模型是一种深度生成模型，在迭代的正向和反向过程中进行训练。前向过程将高斯噪声添加到马尔可夫链中的输入 x0 直到 xT，此时结果为零均值各向同性噪声。这意味着前向过程产生噪声轨迹 q(x1:T |x0)，如下所示：

T
q (x1:T | x0) := ∏ q (xt | xt−1) 。 (6)
t=1

在每个时间步 q(xt|xt−1) 由均值√ 1 − βtxt−1 和协方差 βtI 的正态分布来描述，其中 βt 是预定的或固定的超参数。

相反的过程要求模型学习对零均值高斯进行去噪并生成输入图像的转换。该过程也被定义为马尔可夫链，其中 t 时刻的转移分布为 pθ (xt−1 | xt) := N (xt−1; μθ (xt, t) , Σθ (xt, t))。为了完整起见，逆向过程马尔可夫链由下式给出：

T
pθ (x0:T) := p (xT) ∏ pθ (xt−1 | xt) 。 (7)
t=1

使用证据下界损失函数的简化形式来训练扩散模型，该函数是变分的典型特征

生成模型，如变分自动编码器（VAE）。用于训练的减少损失函数是

$$L = E_q[D_{KL}(q(x_T \mid x_0) \| p(x_T)) + \sum_{t>1} D_{KL}(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t)) - \log p_\theta(x_0 \mid x_1)], \quad (8)$$

其中 DKL(q||p) 表示 Kullback-Leibler 散度，它是分布 q 与分布 p 差异程度的度量。

B. 大型语言模型 (LLM) 示例和历史背景

LLM 拥有数十亿个参数，并接受数万亿个代币的训练。这种大规模使得 GPT-2 [92] 和 BERT [1] 等模型分别在 Winograd Schema 挑战 [93] 和通用语言理解评估 (GLUE) [94] 基准测试中实现最先进的性能。它们的后继者包括 GPT-3 [2]、LLaMA [95] 和 PaLM [96] 在参数数量（通常现在超过 1000 亿）、上下文窗口的大小（现在通常超过 1000 个标记）和训练数据集的大小（现在通常是 10 TB 文本）方面都有了显着增长。 GPT-3 在 Common Crawl 数据集上进行训练。 Common Crawl 包含超过 12 年网络爬行的 PB 级公开数据，包括原始网页数据、元数据和文本摘录。法学硕士也可以是多语言的。例如，ChatGLM-6B和GLM-130B[97]是具有1300亿个参数的双语（英语和中文）预训练语言模型。法学硕士还可以进行微调，即使用特定领域的数据调整模型参数的过程，以使法学硕士的性能与特定用例保持一致。例如，GPT-3 和 GPT-4 [3] 已使用 强化学习（Reinforcement Learning） 和人类反馈 (RLHF) 进行了微调。

C. 视觉变形金刚

视觉变压器（ViT）[98]-[100]是一种用于计算机视觉任务的变压器架构，包括图像分类分割和对象检测。ViT 将图像视为一系列图像块（称为标记）。在图像标记化过程中，图像被划分为固定大小的块。然后将补丁展平为一维向量，这称为线性嵌入。为了捕获图像块之间的空间关系，位置信息被添加到每个标记中。这个过程称为位置嵌入。与位置编码相结合的图像标记被馈送到变压器编码器中，自注意力机制使模型能够捕获输入数据中的长期依赖性和全局模式。在本文中，我们只关注那些具有大量参数的 ViT 模型。ViT-G [101]放大了ViT模型并具有2B参数。此外，ViT-e [102] 有 4B 参数。 ViT-22B [103] 是一个具有 220 亿个参数的视觉变换器模型，用于 PaLM-E 和 PaLI-X [104]，有助于完成机器人任务。

DINO [105] is a self-supervised learning method, for training ViT. DINO is a form of knowledge distillation with no labels. Knowledge distillation is a learning framework where a smaller model (student network) is trained to mimic the behavior of a larger more complex model (teacher network). Both networks share the same architecture with different sets of parameters. Given a fixed teacher network, the student network learns its parameters by minimizing the cross-entropy loss w.r.t. the student network parameters. The neural network architecture is composed of ViT or ResNet [106] backbone and a projection head that includes layers of multi-layer perception (MLP). Self-supervised ViT features learned using DINO contain explicit information about the semantic segmentation of an image including scene layout and object boundaries with such clarity that is not achieved using supervised ViTs or convnets.

DINOv2 [107] provides a variety of pretrained visual models that are trained with different vision transformers (ViT) on the LVD-142M dataset introduced in [107]. It is trained using a discriminative self-supervised method on a compute cluster of 20 nodes equipped with 8 V100-32GB GPUs. DINOv2 provides various visual features at the image (e.g. detection) or pixel level (e.g. segmentation). SAM [59] provides zero-shot promptable image segmentation. It is discussed in more detail in Section IV.

D. Multimodal Vision-Language Models (VLMs)

Multimodal refers to the ability of a model to accept different "modalities" of inputs, for example, images, texts, or audio signals. Visual-language models (VLM) are a type of multimodal model that takes in both images and text. A commonly used VLM in robotics applications is Contrastive Language-Image Pre-training (CLIP) [4]. CLIP offers a method to compare the similarity between textual descriptions and images. CLIP uses internet-scale image-text pairs data to capture the semantic information between images and text. CLIP model architecture contains a text encoder [92] and an image encoder (a modified version of vision transformer ViT) that are trained jointly to maximize the cosine similarity of the image and text embeddings. CLIP uses contrastive learning together with language models and visual feature encoders to incorporate models for zero-shot image classification.

BLIP [108] focuses on multimodal learning by jointly optimizing three objectives during pretraining. These objectives include Image-Text Contrastive Loss, Image-Text Matching Loss, and Language Modeling Loss. The method leverages noisy web data by bootstrapping captions, enhancing the training process. CLIP^2 [109] aims to build well-aligned and instance-based text-image-point proxies. It learns semantic and instance-level aligned point cloud representations using a cross-modal contrastive objective. FILIP [110] focuses on achieving finer-level alignment in multimodal learning. It incorporates a cross-modal late interaction mechanism that utilizes token-wise maximum similarity between visual and textual tokens. This mechanism guides the contrastive objective and improves the alignment between visual and textual information. FLIP [111] proposes a simple and more efficient training method for CLIP. FLIP randomly masks out and removes a significant portion of image patches during training. This approach aims to improve the training efficiency of CLIP while maintaining its performance.

E. Embodied Multimodal Language Models

An embodied agent is an AI system that interacts with a virtual or physical world. Examples include virtual assistance or robots. Embodied language models are foundation models that incorporate real-world sensor and actuation modalities into pretrained large language models. Typical vision-language models are trained on general vision-language tasks such as image captioning or visual question answering. PaLM-E [6] is a multimodal language model that has been trained on not only internet-scale general vision-language data, but also on embodied, robotics data, simultaneously. In order to connect the model to real world sensor modalities, PaLM-E's architecture injects (continuous) inputs such as images, low-level states, or 3D neural scene representations into the language embedding space of a decoder-only language model to enable the model to reason about text and other modalities jointly. The main PaLM-E version is built from the PaLM LLM [96] and a ViT [103]. The ViT transforms an image into a sequence of embedding vectors which are projected into the language embedding space via an affine transformation. The whole model is trained end-to-end, starting from a pre-trained LLM and ViT model. The authors also explore different strategies such as freezing the LLM and just training the ViT, which leads to worse performance. Given multimodal inputs, the output of PaLM-E is text decoded auto-regressively. In order to connect this output to a robot for control, language conditioned short-horizon policies can be used. In this case, PaLM-E acts as a high-level control policy. Experiments show that a single PaLM-E, in addition to being a vision-language generalist, is able to perform many different robotics tasks over multiple robot embodiments. The model exhibits positive transfer, i.e. simultaneously training on internet-scale language, general vision-language, and embodied domains leads to higher performance compared to training the model on single tasks.

F. Visual Generative Models

Web-scale diffusion models such as OpenAI's DALL-E [112] and DALL-E2 [88] provide zero-shot text-to-image generation. They are trained on hundreds of millions of image-caption pairs from the internet. These models learn a language-conditioned distribution over images from which an image can be generated using a given prompt. The DALL-E2 architecture includes a prior that generates a CLIP image embedding from a text caption, and a decoder that generates an image conditioned on the image embedding.

III. Robotics

In this section, we delve into robot decision-making, planning, and control. Within this realm, Large Language Models (LLMs) and Visual Language Models (VLMs) may hold the

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DINO[105]是一种自监督学习方法，用于训练ViT。 DINO 是一种没有标签的知识蒸馏形式。知识蒸馏是一种学习框架，其中较小的模型（学生网络）经过训练以模仿较大且更复杂的模型（教师网络）的行为。两个网络共享相同的架构，但具有不同的参数集。给定固定的教师网络，学生网络通过最小化交叉熵损失来学习其参数。学生网络参数。神经网络架构由 ViT 或 ResNet [106] 主干和包含多层感知（MLP）层的投影头组成。使用 DINO 学习的自监督 ViT 特征包含有关图像语义分割的明确信息，包括场景布局和对象边界，其清晰度是使用监督 ViT 或卷积网络无法实现的。

DINOv2 [107] 提供了各种预训练的视觉模型，这些模型在 [107] 中引入的 LVD-142M 数据集上使用不同的视觉变换器 (ViT) 进行训练。它是在配备 8 个 V100-32GB GPU 的 20 个节点的计算集群上使用判别性自我监督方法进行训练的。 DINOv2 在图像（例如检测）或像素级别（例如分割）提供各种视觉特征。SAM [59] 提供零样本快速图像分割。第四节对此进行了更详细的讨论。
D. 多模态视觉语言模型 (VLM)

多模态是指模型接受不同“模态”输入的能力，例如图像、文本或音频信号。视觉语言模型 (VLM) 是一种同时接收图像和文本的多模态模型。机器人应用中常用的 VLM 是对比语言图像预训练 (CLIP) [4]。 CLIP提供了一种比较文本描述和图像之间相似性的方法。 CLIP 使用互联网规模的图像文本对数据来捕获图像和文本之间的语义信息。 CLIP模型架构包含一个文本编码器[92]和一个图像编码器（视觉变换器ViT的修改版本），它们经过联合训练以最大化图像和文本嵌入的余弦相似度。 CLIP 将对比学习与语言模型和视觉特征编码器结合使用，以合并零样本图像分类的模型。

BLIP [108]通过在预训练期间联合优化三个目标来关注多模态学习。这些目标包括图像-文本对比损失、图像-文本匹配损失和语言建模损失。该方法通过引导字幕来利用嘈杂的网络数据，从而增强训练过程。CLIP2 [109] 旨在构建良好对齐且基于实例的文本图像点智能体。它使用跨模式对比目标来学习语义和实例级对齐的点云表示。 FILIP [110] 专注于在多模态学习中实现更精细的对齐。它采用了跨模式后期交互机制，利用视觉和文本标记之间的标记最大相似性。这种机制指导对比目标并改善视觉和文本信息之间的一致性。 FLIP [111]提出了一种简单且更高效的方法

CLIP训练方法

。 FLIP 在训练过程中随机屏蔽并删除大部分图像块。这种方法旨在提高 CLIP 的训练效率，同时保持其性能。
E. 体现多模态语言模型

具身智能体是一种与虚拟或物理世界交互的人工智能系统。示例包括虚拟协助或机器人。体现的语言模型是 基础模型（Foundation Models），它将现实世界的传感器和驱动模式合并到预训练的大型语言模型中。典型的视觉语言模型是针对一般视觉语言任务（例如图像字幕或视觉问答）进行训练的。 PaLME [6] 是一种多模态语言模型，不仅经过互联网规模的通用视觉语言数据的训练，而且还同时经过具体的机器人数据的训练。为了将模型连接到现实世界的传感器模态，PaLME 的架构将图像、低级状态或 3D 神经场景表示等（连续）输入注入到仅解码器语言模型的语言嵌入空间中，以使模型能够联合推理文本和其他模态。主要的 PaLM-E 版本是由 PaLM LLM [96] 和 ViT [103] 构建的。 ViT 将图像转换为一系列嵌入向量，这些向量通过仿射变换投影到语言嵌入空间中。整个模型从预训练的 LLM 和 ViT 模型开始进行端到端训练。作者还探索了不同的策略，例如冻结 LLM 并仅训练 ViT，这会导致表现更差。给定多模态输入，PaLM-E 的输出是自回归文本解码。为了将此输出连接到机器人进行控制，可以使用语言条件的短视野策略。在这种

情况下，PaLM-E 充当高级控制策略。实验表明，单个 PaLM-E 除了是视觉语言通才之外，还能够在多个机器人实施例中执行许多不同的机器人任务。该模型表现出正向迁移，即与在单一任务上训练模型相比，同时对互联网规模语言、通用视觉语言和具体领域进行训练会带来更高的性能。

F. 视觉生成模型

OpenAI 的 DALLE [112] 和 DALL-E2 [88] 等网络规模扩散模型提供了零样本文本到图像的生成。他们接受了来自互联网的数亿个图像描述对的训练。这些模型学习图像上的语言条件分布，可以使用给定的提示生成图像。 DALL-E2 架构包括从文本标题生成 CLIP 图像嵌入的先验器，以及生成以图像嵌入为条件的图像的解码器。

三．机器人技术

在本节中，我们将深入研究机器人的决策、规划和控制。在这个领域中，大型语言模型（LLM）和视觉语言模型（VLM）可能占据主导地位。

TABLE I
LARGE PRETRAINED MODELS

Model	Architecture	Size	Training Data	What to Pretrain	How to Pretrain	Hardware
CLIP [4]	ViT-L/14@336px and a text encoder [92]	0.307B	400M image-text pairs	zero-shot image classification	contrastive pre-training	fine-tuned CLIP model is trained for 12 days on 256 V100 GPUs
GPT-3 [2]	transformer (slight modification of GPT-2)	175B	Common Crawl (about a trillion words)	text output	autoregressive model	NPA *
GPT-4 [3]	NPA	NPA	NPA	text output	NPA	NPA
PaLI-X [104]	encoder-decoder	55B	10B image-text pairs from WebLI [102] and auxiliary tasks	text and image to text output	autoregressive model	runs on multi-TPU cloud service
DALL-E [112]	decoder-only transformer	12B	250M text-image pairs	zero-shot text-to-image generation	autoregressive model	NPA
DALL-E2 [88]	a prior based on CLIP+ a decoder	3.5B	CLIP and DALL-E [112]	zero-shot text-to-image generation	diffusion	NPA
DINOv2 [107]	ViT-g/14	1.1B	LVD-142M [107]	visual-features (image-level and pixel-level)	discriminative	20 nodes equipped with 8 V100-32GB GPUs
SAM [59]	MAE [113] vision transformer+CLIP [114] text encoder	632M for ViT-H + 63M for CLIP text encoder	SA-1B dataset [59] that includes 1.1B segmentation masks on 11M images	zero-shot promptable image segmentation	supervised learning	256 A100 GPUs for 68 hours

* NPA stands for not publicly available.

potential to serve as valuable tools for enhancing robotic capabilities. For instance, LLMs may facilitate the process of task specification, allowing robots to receive and interpret high-level instructions from humans. VLMs may also promise contributions to this field. VLMs specialize in the analysis of visual data. This visual understanding is a critical component of informed decision-making and complex task execution for robots. Robots can now leverage natural language cues to enhance their performance in tasks involving manipulation, navigation, and interaction. Vision-language goal-conditioned policy learning, whether through imitation learning or reinforcement learning, holds promise for improvement using foundation models. Language models also play a role in offering feedback for policy learning techniques. This feedback loop fosters continual improvement in robotic decision-making, as robots can refine their actions based on the feedback received from an LLM. This section underscores the potential contributions of LLMs and VLMs in robot decision-making. Assessing and comparing the contributions of papers in this section presents greater challenges compared to the other sections like the Perception Section (IV) or the Embodied AI Section (V). This is due to the fact that most papers in this section either rely on hardware experiments, using custom elements in the low-level control and planning stack that are not easily transferred to other hardware or other experimental setups, or they utilize non-physics-based simulators, which allow these low-level parts of the stack to be ignored, but leaving open the issue of non-transferability between different hardware implementations. In Section VI, we discuss the lack of benchmarking and reproducibility that needs to be addressed in future research.

A. Robot Policy Learning for Decision Making and Control

In this section we discuss robot policy learning including language-conditioned imitation learning and language-assisted reinforcement learning.

1) Language-conditioned Imitation Learning for Manipulation: In language-conditioned imitation learning, a goal-conditioned policy $\pi_\theta(a_t|s_t, l)$ is learned that outputs actions $a_t \in \mathcal{A}$ conditioned on the current state $s_t \in \mathcal{S}$ and language instruction $l \in \mathcal{L}$. The loss function is defined as the maximum likelihood goal conditioned imitation objective:

$$\mathcal{L}_{\text{GCIL}} = \mathbb{E}_{(\tau,l)\sim\mathcal{D}} \sum_{t=0}^{|\tau|} \log \pi_\theta(a_t|s_t, l), \tag{9}$$

where $\mathcal{D}$ is the language-annotated demonstration dataset $\mathcal{D} = \{\tau_i\}_i^N$. Demonstrations can be represented as trajectories, or sequences of images, RGB-D voxel observations, etc. Language instructions are paired with demonstrations to be used as the training dataset. Each language-annotated demonstration τ_i consists of $\tau_i = \{(s_1, l_1, a_1), (s_2, l_2, a_2), ...\}$. At test time, the robot is given a series of instructions and the language-conditioned visuomotor policy π_θ provides actions a_t in a closed loop given the instruction at each time step. The main challenges in this domain are: (i) obtaining a sufficient volume of demonstrations and conditioning labels to train a policy, (ii) distribution shift under the closed-loop policy—the feedback of the policy can lead the robot into regions of the state space that are not well-covered in the training data, negatively impacting performance. (All the following papers in this subsection focus on robot manipulation tasks.)

Since generating language-annotated data by pairing demonstrations with language instruction is an expensive pro-

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表 I 大型预训练模型

模型	架构	大小	训练数据	预训练什么	如何预训练	硬件
CLIP [4]	ViT-L/14@336px 和
文本编码器 [92]	0.307B	400M 图像文本对	零样本图像
分类	对比
预训练	微调
CLIP 模型
在 256 个
V100 GPU 上训练 12
天
GPT-3 [2]	变压器（GPT-2 的轻微
修改）	175B	普通爬行（约
一万亿字）	文本输出	自回归
模型	NPA *
GPT-4 [3]	NPA	NPA	NPA	文本输出	NPA	NPA
PaLI-X [104]	编码器-解码器	55B	10B 图像-文本对
来自 WebLI [102] 和
辅助任务	文本和图像到
文本输出	自回归
模型	在
多 TPU
云服务上运行
DALL-E [112]	仅解码器
变压器	12B	250M 文本图像对	零样本
文本到图像
生成	自回归
模型	NPA
DALL-E2 [88]	基于
CLIP+解码器的先验	3.5B	CLIP和
DALL-E [112]	零样本
文本到图像
生成	扩散	NPA
DINOv2 [107]	ViT-g/14	1.1B	LVD-142M [107]	视觉特征
（图像级和
像素级）|判别性|20 个节点
配备
8 V100-32GB

GPU|
|SAM [59]|MAE [113] 视觉
变压器+CLIP
[114] 文本编码器|632M 用于
ViT-H + 63M
用于 CLIP 文本
编码器|SA-1B 数据集 [59]
包括 1.1B
分割掩模
11M 图像|零镜头
提示图像
分割|监督学习|256 A100
GPU 68 小时|

> * NPA 代表未公开可用。

有潜力成为增强机器人能力的有价值的工具。例如，法学硕士可以促进任务规范的过程，允许机器人接收和解释来自人类的高级指令。 VLM 也可能承诺对该领域做出贡献。 VLM 专注于视觉数据分析。这种视觉理解是机器人做出明智决策和执行复杂任务的关键组成部分。机器人现在可以利用自然语言提示来提高其在涉及操作、导航和交互的任务中的表现。视觉语言目标条件策略学习，无论是通过 模仿学习（Imitation Learning） 还是 强化学习（Reinforcement Learning），都有望使用 基础模型（Foundation Models） 进行改进。语言模型在为策略学习技术提供反馈方面也发挥着作用。这种反馈循环促进了机器人决策的持续改进，因为机器人可以根据法学硕士收到的反馈来完善其行为。本节强调了 LLM 和 VLM 在机器人决策中的潜在贡献。与感知部分 (IV) 或 具身智能（Embodied AI） 部分 (V) 等其他部分相比，评估和比较本部分论文的贡献提出了更大的挑战。这是因为本节中的大多数论文要么依赖于硬件实验，使用低级控制和规划堆栈中的自定义元素，这些元素不容易转移到其他硬件或其他实验设置，要么使用非基于物理的模拟器，这允许忽略堆栈的这些低级部分，但留下了不同硬件实现之间不可转移性的问题

。在第六节中，我们讨论了未来研究中需要解决的缺乏基准测试和可重复性的问题。

A. 用于决策和控制的机器人策略学习

在本节中，我们讨论机器人策略学习，包括语言条件 模仿学习（Imitation Learning） 和语言辅助 强化学习（Reinforcement Learning）。

1）用于操作的语言条件 模仿学习（Imitation Learning）：在语言条件 模仿学习（Imitation Learning） 中，学习目标条件策略 πθ(at|st, l)，以当前状态 st εS 和语言指令 l εL 为条件输出 εA 处的动作。损失函数定义为条件模仿目标的最大似然目标：

|τ |
LGCIL = E(τ,l)~D logπθ(at|st, l), (9)
t=0

其中 D 是语言注释的演示数据集 D = {τi}N i。演示可以表示轨迹或图像序列、RGB-D 体素观察结果等。语言指令与演示配对，用作训练数据集。每个语言注释的演示 τi 由 τi = {(s1, l1, a1), (s2, l2, a2), ...} 组成。

在测试时，机器人会收到一系列指令，并且语言条件视觉运动策略 πθ 在给定每个时间步指令的闭环中提供动作。该领域的主要挑战是：（i）获得足够数量的演示和调节标签来训练策略，（ii）闭环策略下的分布转移 - 策略的反馈可能导致机器人进入训练数据未充分覆盖的状态空间区域，从而对性能产生负面影响。（本小节中的所有以下论文都重点关注机器人操作任务。）

由于通过将演示与语言指令配对来生成语言注释数据是一种昂贵的过程

cess, the authors in Play-LMP [26] propose learning from teleoperated play data. In this setting, reusable latent plan representations are learned from unlabeled play data. Also, a goal-conditioned policy is learned to decode the inferred plan to perform the task specified by the user. In addition, the distributional shift in imitation learning is analyzed and it is shown in this setting that the play data is more robust with respect to perturbation compared to expert positive demonstrations. Note that language goal l in (9) can be substituted with any other type of goal for example goal image, which is another common choice of goal in goal-conditioned imitation learning.

In a follow-up work [28], the authors present multi-context imitation (MCIL) which uses language-conditioned imitation learning over unstructured data. The multi-Context imitation framework is based on relabeled imitation learning and labeled instruction following. MCIL assumes access to multiple contextual imitation datasets, for example, goal image demonstrations, language goal demonstrations, or one-hot task demonstrations. MCIL trains a single latent goal-conditioned policy over all datasets simultaneously by encoding contexts in the shared latent space using the associated encoder for each context. Then a goal-conditioned imitation loss is computed by averaging over all datasets. The policy and goal-encoders are trained end-to-end. Another approach to tackle the data annotation challenge in language-conditioned imitation learning involves utilizing foundation models to offer feedback by labeling demonstrations. In [115], the authors propose to use pretrained foundation models to provide feedback. To deploy a trained policy to a new task or new environment, the policy is played using randomly generated instructions, and a pretrained foundation model provides feedback by labeling the demonstration. Also, this paired instruction-demonstration data can be used for policy fine-tuning. CLIPort [25] also presents a language-conditioned imitation learning for vision-based manipulation. A two-stream architecture is presented that combines the semantic understanding of CLIP with the spatial precision of Transporter [116]. This end-to-end framework solves language-specified manipulation tasks without any explicit representation of the object poses or instance segmentation. CLIPort grounds semantic concepts in precise spatial reasoning, but it is limited to 2D observation and action spaces. To address this limitation, the authors of PerAct (Perceiver-Actor) [27] propose to represent observation and action spaces with 3D voxels and employ the 3D structure of voxel patches for efficient language-conditioned behavioral cloning with transformers to imitate 6-DoF manipulation tasks from just a few demonstrations. While 2D behavioral cloning methods such as CLIPort are limited to single-view observations, 3D approaches such as PerAct allow for multi-view observations as well as 6-DoF action spaces. PerAct uses only CLIP's language encoder to encode the language goal. PerAct takes language goals and RGB-D voxel observations as inputs to a Perceiver Transformer and outputs discretized actions by detecting the next best voxel action. PerAct is trained through supervised learning with discrete-time input actions from the demonstration dataset. The demonstration dataset includes voxel observations paired with language goals and keyframe action sequences. An action consists of a 6-DoF pose, gripper open state, and collision avoidance action. During training, a tuple is randomly sampled and the agent predicts the keyframe action given the observation and goal.

Grounding semantic representations into a spatial environment is essential for effective robot interaction. CLIPort and PerAct utilize CLIP (which is trained based on contrastive learning) for semantic reasoning and Transporter and Perceiver for spatial reasoning.

Voltron [29] presents a framework for language-driven representation learning in robotics. Voltron captures semantic, spatial, and temporal representations that are learned from videos and captions. Contrastive learning captures semantic representations but loses spatial relationships, and in contrast, masked autoencoding captures spatial and not semantic representations. Voltron trades off language-conditioned visual reconstruction for local spatial representations and visually-grounded language generation to capture semantic representations. This framework includes grasp affordance prediction, single-task visuomotor control, referring expression grounding, language-conditioned imitation, and intent-scoring tasks. Voltron models take videos and their associated language captions as input to a multimodal encoder whose outputs are then decoded to reconstruct one or more frames from a masked context. Voltron starts with a masked autoencoding backbone and adds a dynamic component to the model by conditioning the MAE encoder on a language prefix. Temporal information is captured by conditioning on multiple frames.

Deploying robot policy learning techniques that leverage language-conditioned imitation learning with real robots presents ongoing challenges. These models rely on end-to-end learning, where the policy maps pixels or voxels to actions. As they are trained through supervised learning on demonstration datasets, they are susceptible to issues related to generalization and distribution shifts. To improve robustness and adaptability, techniques such as data augmentation and domain adaptation can make the policies more robust to the distribution shift.

CACTI [14] is a novel framework designed to enhance scalability in robot learning using foundation models such as Stable Diffusion [117]. CACTI introduces the four stages of data collection, data augmentation, visual representation learning, and imitation policy training. In the data collection stage, limited in-domain expert demonstration data is collected. In the data augmentation stage, CACTI employs visual generative models such as Stable Diffusion [117] to boost visual diversity by augmenting the data with scene and layout variations. In the visual representation learning stage, CACTI leverages pretrained zero-shot visual representation models trained on out-of-domain data to improve training efficiency. Finally, in the imitation policy training stage, a general multi-task policy is learned using imitation learning on the augmented dataset with compressed visual representations as input. CACTI is trained for multi-task and multi-scene manipulation in kitchen environments, both in simulation and the real world. The use of these techniques enhances the generalization ability of the framework and enables it to learn from a wide range of environments.

Beyond language, recent works have investigated other

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过程中，Play-LMP [26] 的作者提出从远程操作的游戏数据中学习。在这种设置中，可重用的潜在计划表示是从未标记的游戏数据中学习的。此外，还学习目标条件策略来解码推断的计划，以执行用户指定的任务。此外，还分析了 模仿学习（Imitation Learning） 中的分布变化，并表明在此设置中，与专家的积极演示相比，游戏数据在扰动方面更加稳健。请注意，（9）中的语言目标 l 可以替换为任何其他类型的目标，例如目标图像，这是目标条件 模仿学习（Imitation Learning） 中目标的另一种常见选择。

在后续工作 [28] 中，作者提出了多上下文模仿（MCIL），它在非结构化数据上使用语言条件 模仿学习（Imitation Learning）。多上下文模仿框架基于重新标记的模仿学习（Imitation Learning）和标记的指令跟随。 MCIL 假设可以访问多个上下文模仿数据集，例如目标图像演示、语言目标演示或单热点任务演示。MCIL 通过使用每个上下文的关联编码器在共享潜在空间中编码上下文，同时在所有数据集上训练单个潜在目标条件策略。然后通过对所有数据集进行平均来计算目标条件模仿损失。策略和目标编码器经过端到端训练。解决语言条件 模仿学习（Imitation Learning） 中数据注释挑战的另一种方法是利用 基础模型（Foundation Models） 通过标记演示来提供反馈。在[115]中，作者建议使用预训练的 基础模型（Foundation Models） 来提供反馈。要将经过训练的策略部署到新任务或新环境，可以使用随机生成的指令来执行该策略，并且预训练的 基础模型（Foundation Model） 通过标记演示来提供反馈。此外，该配对的指令演示数据可用于策略微调。 CLIPort [25] 还提出了一种用于基于视觉的操作的语言条件 模仿学习（Imitation Learning）。提出了一种双流架构，它将 CLIP 的语义理解与 Transporter 的空间精度结合起来[116]。这个端到端框架解决了语言指定的操作任务，无需任何对象姿势或实例分割的显式表示。CLIPort 将语义概念建立在精确的空间推理中，但仅限于 2D 观察和动作空间。为了解决这个限制，PerAct (Perceiver-Actor) [27] 的作者提出用 3D 体素表示观察和动作空间，并利用体素补丁的 3D 结构通过 Transformer 进行高效的语言条件行为克隆，从而通过几个演示来模仿 6-DoF 操作任务。虽然 CLIPort 等 2D 行为克隆方法仅限于单视图观察，但 PerAct 等 3D 方法允许多视图观察以及 6-DoF 动作空间。PerAct 仅使用 CLIP 的语言编码器来编码语言目标。 PerAct 将语言目标和 RGB-D 体素观察结果作为 Perceiver Transformer 的输入，并通过检测下一个最佳体素动作来输出离散动作

。 PerAct 通过监督学习和来自演示数据集的离散时间输入动作进行训练。演示数据集包括与语言目标配对的体素观察结果

和关键帧动作序列。一个动作由 6-DoF 姿势、夹具打开状态和避免碰撞动作组成。在训练期间，对元组进行随机采样，并且智能体根据观察结果和目标预测关键帧动作。

落地关联（Grounding） 空间环境中的语义表示对于有效的机器人交互至关重要。 CLIPort 和 PerAct 利用 CLIP（基于对比学习进行训练）进行语义推理，利用 Transporter 和 Perceiver 进行空间推理。

Voltron [29] 提出了机器人技术中语言驱动的表示学习的框架。 Voltron 捕获从视频和字幕中学习到的语义、空间和时间表示。对比学习捕获语义表示，但丢失空间关系，相反，屏蔽自动编码捕获空间表示，而不是语义表示。 Voltron 权衡了局部空间表征的语言条件视觉重建和捕捉语义表征的视觉基础语言生成。该框架包括掌握 可供性（Affordance） 预测、单任务视觉运动控制、引用表达 落地关联（Grounding）、语言条件模仿和意图评分任务。 Voltron 模型将视频及其相关语言字幕作为多模态编码器的输入，然后对多模态编码器的输出进行解码，以从屏蔽上下文中重建一个或多个帧。 Voltron 从屏蔽自动编码主干开始，并通过根据语言前缀调节 MAE 编码器向模型添加动态组件。通过对多个帧进行调节来捕获时间信息。

部署利用语言条件 模仿学习（Imitation Learning） 和真实机器人的机器人策略学习技术带来了持续的挑战。这些模型依赖于端到端学习，其中策略将像素或体素映射到动作。由于它们是通过示范数据集的监督学习进行训练的，因此它们很容易受到与泛化和分布变化相关的问题的影响。为了提高鲁棒性和适应性，数据增强和域适应等技术可以使策略对分布变化更加鲁棒。

CACTI [14] 是一种新颖的框架，旨在使用 基础模型（Foundation Models）（例如稳定扩散 [117]）增强机器人学习的可扩展性。 CACTI介绍了数据收集、数据增强、视觉表示学习和模仿策略训练四个阶段。在数据收集阶段，收集有限的领域内专家论证数据。在数据增强阶段，CACTI 采用稳定扩散 [117] 等视觉生成模型，通过场景和布局变化来增强数据，从而提高视觉多样性。在视觉表示学习阶段，CACTI利用在域外数据上训练的预训练零样本视觉表示模型来提高训练效率。最后，在模仿策略训练阶段，使用 模仿学习（Imitation Learning） 在增强数据集上以压缩视觉表示作为输入来学习通用的多任务策略

。 CACTI 经过训练，可以在模拟和现实世界中的厨房环境中进行多任务和多场景操作。这些技术的使用增强了框架的泛化能力，并使其能够从广泛的环境中学习。

除了语言之外，最近的作品还研究了其他

forms of task specification. Notably, MimicPlay [118] presents a hierarchical imitation learning algorithm that learns high-level plans in latent spaces from human play data and low-level motor commands from a small number of teleoperated demonstrations. By harnessing the complementary strengths of these two data sources, this algorithm can significantly reduce the cost of training visuomotor policies for long-horizon manipulation tasks. Once trained, it is capable of performing new tasks based on one human video demonstration at test time. MUTEX [119] further explores learning a unified policy across multimodal task specifications in video, image, text, and audio, showing improved policy performances over single-modality baselines through cross-modal learning.

2) Language-Assisted Reinforcement Learning: Reinforcement learning (RL) is a family of methods that enable a robot to optimize a policy through interaction with its environment by optimizing a reward function. These interactions are usually in a simulation environment, sometimes augmented with data from physical robot hardware for sim-to-real transfer. RL has close ties to optimal control. Unlike imitation learning, RL does not require human demonstrations, and (in theory) has the potential to attain super-human performance. In the RL problem, the expected return of a policy is maximized using the collected roll-outs from interactions with the environment. The feedback received from the environment in the form of a reward signal guides the robot to learn which actions lead to favorable results and which do not. In this section, we discuss works that have incorporated foundation models (LLM, VLMs, etc.) into RL problems.

Fast and flexible adaptation is a desired capability of artificial agents and is essential for progress toward general intelligence. In Adaptive Agent (AdA) [30] the authors present an RL foundation model that is an agent pretrained on diverse tasks and is designed to quickly adapt to open-ended embodied 3D problems by using fast in-context learning from feedback. This work considers navigation, coordination, and division of labor tasks. Given a few episodes within an unseen environment at test time, the agent engages in trial-and-error exploration to refine its policy toward optimal performance. In AdA a transformer architecture is trained using model-based RL^2 [120] to train agents with large-scale attention-based memory, which is required for adaptation. Transformer-XL [121] with some modification is used to enable long and variable-length context windows to increase the model memory to capture long-term dependencies. The agent collects diverse data in the XLand environment that includes 10^{40} possible tasks [122], in an automated curriculum. In addition, distillation is used to enable scaling to models with more than 500M parameters.

Palo et al. [15] propose an approach to enhance reinforcement learning by integrating Large Language Models (LLMs) and Visual-Language Models (VLMs) to create a more unified RL framework. This work considers robot manipulation tasks. Their approach addresses core RL challenges related to exploration, experience reuse and transfer, skills scheduling, and learning from observation. The authors use an LLM to decompose complex tasks into simpler sub-tasks, which are then utilized as inputs for a transformer-based agent to interact with the environment. The agent is trained using a combination of supervised and reinforcement learning, enabling it to predict the optimal sub-task to execute based on the current state of the environment.

B. Language-Image Goal-Conditioned Value Learning

In value learning, the aim is to construct a value function that aligns goals in different modalities and preserves temporal coherence due to the recursive nature of the value function. Reusable Representation for Robotic Manipulations (R3M) [31] provides pretrained visual representation for robot manipulation using diverse human video datasets such as Ego4D and can be used as a frozen perception module for policy learning in robot manipulation tasks. R3M's pretrained visual representation is demonstrated on Franka Emika Panda's arm and enables different downstream manipulation tasks. R3M is trained using time-contrastive learning to capture temporal dependencies, video-language alignment to capture semantic features of the scene (such as objects and their relationships) and $L1$ penalty to encourage sparse and compact representation. For a batch of videos, using time-contrastive loss, an encoder is trained to generate a representation wherein the distance between images that are temporally closer is minimized compared to images that are farther apart in time or from different videos.

Similar to R3M, Value-Implicit Pretraining (VIP) [36] employs time-contrastive learning to capture temporal dependencies in videos, but it does not require video-language alignment. VIP is also focused on robot manipulation tasks. VIP is a self-supervised approach for learning visual goal-conditioned value functions and representations from videos. VIP learns visual goal-based rewards for downstream tasks and can be used for zero-shot reward specification. The reward model is derived from pretrained visual representations. Pretraining involves using unlabeled human videos. Human videos do not contain any action information to be used for robot policy learning, therefore the learned value function does not explicitly depend on actions. VIP introduces a novel time contrastive objective that generates a temporally smooth embedding. The value function is implicitly defined via distance embedding. The proposed implicit time contrastive learning attracts the representation of the initial and goal frames in the same trajectory and repels the representation of intermediate frames by recursive one-step temporal difference minimization. This representation captures long-term temporal dependencies across task frames and local temporal smoothness among adjacent frames.

Language-Image Value Learning (LIV) [37] is a control-centric vision-language representation. LIV generalizes the prior work VIP by learning multi-modal vision-language value functions and representations using language-aligned videos. For tasks specified as language goals or image goals, a multi-model representation is trained that encodes a universal value function. LIV is also focused on robot manipulation tasks. LIV is a pretrained control-centric vision-language representation based on large human video datasets such as EPIC-KITCHENS [123]. The representations are kept frozen during policy learning. A simple MLP is used on top of pretrained

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任务规范的形式。值得注意的是，MimicPlay [118] 提出了一种分层 模仿学习（Imitation Learning） 算法，该算法从人类游戏数据中学习潜在空间中的高级计划，并从少量远程操作演示中学习低级运动命令。通过利用这两个数据源的互补优势，该算法可以显着降低 长时程（Long-Horizon） 操作任务的视觉运动策略训练成本。经过训练后，它能够在测试时根据人类视频演示执行新任务。 MUTEX [119] 进一步探索了跨视频、图像、文本和音频的多模态任务规范学习统一策略，显示通过跨模态学习比单模态基线提高了策略性能。

2）语言辅助强化学习（Reinforcement Learning）：强化学习（Reinforcement Learning）（RL）是一系列方法，使机器人能够通过优化奖励函数与环境交互来优化策略。这些交互通常在模拟环境中进行，有时会使用来自物理机器人硬件的数据进行增强，以进行 仿真到现实（Sim-to-Real） 传输。强化学习与最优控制有着密切的联系。与 模仿学习（Imitation Learning） 不同，强化学习不需要人类演示，并且（理论上）有潜力实现超人类的表现。在强化学习问题中，使用从与环境的交互中收集的推出来最大化策略的预期回报。从环境中以奖励信号的形式收到的反馈引导机器人了解哪些行为会带来有利的结果，哪些行为不会。在本节中，我们讨论将 基础模型（Foundation Models）（LLM、VLM 等）纳入 RL 问题的工作。

快速而灵活的适应是人工智能所需的能力，对于通用智能的进步至关重要。在自适应智能体 (AdA) [30] 中，作者提出了一种 RL 基础模型（Foundation Model），它是针对不同任务进行预训练的智能体，旨在通过使用反馈中的快速上下文学习来快速适应开放式的具体 3D 问题。这项工作考虑了导航、协调和分工任务。考虑到测试时在未见过的环境中发生的一些情况，智能体会进行试错探索，以完善其策略以实现最佳性能。在 AdA 中，变压器架构使用基于模型的 RL2 [120] 进行训练，以训练具有大规模基于注意力的记忆的智能体，这是适应所需的。经过一些修改的 TransformerXL [121] 用于启用长且可变长度的上下文窗口，以增加模型内存以捕获长期依赖性。该智能体在 XLand 环境中收集各种数据，其中包括自动化课程中的 1040 可能任务 [122]。此外，蒸馏还用于扩展具有超过 500M 参数的模型。

帕洛等人。 [15]提出了一种通过集成大型语言模型（LLM）和视觉语言模型（VLM）来增强 强化学习（Reinforcement Learning） 的方法，以创建更统一的 RL 框架。这项工作考虑了机器人操纵任务。他们的方法解决了与探索、经验重用和转移、技能调度以及从观察中学习相关的核心强化学习挑战

。作者使用 LLM 将复杂的任务分解为更简单的子任务，然后将这些子任务用作基于变压器的智能体进行交互的输入

与环境。该智能体使用监督和 强化学习（Reinforcement Learning） 的组合进行训练，使其能够根据当前环境状态预测要执行的最佳子任务。
B. 语言图像目标条件价值学习

在价值学习中，目标是构建一个价值函数，该价值函数能够以不同的方式调整目标，并由于价值函数的递归性质而保持时间连贯性。机器人操作的可重用表示（R3M）[31]使用不同的人类视频数据集（例如 Ego4D）为机器人操作提供预训练的视觉表示，并且可以用作机器人操作任务中策略学习的冻结感知模块。R3M 的预训练视觉表示在 Franka Emika Panda 的手臂上进行了演示，并支持不同的下游操作任务。R3M 的训练使用时间对比学习来捕获时间依赖性、视频语言对齐来捕获场景的语义特征（例如对象及其关系）以及 L1 惩罚来鼓励稀疏和紧凑的表示。对于一批视频，使用时间对比损失，训练编码器来生成表示，其中与时间上相距较远的图像或来自不同视频的图像相比，时间上较近的图像之间的距离被最小化。

与 R3M 类似，隐式价值预训练（VIP）[36] 采用时间对比学习来捕获视频中的时间依赖性，但它不需要视频语言对齐。 VIP 还专注于机器人操纵任务。 VIP 是一种自监督方法，用于从视频中学习视觉目标条件价值函数和表示。 VIP 学习下游任务的基于视觉目标的奖励，并可用于零样本奖励规范。奖励模型源自预先训练的视觉表示。预训练涉及使用未标记的人类视频。人类视频不包含任何用于机器人策略学习的动作信息，因此学习到的价值函数不明确依赖于动作。 VIP 引入了一种新颖的时间对比目标，可以生成时间平滑的嵌入。价值函数是通过距离嵌入隐式定义的。所提出的隐式时间对比学习通过递归一步时间差异最小化吸引同一轨迹中的初始帧和目标帧的表示并排斥中间帧的表示。这种表示捕获任务帧之间的长期时间依赖性以及相邻帧之间的局部时间平滑度。

语言图像价值学习（LIV）[37]是一种以控制为中心的视觉语言表示。 LIV 通过使用语言对齐视频学习多模态视觉语言价值函数和表示来概括先前的工作 VIP。对于指定为语言目标或图像目标的任务，训练编码通用值函数的多模型表示。 LIV也专注于机器人操控任务

。 LIV 是一种基于大型人类视频数据集（例如 EPICKITCHENS [123]）的预训练的以控制为中心的视觉语言表示。在策略学习期间，这些表示保持冻结。在预训练之上使用简单的 MLP

representations for the policy network. Policy learning is decoupled from language-visual representation pretraining. The LIV model is pretrained on arbitrary video activity datasets with text annotation, and the model can be fine-tuned on small datasets of in-domain robot data to ground language in a context-specific way. LIV uses a generalization of the mutual information-based image-text contrastive representation learning objective as used in CLIP, so LIV can be considered as a combination of CLIP and VIP. Both VIP and LIV learn a self-supervised goal-conditioned value-function objective using contrastive learning. The LIV extends the VIP framework to multi-modal goal specifications. LOREL [38] learns a language-conditioned reward from offline data and uses it during model predictive control to complete language-specified tasks.

Value functions can be used to help ground semantic information obtained from an LLM to the physical environment in which a robot is operating. By leveraging value functions, a robot can associate the information processed by the LLM with specific locations and objects in its surroundings. In SayCan [32], researchers investigate the integration of large language models with the physical world through learning. They use the language model to provide task-grounding (Say), enabling the determination of useful sub-goals based on high-level instructions, and a learned affordance function to achieve world-grounding (Can), enabling the identification of feasible actions to execute the plan. Inner Monologue [33] studies the role of grounded environment feedback provided to the LLM, thus closing the loop with the environment. The feedback is used for robot planning with large language models by leveraging a collection of perception models (e.g., scene descriptors and success detectors) in tandem with pretrained language-conditioned robot skills. Feedback includes task-specific feedback, such as success detection, and scene-specific feedback (either "passive" or "active"). In both SayCan and Inner Monologue robot manipulation and navigation tasks are considered using a real-world mobile manipulator robot from Everyday Robots. Text2Motion [124] is a language-based planning framework for long-horizon robot manipulation. Similar to SayCan and Inner Monologue, Text2Motion computes a score (S_{LMM}) associated with each skill at each time step. The task planning problem is to find a sequence of skills by maximizing the likelihood of a skill sequence given a language instruction and the initial state. In Text2Motion, the authors propose to verify that the generated long-horizon plans are symbolically correct and geometrically feasible. Hence, a geometric feasibility score (S_{geo}) is defined as the probability that all the skills in the sequence achieve rewards. To compute the overall score, the LLM score is multiplied by the geometric feasibility score ($S_{\text{Skill}} = S_{\text{LMM}} \cdot S_{\text{geo}}$).

VoxPoser [34] builds 3D value maps to ground affordances and constraints into the perceptual space. VoxPser considers robot manipulation tasks. Given the RGB-D observation of the environment and language instruction, VoxPoser utilizes large language models to generate code, which interacts with vision-language models to extract a sequence of 3D affordance maps and constraint maps. These maps are composed together to create 3D value maps. The value maps are then utilized as objective functions to guide motion planners to synthesize trajectories for everyday manipulation tasks without requiring any prior training or instruction.

In [35], reward shaping using CLIP is presented. This work considers robot manipulation tasks. The proposed model utilizes CLIP to ground objects in a scene described by the goal text paired with spatial relationship rules to shape the reward by using raw pixels as input. They use developments in building large-scale visuo-lingual models like CLIP to devise a framework that generates the task reward signal from just the goal text description and raw pixel observations. This signal is then used to learn the task policy.

In [125], Hierarchical Universal Language Conditioned Policies 2.0 (HULC++) is presented. This work considers robot manipulation tasks. A self-supervised visuo-lingual affordance model is used to learn general-purposed language-conditioned robot skills from unstructured offline data in the real world. This method requires annotating as little as 1% of the total data with language. The visuo-lingual affordance model has an encoder-decoder architecture with two decoder heads. Both heads share the same encoder and are conditioned on the input language instruction. One head predicts a distribution over the image, in which each pixel likelihood is an afforded point. The other head predicts a Gaussian distribution from which the corresponding predicted depth is sampled. Given visual observations and language instructions as input, the affordance model outputs a pixel-wise heat map that represents affordance regions and the corresponding depth map.

C. Robot Task Planning using Large Language Models

LLMs can be used to provide high-level task planning for performing complex long-horizon robot tasks.

1) Language Instructions for Task Specification: As discussed above, SayCan [32] uses an LLM for high-level task planning in language, though with a learned value function to ground these instructions in the environment.

Temporal logic is useful for imposing temporal specifications in robotic systems. In [39], translation from natural language (NL) to temporal logic (TL) is proposed. A dataset with 28k NL-TL pairs is created and the T5 [126] model is finetuned using the dataset. LLMs are often used to plan task sub-goals. This work considers robot navigation tasks. In [40], instead of direct task planning, a few-shot translation from a natural language task description to an intermediary task representation is performed. This representation is used by a Task and Motion Planning (TAMP) algorithm to jointly optimize task and motion plans. Autoregressive re-prompting is used to correct synthetic and semantic errors. This work also considers robot navigation tasks.

2) Code Generation using Language Models for Task Planning: Classical task planning requires extensive domain knowledge and the search space is large [127], [128]. LLMs can be used to generate sequences of tasks required to achieve a high-level task. In ProgPrompt [41], the authors introduce a prompting method that uses LLMs to generate sequences of actions directly with no additional domain knowledge. The

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策略网络的代表。策略学习与语言视觉表示预训练分离。 LIV 模型在带有文本注释的任意视频活动数据集上进行预训练，并且该模型可以在域内机器人数据的小数据集上以特定于上下文的方式微调为基础语言。 LIV 使用 CLIP 中基于互信息的图像文本对比表示学习目标的推广，因此 LIV 可以被视为 CLIP 和 VIP 的组合。 VIP 和 LIV 都使用对比学习来学习自我监督的目标条件价值函数目标。 LIV 将 VIP 框架扩展到多模式目标规范。 LOREL [38] 从离线数据中学习语言条件奖励，并在模型预测控制期间使用它来完成语言指定的任务。

价值函数可用于帮助将从法学硕士获得的语义信息转化为机器人运行的物理环境。通过利用价值函数，机器人可以将 LLM 处理的信息与其周围的特定位置和物体相关联。在 SayCan [32] 中，研究人员通过学习研究了大型语言模型与物理世界的集成。他们使用语言模型来提供任务 落地关联（Grounding）（Say），从而能够根据高级指令确定有用的子目标，并使用学习的 可供性（Affordance） 函数来实现世界 落地关联（Grounding）（Can），从而能够识别执行计划的可行操作。内心独白[33]研究了提供给法学硕士的接地环境反馈的作用，从而实现了与环境的闭环。通过利用感知模型（例如场景描述符和成功检测器）的集合以及预先训练的语言条件机器人技能，反馈用于使用大型语言模型进行机器人规划。反馈包括特定于任务的反馈，例如成功检测和特定于场景的反馈（“被动”或“主动”）。在 SayCan 和 Inner Monologue 中，机器人操纵和导航任务都考虑使用来自 Everyday Robots 的现实世界移动操纵机器人。 Text2Motion [124] 是用于 长时程（Long-Horizon） 机器人操作的基于语言的规划框架。与 SayCan 和 Inner Monologue 类似，Text2Motion 在每个时间步计算与每个技能相关的分数 (SLMM)。任务规划问题是在给定语言指令和初始状态的情况下，通过最大化技能序列的可能性来找到技能序列。在 Text2Motion 中，作者建议验证生成的 长时程（Long-Horizon） 计划在符号上是否正确且在几何上可行。因此，几何可行性得分（Sgeo）被定义为序列中所有技能获得奖励的概率。为了计算总体分数，LLM 分数乘以几何可行性分数 (SSkill = SLMM · Sgeo)。

VoxPoser [34] 构建 3D 值映射到 可供性（Affordances） 和感知空间的约束。 VoxPser 考虑机器人操作任务

。考虑到对环境和语言指令的 RGB-D 观察，VoxPoser 利用大型语言模型来生成代码，该代码与视觉语言模型交互以提取一系列 3D 可供性（Affordance） 映射和约束映射。这些地图组合在一起以创建 3D 值地图。然后利用价值图

作为目标函数来指导运动规划者合成日常操作任务的轨迹，而不需要任何事先的训练或指导。

在[35]中，提出了使用 CLIP 的奖励塑造。这项工作考虑了机器人操纵任务。所提出的模型利用 CLIP 将目标文本描述的场景中的物体与空间关系规则配对，通过使用原始像素作为输入来塑造奖励。他们利用构建大规模视觉语言模型（如 CLIP）的进展来设计一个框架，该框架仅根据目标文本描述和原始像素观察生成任务奖励信号。然后使用该信号来学习任务策略。

在[125]中，提出了分层通用语言条件策略2.0（HULC++）。这项工作考虑了机器人操纵任务。自监督视觉语言 可供性（Affordance） 模型用于从现实世界中的非结构化离线数据中学习通用语言条件机器人技能。该方法只需要用语言注释总数据的 1%。视觉语言 可供性（Affordance） 模型具有带有两个解码器头的编码器-解码器架构。两个头共享相同的编码器，并以输入语言指令为条件。一个头预测图像上的分布，其中每个像素可能性都是一个提供的点。另一个头预测高斯分布，从中采样相应的预测深度。给定视觉观察和语言指令作为输入，可供性（Affordance） 模型输出代表 可供性（Affordance） 区域和相应深度图的像

素级热图。

C. 使用大型语言模型的机器人任务规划

LLM 可用于提供高级任务规划，以执行复杂的 长时程（Long-Horizon） 机器人任务。

1）用于任务规范的语言指令：如上所述，SayCan [32] 使用 LLM 进行语言的高级任务规划，尽管使用学习值函数将这些指令扎根在环境中。

时间逻辑对于在机器人系统中施加时间规范非常有用。在[39]中，提出了从自然语言（NL）到时序逻辑（TL）的翻译。创建具有 28k NL-TL 对的数据集，并使用该数据集对 T5 [126] 模型进行微调。法学硕士通常用于规划任务子目标。这项工作考虑机器人导航任务。在[40]中，不是直接任务规划，而是执行从自然语言任务描述到中间任务表示的几次翻译。任务和运动规划 (TAMP) 算法使用此表示来联合优化任务和运动计划。自回归重新提示用于纠正合成和语义错误。这项工作还考虑了机器人导航任务。

2）使用语言模型进行任务规划的代码生成：经典的任务规划需要广泛的领域知识并且搜索空间很大[127]，[128]

。 LLM 可用于生成实现高级任务所需的任务序列。在 ProgPrompt [41] 中，作者介绍了一种提示方法，该方法使用 LLM 直接生成动作序列，无需额外的领域知识。这

prompt to the LLM includes specifications of the available actions, objects in the environment, and example programs that can be executed. VirtualHome [129] is used as a simulator for demonstration.

Code-as-Policies [42] explores the use of code-writing LLMs to generate robot policy code based on natural language commands. This work considers robot manipulation and navigation tasks using a real-world mobile manipulator robot from Everyday Robots. The study demonstrates that LLMs can be repurposed to write policy code by expressing functions or feedback loops that process perception outputs and invoke control primitive APIs. To achieve this, the authors utilize few-shot prompting, where example language commands formatted as comments are provided along with the corresponding policy code. Without any additional training on this data, they enable the models to autonomously compose API calls and generate new policy code when given new commands. The approach leverages classic logic structures and references third-party libraries like NumPy and Shapely to perform arithmetic operations. By chaining these structures and using contextual information (behavioral commonsense), the LLMs can generate robot policies that exhibit spatial-geometric reasoning, generalize to new instructions, and provide precise values (e.g., velocities) for ambiguous descriptions such as "faster." The concept of "code as policies" formalizes the generation of robot policies using language model-generated programs (LMPs). These policies can represent reactive policies like impedance controllers, as well as waypoint-based policies such as vision-based pick and place or trajectory-based control. The effectiveness of this approach is demonstrated on multiple real robot platforms. A crucial aspect of this approach is the hierarchical code generation process, which involves recursively defining undefined functions. This enables the LLMs to generate more complex code structures to fulfill the desired policy requirements.

In [43], the authors provide design principles for using ChatGPT in robotics and demonstrate how LLMs can help robotic capabilities rapidly generalize to different form factors. This work considers robot manipulation and aerial navigation tasks. First, a high-level robot function library that maps to multiple atomic tasks executable by the robot is defined. Then, a prompt is crafted that includes these functions, and the required constraints along the task description. ChatGPT then provides executable code specific to the given robot configuration and task. The generated code can then be evaluated by a user and appropriate feedback with modified prompts to LLMs further help refine and generate programs that are safe and deployable on the physical robot. The study demonstrated that such a methodology can be applied to multiple form factors both in simulation and in the real world.

D. In-context Learning (ICL) for Decision-Making

In-context Learning (ICL) [130] operates without the need for parameter optimization, relying instead on a set of examples included in the prompt (the concept of prompting). This learning approach is intimately linked with prompt engineering and finds extensive use in natural language processing. The method of Chain-of-Thought [131] is a prominent technique within in-context learning. It involves executing a sequence of intermediate steps to arrive at the final solution for complex, multi-step problems. This technique allows models to produce step-by-step explanations that parallel human cognitive processes. However, despite its numerous benefits, ICL also faces certain challenges, including issues related to ambiguity and interpretation, domain-specific knowledge, transparency, and explainability. In-context learning has had a significant impact on the field of LLMs in a broad sense, and many robotics works have used it to apply LLMs to specific domains. Investigating this, Mirchandani and colleagues [132] illustrate that Large Language Models (LLMs) possess remarkable pattern recognition abilities. They reveal that, through in-context learning, LLMs can effectively handle general patterns that extend beyond standard language-based prompts. This capability allows for the application of LLMs in scenarios such as offline trajectory optimization and online, in-context reinforcement learning. Additionally, Jia and the team in their work on Chain-of-Thought Predictive Control [133] suggest a method to identify specific brief sequences within demonstrations, termed as 'chain-of-thought'. They focus on understanding and representing the hierarchical structure of these sequences, highlighting the achievement of subgoals within tasks. This work considers robot policy learning from demonstrations for contact-rich object manipulation tasks.

E. Robot Transformers

Foundation models can be used for end-to-end control of robots by providing an integrated framework that combines perception, decision-making, and action generation.

Xiao et al. [48] demonstrate the effectiveness of self-supervised visual pretraining using real-world images for learning motor control tasks directly from pixel inputs. This work is focused on robot manipulation tasks. They show that without any task-specific fine-tuning of the pretrained encoder, the visual representations can be utilized for various motor control tasks. This approach highlights the potential of leveraging self-supervised learning from real-world images to acquire general visual representations that can be applied across different motor control tasks. Similarly, Radosavovic et al. [49] investigate the use of self-supervised visual pretraining on diverse, in-the-wild videos for real-world robotic tasks. This work considers robot manipulation tasks. They find that the pretrained representations obtained from such videos are effective in a range of real-world robotic tasks, considering different robotic embodiments. This suggests that the learned visual representations generalize well across various tasks and robot platforms, demonstrating the broad applicability of self-supervised pretraining for real-world robotic applications. Both studies emphasize the advantages of self-supervised visual pretraining, where models are trained on large amounts of unlabeled data to learn useful visual representations. By leveraging real-world images and videos, these approaches enable learning from diverse and unstructured visual data, leading to more robust and transferable representations for motor control tasks in robotic systems.

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LLM 的提示包括可用操作、环境中的对象以及可以执行的示例程序的规范。 VirtualHome[129]用作模拟器进行演示。

代码即策略 [42] 探索使用代码编写 LLM 来基于自然语言命令生成机器人策略代码。这项工作考虑使用来自 Everyday Robots 的现实世界移动机械手机器人来执行机器人操纵和导航任务。该研究表明，法学硕士可以通过表达处理感知输出和调用控制原语 API 的函数或反馈循环来重新编写策略代码。为了实现这一目标，作者利用了少镜头提示，其中提供了格式化为注释的示例语言命令以及相应的策略代码。无需对这些数据进行任何额外的训练，它们就可以使模型自主地编写 API 调用，并在给出新命令时生成新的策略代码。该方法利用经典的逻辑结构并引用 NumPy 和 Shapely 等第三方库来执行算术运算。通过链接这些结构并使用上下文信息（行为常识），法学硕士可以生成展示空间几何推理的机器人策略，推广到新指令，并为“更快”等模糊描述提供精确值（例如速度）。 “代码即策略”的概念使用语言模型生成程序（LMP）形式化地生成机器人策略。这些策略可以表示阻抗控制器等反应性策略，以及基于路径点的策略，例如基于视觉的拾取和放置或基于轨迹的控制。这种方法的有效性在多个真实的机器人平台上得到了证明。这种方法的一个关键方面是分层代码生成过程，其中涉及递归定义未定义的函数。这使得法学硕士能够生成更复杂的代码结构来满足所需的策略要求。

在[43]中，作者提供了在机器人技术中使用 ChatGPT 的设计原则，并演示了 LLM 如何帮助机器人功能快速推广到不同的外形尺寸。这项工作考虑了机器人操纵和空中导航任务。首先，定义映射到机器人可执行的多个原子任务的高级机器人函数库。然后，精心制作一个提示，其中包括这些功能以及任务描述中所需的约束。然后，ChatGPT 提供特定于给定机器人配置和任务的可执行代码。然后，用户可以评估生成的代码，并通过修改后的提示向法学硕士提供适当的反馈，进一步帮助完善和生成安全且可部署在物理机器人上的程序。该研究表明，这种方法可以应用于模拟和现实世界中的多种外形尺寸。

D. 用于决策的情境学习 (ICL)

上下文学习（ICL）[130]的运行不需要参数优化，而是依赖于提示中包含的一组示例（提示的概念）。这种学习方法与即时工程密切相关，并在自然语言处理中得到广泛应用。这

思想链方法[131]是上下文学习中的一项重要技术。它涉及执行一系列中间步骤以获得复杂的多步骤问题的最终解决方案。这项技术允许模型产生与人类认知过程平行的逐步解释。然而，尽管 ICL 具有众多优势，但也面临着某些挑战，包括与歧义和解释、特定领域知识、透明度和可解释性相关的问题。情境学习在广义上对法学硕士领域产生了重大影响，许多机器人学著作都利用它来将法学硕士应用到特定领域。
Mirchandani 和同事[132]对此进行了调查，表明大型语言模型（LLM）拥有卓越的模式识别能力。他们表明，通过情境学习，法学硕士可以有效地处理超出基于标准语言的提示的一般模式。此功能允许在离线轨迹优化和在线上下文 强化学习（Reinforcement Learning） 等场景中应用法学硕士。此外，贾和团队在思想链预测控制[133]的工作中提出了一种识别演示中特定简短序列的方法，称为“思想链”。他们专注于理解和表示这些序列的层次结构，突出任务中子目标的实现。这项工作考虑了从接触丰富的对象操作任务的演示中学习机器人策略。

E. 机器人变形金刚

基础模型（Foundation Models）通过提供结合感知、决策和动作生成的集成框架，可用于机器人的端到端控制。

肖等人。 [48]展示了使用真实世界图像直接从像素输入学习运动控制任务的自监督视觉预训练的有效性。这项工作的重点是机器人操纵任务。他们表明，无需对预训练编码器进行任何特定于任务的微调，视觉表示即可用于各种电机控制任务。这种方法凸显了利用现实世界图像的自我监督学习来获取可应用于不同运动控制任务的通用视觉表示的潜力。同样，Radosavovic 等人。 [49]研究了在各种野外视频上使用自监督视觉预训练来执行现实世界的机器人任务。这项工作考虑了机器人操纵任务。他们发现，考虑到不同的机器人实施例，从此类视频获得的预训练表示在一系列现实世界的机器人任务中是有效的。这表明学习到的视觉表示可以很好地推广到各种任务和机器人平台，证明自监督预训练对于现实世界机器人应用的广泛适用性。两项研究都强调了自监督视觉预训练的优势，即模型在大量未标记数据上进行训练以学习有用的视觉表示

。通过利用现实世界的图像和视频，这些方法能够从多样化和非结构化的视觉数据中进行学习，从而为机器人系统中的电机控制任务提供更强大和可转移的表示。

Another example of a Transformer-based policy model is the work on Robotics Transformer (RT-1) [44], where the authors demonstrate a model that shows promising scalability properties. To train the model, the authors use a large dataset of over 130k real-world robotic experiences, comprising more than 700 tasks, that was collected over 17 months using a fleet of 13 robots. RT-1 receives images and natural language instructions as inputs and outputs discretized base and arm actions. It can generalize to new tasks, maintain robustness in changing environments, and execute long-horizon instructions. The authors also demonstrate the model's capability to effectively absorb data from diverse domains, including simulations and different robots.

The follow-up work, called Robotic Transformer 2 (RT-2) [45], demonstrates a vision-language-action (VLA) model that takes a step further by learning from both web and robotics data. The model effectively utilizes this data to generate generalized actions for robotic control. To do so, the authors use pre-existing vision-language models and directly co-fine-tune them on robot trajectories resulting in a single model that operates as a language model, a vision-language model, and a robot policy. To make co-fine-tuning possible, the actions are represented as simple text strings which are then tokenized using an LLM tokenizer into text tokens. The resulting model, RT-2, enables vision-language models to output low-level closed-loop control. Similarly to RT-1, actions are produced based on robot instructions paired with camera observations and the action space includes 6-DoF positional and rotational displacement of the robot end-effector, gripper extension, and episode termination command. Via extensive experiments, the authors show that utilizing VLMs aids in the enhancement of generalization across visual and semantic concepts and enables the robots to respond to the so-called chain of thought prompting, where the agent performs more complex, multi-stage semantic reasoning. Both RT-1 and RT-2 consider robot manipulation and navigation tasks using a real-world mobile manipulator robot from Everyday Robots. One key limitation of RT-2 and other related works in robotics is the fact that the range of physical skills exhibited by the robot is limited to the distribution of skills observed within the robot's data. While one way to approach this limitation is to collect more diverse and dexterous robotic data, there might be other intriguing research directions such as using motion data in human videos, robotic simulations, or other robotic embodiments.

The next work utilizing the Transformer architecture indeed focuses on learning from data that combines multiple robotic embodiments. In RT-X [46], the authors provide a number of datasets in a standardized data format and models to make it possible to explore the possibility of training large cross-embodied robotic models in the context of robotic manipulation. In particular, they assembled a dataset from 22 different robots collected through a collaboration between 21 institutions, demonstrating 527 skills (160266 tasks). With this unified dataset, RT-X demonstrates that RT-1- and RT-2-based models trained on this multi-embodiment, diverse data exhibit positive transfer across robotic domains and improve the capabilities of multiple robots by leveraging experience from other platforms.

Other works have investigated general pretrained transformers for robot control, trained with self-supervised trajectory data from multiple robots. For example, Perception-Action Causal Transformer (PACT) [47] is a generative transformer architecture that builds representations from robot data with self-supervision. This work considers robot navigation tasks. PACT pretrains a representation useful for multiple tasks on a given robot. Similar to how large language models learn from extensive text data, PACT is trained on abundant safe state-action data (trajectories) from a robot, learning to predict appropriate safe actions. By predicting states and actions over time in an autoregressive manner, the model implicitly captures dynamics and behaviors specific to a robot. PACT was tested in experiments involving mobile agents: a wheeled robot with a LiDAR sensor (MuSHR) and a simulated agent using first-person RGB images (Habitat). The results show that this robot-specific representation can serve as a starting point for tasks like safe navigation, localization, and mapping. Additionally, the experiments demonstrated that fine-tuning smaller task-specific networks on the pre-trained model leads to significantly better performance compared to training a single model from scratch for all tasks simultaneously, and comparable performance to training a separate large model for each task independently.

Another work in this space is Self-supervised Multi-task pretrAining with contRol Transformer (SMART) [134], which introduces a self-supervised multi-task pertaining to control transformers, providing a pretraining-finetuning approach tailored for sequential decision-making tasks. During the pre-training phase, SMART captures information essential for both short-term and long-term control, facilitating transferability across various tasks. Subsequently, the finetuning process can adapt to a wide variety of tasks spanning diverse domains. Experimentation underscores SMART's ability to enhance learning efficiency across tasks and domains. This work considers cart pole-swing-up, cart pole-balance, hopper-hop, hopper-stand, cheetah-run, walker-stand walker-run, and walker-walk tasks. The approach demonstrates robustness against distribution shifts and proves effective with low-quality pretraining datasets.

Some works have investigated transformer models in conjunction with classical planning and control layers as part of a modular robot control architecture. For example, in [50], a multi-modal transformer (LATTE) is presented that allows a user to reshape robot trajectories using language instructions. This work considers both robot manipulation and navigation tasks. LATTE transformer takes as input geometrical features of an initial trajectory guess along with the obstacle map configuration, language instructions from a user, and images of each object in the environment. The model's output is modified for each waypoint in the trajectory so that the final robot motion can adhere to the user's language instructions. The initial trajectory plan can be generated using any geometric planner such as A*, RRT*, or model predictive control. Subsequently, this plan is enriched with the semantic objectives within the model. LATTE leverages pretrained language and visual-language models to harness semantic representations of the world.

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基于 Transformer 的策略模型的另一个例子是 Robotics Transformer (RT-1) [44] 的工作，其中作者展示了一个具有良好可扩展性的模型。为了训练该模型，作者使用了包含超过 13 万个真实机器人体验的大型数据集，其中包含 700 多项任务，这些数据是使用 13 个机器人组成的车队在 17 个月内收集的。 RT-1 接收图像和自然语言指令作为输入，并输出离散的底座和手臂动作。它可以推广到新任务，在不断变化的环境中保持鲁棒性，并执行 长时程（Long-Horizon） 指令。作者还展示了该模型有效吸收来自不同领域（包括模拟和不同机器人）数据的能力。

后续工作称为 Robotic Transformer 2 (RT2) [45]，演示了 视觉语言动作（Vision-Language-Action） (VLA) 模型，该模型通过从网络和机器人数据中学习而更进一步。该模型有效地利用这些数据来生成机器人控制的通用动作。为此，作者使用预先存在的视觉语言模型，并直接在机器人轨迹上对它们进行协同微调，从而形成一个可作为语言模型、视觉语言模型和机器人策略运行的单一模型。为了使协同微调成为可能，这些操作被表示为简单的文本字符串，然后使用 LLM 标记器将其标记为文本标记。由此产生的模型 RT-2 使视觉语言模型能够输出低级闭环控制。与 RT-1 类似，动作是根据与摄像机观察配对的机器人指令产生的，动作空间包括机器人末端执行器的 6-DoF 位置和旋转位移、夹具延伸和情节终止命令。通过大量的实验，作者表明，利用 VLM 有助于增强视觉和语义概念的泛化，并使机器人能够响应所谓的思维提示链，其中智能体执行更复杂的多阶段语义推理。 RT-1 和 RT-2 都考虑使用 Everyday Robots 的真实移动机械手机器人来执行机器人操纵和导航任务。 RT-2 和机器人技术中其他相关工作的一个关键限制是，机器人表现出的物理技能范围仅限于在机器人数据中观察到的技能分布。虽然解决这一限制的一种方法是收集更加多样化和灵巧的机器人数据，但可能还有其他有趣的研究方向，例如在人类视频、机器人模拟或其他机器人实施例中使用运动数据。

利用 Transformer 架构的下一项工作确实侧重于从组合多个机器人实施例的数据中学习。在 RT-X [46] 中，作者提供了许多标准化数据格式和模型的数据集，使得探索在机器人操作的背景下训练大型跨实体机器人模型的可能性成为可能。特别是，他们通过 21 个机构之间的合作收集了 22 个不同机器人的数据集，展示了 527 项技能（160266 项任务）

。借助这个统一的数据集，RT-X 证明了在这种多实施例、多样化的数据上训练的基于 RT-1 和 RT-2 的模型表现出跨机器人领域的积极转移，并通过利用其他平台的经验来提高多个机器人的能力。

其他工作研究了用于机器人控制的通用预训练变压器，并使用来自多个机器人的自监督轨迹数据进行训练。例如，感知-动作因果变换器（PACT）[47]是一种生成变换器架构，它通过自我监督从机器人数据中构建表示。这项工作考虑机器人导航任务。 PACT 预训练对给定机器人上的多个任务有用的表示。与大型语言模型从大量文本数据中学习的方式类似，PACT 接受来自机器人的丰富安全状态动作数据（轨迹）的训练，学习预测适当的安全动作。通过以自回归方式预测随时间变化的状态和动作，该模型隐式地捕获了机器人特有的动态和行为。 PACT 在涉及移动智能体的实验中进行了测试：带有 LiDAR 传感器的轮式机器人 (MuSHR) 和使用第一人称 RGB 图像的模拟智能体 (Habitat)。结果表明，这种机器人特定的表示可以作为安全导航、定位和地图绘制等任务的起点。此外，实验表明，与同时为所有任务从头开始训练单个模型相比，在预训练模型上微调较小的特定任务网络可带来显着更好的性能，并且与为每个任务独立训练单独的大型模型的性能相当。

该领域的另一项工作是带有控制变压器的自监督多任务预训练（SMART）[134]，它引入了与控制变压器相关的自监督多任务，提供了一种针对顺序决策任务量身定制的预训练微调方法。在预训练阶段，

SMART 捕获短期和长期控制所必需的信息，从而促进跨各种任务的可转移性。随后，微调过程可以适应跨越不同领域的各种任务。实验强调了 SMART 提高跨任务和领域学习效率的能力。这项工作考虑了车杆摆动、车杆平衡、漏斗跳跃、漏斗站立、猎豹奔跑、步行者站立步行者跑步和步行者行走任务。该方法展示了针对分布变化的鲁棒性，并证明对于低质量的预训练数据集是有效的。

一些工作研究了变压器模型与经典规划和控制层的结合，作为模块化机器人控制架构的一部分。例如，在[50]中，提出了一种多模态转换器（LATTE），它允许用户使用语言指令重塑机器人轨迹。这项工作考虑了机器人操纵和导航任务。 LATTE 转换器将初始轨迹猜测的几何特征以及障碍物地图配置、用户的语言指令以及环境中每个对象的图像作为输入

。针对轨迹中的每个路点修改模型的输出，以便最终的机器人运动可以遵循用户的语言指令。初始轨迹规划可以使用任何几何规划器（例如 A* 、 RRT* 或模型预测控制）生成。随后，该计划通过模型内的语义目标得到丰富。 LATTE 利用预训练的语言和视觉语言模型来表达世界的 运行支撑框架（Harness） 语义。

F. Open-Vocabulary Robot Navigation and Manipulation

1) Open-Vocabulary Navigation: Open-vocabulary navigation addresses the challenge of navigating through unseen environments. The open-vocabulary capability signifies that the robot possesses the capacity to comprehend and respond to language cues, instructions, or semantic information, without being restricted to a predefined dataset. In this section, we explore papers that examine the integration of LLMs, VLMs, or a combination of both in a plug-and-play manner for robot navigation tasks. Additionally, we discuss papers that take a different approach by constructing foundation models explicitly tailored for robot navigation tasks.

In VLN-BERT [135], the authors present a visual-linguistic transformer-based model that leverages multi-modal visual and language representations for visual navigation using web data. The model is designed to score the compatibility between an instruction, such as "...stop at the brown sofa," and a sequence of panoramic RGB images captured by the agent.

Similarly, LM-Nav [136] considers visual navigation tasks. LM-Nav is a system that utilizes pretrained models of images and language to provide a textual interface to visual navigation. LM-Nav demonstrates visual navigation in a real-world outdoor environment from natural language instructions. LM-Nav utilizes an LLM (GPT-3 [2]), a VLM (CLIP [4]), and a VNM (Visual Navigation Model). First, LM-Nav constructs a topological graph of the environment via the VNM estimating the distance between images. The LLM is then used to translate the natural instructions to sequences of intermediate language landmarks. The VLM is used to ground the visual observations in landmark descriptions via a joint probability distribution over landmarks and images. Using the VLM's probability distribution, the LLM instructions, and the VNM's graph connectivity, the optimal path is planned using the search algorithm. Then the plan is executed by the goal-conditioned policy of VNM.

While LM-Nav makes use of LLMs and VLMs as plug-and-play for visual navigation tasks, the authors of ViNT [137] propose to build a foundation model for visual navigation tasks. ViNT is an image goal-conditioned navigation policy trained on diverse training data and can control different robots in zero-shot. It can be fine-tuned to be adapted for different robotic platforms and various downstream tasks. ViNT is trained on various navigation datasets from different robotic platforms. It is trained with goal-reaching objectives and utilizes a Transformer-based architecture to learn navigational affordances. ViNT encodes visual observations and visual goals using an EfficientNet CNN and predicts temporal distance and normalized actions in an embodiment-agnostic manner. Additionally, ViNT can be augmented with diffusion-based sub-goal proposals to help explore environments not encountered during training. An image-to-image diffusion generates sub-goal images, which the ViNT then navigates toward while building a topological map in the background.

Another work that considers zero-shot navigation tasks is Audio Visual Language Maps (AVLMaps) [138]. AVLMaps presents a 3D spatial map representation for cross-modal information from audio, visual, and language cues. AVLMaps receives multi-modal prompts and performs zero-shot navigation tasks in the real world. The inputs are depth and RGB images, camera pose, and audio. Visual features are encoded using pretrained foundation models. Visual localization features (using NetVLAD [139], SuperPoint [140]), visual-language features (using LSeg [58]), and audio-language features (using AudioCLIP [141]) are computed and predictions from different modalities are combined into 3D heatmaps. The pixel-wise joint probability of the heatmap is computed and used for planning. Additionally, navigation policies are generated as executable codes with the help of GPT-3. Finally, 3D heatmaps are predicted indicating the location of multimodal concepts such as objects, sounds, and images.

Many roboticists may wonder about the comparative strengths of classical modular robot navigation systems versus end-to-end learned systems. Semantic navigation [142] seeks to address this question by presenting an empirical analysis of semantic visual navigation methods. The study compares representative approaches from classical, modular, and end-to-end learning paradigms across six different homes, without any prior knowledge, maps, or instrumentation. The findings of the study reveal that modular learning methods perform well in real-world scenarios. In contrast, the end-to-end learning approaches face challenges due to a significant domain gap between simulated and real-world images. This domain gap hinders the effectiveness of end-to-end learning methods in real-world navigation tasks. For practitioners, the study emphasizes that modular learning is a reliable approach to object navigation. The modularity and abstraction in policy design enable successful transfer from simulation to reality, making modular learning an effective choice for practical implementations. For researchers, the study also highlights two critical issues that limit the reliability of current simulators as evaluation benchmarks. Firstly, there exists a substantial Sim-to-Real gap in images, which hampers the transferability of learned policies from simulation to the real world. Secondly, there is a disconnect between simulation and real-world error modes, which further complicates the evaluation process.

Another line of work in open-vocabulary navigation is object navigation tasks. In this task, the robot must be able to find the object described by humans and navigate towards the object. The navigation task is decomposed into exploration when the language target is not detected and exploitation when the target is detected and the robot navigates toward the target. As the robot moves in the environment, it creates a top-down map using RGB-D observations and poses estimates. In [143], the authors introduce a zero-shot object navigation setting that uses an open-vocabulary classifier such as CLIP [4] to compute the cosine similarity between an image and a user-specified description.

Common datasets and benchmarks for these types of problems are Matterport3D [144], [145], Gibson [146] and Habitat [147]. L3MVN [148] enhances visual target navigation by constructing an environment map and selecting long-term goals using the inference capabilities of large language models. The system can determine appropriate long-term goals for navigation by leveraging pretrained language models such as RoBERTa-large [149], enabling efficient exploration and

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F. 开放词汇机器人导航和操作

1) 开放词汇导航：开放词汇导航解决了在看不见的环境中导航的挑战。开放词汇能力意味着机器人具有理解和响应语言提示、指令或语义信息的能力，而不受预定义数据集的限制。在本节中，我们将探讨一些论文，这些论文以即插即用的方式检查 LLM、VLM 或两者的组合在机器人导航任务中的集成。此外，我们还讨论了采用不同方法构建专为机器人导航任务定制的 基础模型（Foundation Models） 的论文。

在 VLN-BERT [135] 中，作者提出了一种基于视觉语言转换器的模型，该模型利用多模态视觉和语言表示来使用网络数据进行视觉导航。该模型旨在对指令（例如“……停在棕色沙发处”）与智能体捕获的一系列全景 RGB 图像之间的兼容性进行评分。

类似地，LM-Nav [136]考虑视觉导航任务。 LM-Nav 是一个利用预训练的图像和语言模型为视觉导航提供文本界面的系统。 LM-Nav 通过自然语言指令演示了真实户外环境中的视觉导航。 LMNav 利用 LLM (GPT-3 [2])、VLM (CLIP [4]) 和 VNM（视觉导航模型）。首先，LM-Nav 通过 VNM 估计图像之间的距离构建环境拓扑图。然后，法学硕士用于将自然指令翻译成中间语言地标序列。 VLM 用于通过地标和图像上的联合概率分布来将视觉观察结果纳入地标描述中。利用 VLM 的概率分布、LLM 指令和 VNM 的图连接性，使用搜索算法规划最佳路径。然后该计划由VNM的目标条件策略执行。

虽然 LM-Nav 使用 LLM 和 VLM 作为视觉导航任务的即插即用，但 ViNT [137] 的作者建议为视觉导航任务构建 基础模型（Foundation Model）。 ViNT 是一种基于不同训练数据训练的图像目标条件导航策略，可以零样本控制不同的机器人。它可以进行微调以适应不同的机器人平台和各种下游任务。 ViNT 在来自不同机器人平台的各种导航数据集上进行了训练。它接受了旨在达到目标的训练，并利用基于 Transformer 的架构来学习导航 可供性（Affordances）。 ViNT 使用 EfficientNet CNN 对视觉观察和视觉目标进行编码，并以与实施例无关的方式预测时间距离和标准化动作。此外，ViNT 可以通过基于扩散的子目标建议进行增强，以帮助探索训练期间未遇到的环境。图像到图像的扩散会生成子目标图像，然后 ViNT 会朝着这些图像导航，同时在后台构建拓扑图。

另一项考虑零样本导航任务的工作是视听语言地图（AVLMaps）[138]

。 AVLMaps 为来自音频、视觉和语言提示的跨模式信息提供了 3D 空间地图表示。 AVLMaps 重新

接收多模态提示并在现实世界中执行零射击导航任务。输入是深度和 RGB 图像、相机姿势和音频。视觉特征使用预训练的 基础模型（Foundation Models） 进行编码。计算视觉定位特征（使用 NetVLAD [139]、SuperPoint [140]）、视觉语言特征（使用 LSeg [58]）和音频语言特征（使用 AudioCLIP [141]），并将来自不同模态的预测组合成 3D 热图。计算热图的逐像素联合概率并用于规划。此外，导航策略在 GPT-3 的帮助下生成为可执行代码。最后，预测 3D 热图，指示对象、声音和图像等多模态概念的位置。

许多机器人专家可能想知道经典模块化机器人导航系统与端到端学习系统的比较优势。语义导航[142]试图通过对语义视觉导航方法的实证分析来解决这个问题。该研究比较了六个不同家庭中经典、模块化和端到端学习范式的代表性方法，没有任何先验知识、地图或仪器。研究结果表明，模块化学习方法在现实场景中表现良好。相比之下，由于模拟图像和真实图像之间存在巨大的域差距，端到端学习方法面临着挑战。这种领域差距阻碍了端到端学习方法在现实世界导航任务中的有效性。对于从业者来说，该研究强调模块化学习是对象导航的可靠方法。策略设计中的模块化和抽象性能够成功地从模拟转移到现实，使模块化学习成为实际实施的有效选择。对于研究人员来说，该研究还强调了限制当前模拟器作为评估基准的可靠性

的两个关键问题。首先，图像中存在巨大的 Simto-Real 差距，这阻碍了学习策略从模拟到现实世界的可迁移性。其次，模拟和现实世界的错误模式之间存在脱节，这使得评估过程进一步复杂化。

开放词汇导航的另一项工作是对象导航任务。在此任务中，机器人必须能够找到人类描述的物体并导航至该物体。导航任务被分解为未检测到语言目标时的探索和检测到目标并且机器人向目标导航时的利用。当机器人在环境中移动时，它会使用 RGB-D 观察结果创建自上而下的地图并进行估计。在[143]中，作者介绍了一种零样本对象导航设置，该设置使用开放词汇分类器（例如 CLIP [4]）来计算图像和用户指定的描述之间的余弦相似度。

此类问题的常见数据集和基准包括 Matterport3D [144]、[145]、Gibson [146] 和 Habitat [147]

。 L3MVN [148]通过构建环境地图并使用大型语言模型的推理能力选择长期目标来增强视觉目标导航。该系统可以通过利用 RoBERTa-large [149] 等预训练语言模型来确定适当的长期导航目标，从而实现高效的探索和

searching. Chen et al. [150] presents a training-free and modular system for object goal navigation, which constructs a structured scene representation through active exploration. The system utilizes semantic information in the scene graphs to deduce the location of the target object and integrates semantics with the geometric frontiers to enable the agent to navigate effectively to the most promising areas for object search while avoiding detours in unfamiliar environments. HomeRobot [151] introduces a benchmark for the Open-Vocabulary Mobile Manipulation (OVMM) task. OVMM task is the problem of finding an object in any unseen environment, navigating towards the object, picking it up, and navigating towards a goal location to place the object. HomeRobot provides a benchmark in simulation and the real world for OVMM tasks.

2) Open-Vocabulary Manipulation: Open-vocabulary manipulation refers to the problem of manipulating any object in a previously unseen environment. VisuoMotor Attention Agent (VIMA) [152] learns robot manipulation from multi-modal prompts. VIMA is a transformer-based agent that predicts motor commands conditioned on a task prompt and a history of interactions. VIMA It introduces a new form of task specifications that combines textual and visual tokens. Multi-modal prompting converts different robot manipulation tasks, such as visual goal-reaching, learning from visual demonstrations, and novel concept grounding into one sequence modeling problem. It offers the training of a unified policy across diverse tasks, potentially allowing for zero-shot generalization to previously unseen ones. VIMA-BENCH is introduced as a benchmark for multi-modal robot learning. The VIMA-BENCH simulator supports collections of objects and textures that can be utilized in multi-modal prompting. RoboCat [153] is a self-improving AI agent. It uses a 1.18B-parameter decoder-only transformer. It learns to operate different robotic arms, solves tasks from as few as 100 demonstrations, and improves from self-generated data. RoboCat is based on Gato [154] architecture and is trained with a self-improvement cycle.

For robots to operate effectively in the real world they must be able to manipulate previously unseen objects. Liu et al. present StructDiffusion [155], which seeks to enable robots to use partial viewpoint clouds and natural language instructions to construct a goal configuration for objects that were previously seen or unseen. They accomplish this by first using segmentation to break up the scene into objects. Then they use a multi-model transformer to combine word and point cloud embeddings and output a 6-DoF goal pose prediction. The predictions are iteratively refined via diffusion and a discriminator that is trained to determine if a sampled configuration is feasible. Manipulation of Open-World Objects (MOO) [156] leverages a pretrained vision-language model to extract object-centric information from the language command and the image and conditions the robot policy on the current image, the instructions, and the extracted object information in a form of a single-pixel overlaid onto the image. MOO uses Owl-ViT for object detection and RT-1 for language-conditioned policy learning.

Another task in robot manipulation involves autonomous scene rearrangement and in-painting. DALL-E-Bot [157] performs zero-shot autonomous rearrangement in the scene in a human-like way using pretrained image diffusion model DALL-E2 [88]. DALL-E-Bot autonomous object rearrangement does not require any further data collection or training. First, the initial observation image (of the disorganized scene) is converted into a per-object representation including a segmentation mask using Mask R-CNN [158], an object caption, and a CLIP visual feature vector. Then a text prompt is generated by describing the object in the scene and is given to DALL-E to create a goal image for the rearrangement task (the objects should be rearranged in a human-like way). Next, the objects in the initial and generated images are matched using their CLIP visual features. Poses are estimated by aligning their segmentation masks. The robot rearranges the scene based on the estimated poses to create the generated arrangement.

In Table II some robotic-specific foundation models are reported along with information about their size and architecture, pretrained task, inference time, and hardware setup.

IV. Perception

Robots interacting with their surrounding environments receive raw sensory information in different modalities such as images, video, audio, and language. This high-dimensional data is crucial for robots to understand, reason, and interact in their environments. Foundation models, including those that have been developed in the vision and NLP domains, are promising tools for converting these high-dimensional inputs into abstract, structured representations that can be more easily interpreted and manipulated. Particularly, multi-modal foundation models enable robots to integrate different sensory inputs into a unified representation encompassing semantic, spatial, temporal, and affordance information. These multi-modal models reflect cross-modal interactions, often by aligning elements across modalities to ensure coherence and correspondence. For example, text and image data are aligned for image captioning tasks. This section will explore a range of tasks related to robot perception that are improved through aligning modalities using foundation models, with a focus on vision and language. There is an extensive body of literature studying multi-modality in the machine learning community, and an interested reader is referred to the survey paper [161] that presents a taxonomy of multi-modal learning. We focus on applications of multi-modal models to robotics.

A. Open-Vocabulary Object Detection and 3D Classification

1) Object Detection: Zero-shot object detection allows robots to identify and locate objects they have never encountered previously. Grounded Language-Image Pre-training (GLIP) [52] integrates object detection and grounding by redefining object detection as phrase grounding. This reformulation enables the learning of a visual representation that is both language-aware and semantically rich at the object level. In this framework, the input to the detection model comprises not only an image but also a text prompt that describes all the potential categories for the detection task. To train GLIP, a dataset of 27 million grounding instances was compiled,

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寻找。陈等人。 [150]提出了一种用于对象目标导航的免训练和模块化系统，该系统通过主动探索构建结构化场景表示。该系统利用场景图中的语义信息来推断目标物体的位置，并将语义与几何边界相结合，使智能体能够有效地导航到最有希望进行物体搜索的区域，同时避免在不熟悉的环境中走弯路。
HomeRobot [151] 引入了 OpenVocabulary Mobile Manipulation (OVMM) 任务的基准。 OVMM 任务是在任何看不见的环境中找到对象、导航到对象、拾取对象、导航到目标位置以放置对象的问题。
HomeRobot 为 OVMM 任务提供了模拟和现实世界的基准。

2）开放词汇操作：开放词汇操作是指在以前未见过的环境中操作任何对象的问题。 VisuoMotor Attention Agent (VIMA) [152] 从多模式提示中学习机器人操作。 VIMA 是一种基于变压器的智能体，可以根据任务提示和交互历史来预测运动命令。 VIMA 它引入了一种新形式的任务规范，结合了文本和视觉标记。多模态提示将不同的机器人操作任务（例如视觉目标达成、从视觉演示中学习以及新概念 落地关联（Grounding））转换为一个序列建模问题。它提供了跨不同任务的统一策略的训练，有可能允许零样本泛化到以前未见过的任务。 VIMA-BENCH 是作为多模态机器人学习的基准而引入的。 VIMA-BENCH 模拟器支持可用于多模式提示的对象和纹理集合。 RoboCat [153] 是一个 自我改进（Self-Improving） AI 智能体。它使用 1.18B 参数解码器专用变压器。它可以学习操作不同的机械臂，通过少至 100 次演示来解决任务，并通过自行生成的数据进行改进。 RoboCat 基于 Gato [154] 架构，并使用 自我改进（Self-Improvement） 循环进行训练。

为了使机器人在现实世界中有效运行，它们必须能够操纵以前看不见的物体。刘等人。目前的 StructDiffusion [155] 旨在使机器人能够使用部分视点云和自然语言指令来为之前看到或未看到的对象构建目标配置。他们首先使用分割将场景分解为对象来实现这一目标。然后，他们使用多模型转换器来组合词和点云嵌入并输出 6-DoF 目标姿势预测。通过扩散和鉴别器迭代地细化预测，鉴别器经过训练以确定采样的配置是否可行。开放世界对象的操纵（MOO）[156]利用预训练的视觉语言模型从语言命令和图像中提取以对象为中心的信息，并以覆盖在图像上的单像素的形式调整当前图像、指令和提取的对象信息上的机器人策略。 MOO 使用 Owl-ViT 进行对象检测，使用 RT-1 进行语言条件策略学习。

机器人操纵的另一项任务涉及自主场景重新排列和修复。 DALL-E-Bot [157] 每-

使用预训练的图像扩散模型 DALL-E2 [88] 以类似人类的方式在场景中形成零样本自主重排。
DALL-E-Bot 自主对象重新排列不需要任何进一步的数据收集或训练。首先，（杂乱场景的）初始观察图像被转换为每个对象的表示，包括使用 Mask R-CNN [158] 的分割掩模、对象标题和 CLIP 视觉特征向量。然后通过描述场景中的对象生成文本提示，并将其交给 DALL-E 来为重新排列任务创建目标图像（对象应该以类似人类的方式重新排列）。接下来，使用 CLIP 视觉特征来匹配初始图像和生成图像中的对象。通过对齐分割掩模来估计姿势。机器人根据估计的姿势重新排列场景以创建生成的排列。

在表 II 中，报告了一些特定于机器人的 基础模型（Foundation Models） 以及有关其大小和架构、预训练任务、推理时间和硬件设置的信息。
四．感知

与周围环境交互的机器人以不同方式接收原始感官信息，例如图像、视频、音频和语言。这种高维数据对于机器人在其环境中理解、推理和交互至关重要

。 基础模型（Foundation Models），包括那些在视觉和 NLP 领域开发的工具，是将这些高维输入转换为更容易解释和操作的抽象结构化表示的有前景的工具。特别是，多模态 基础模型（Foundation Models）使机器人能够将不同的感官输入集成到包含语义、空间、时间和 可供性（Affordance） 信息的统一表示中。这些多模态模型反映了跨模态交互，通常通过跨模态对齐元素来确保连贯性和对应性。例如，文本和图像数据针对图像字幕任务进行对齐

。本节将探讨一系列与机器人感知相关的任务，这些任务通过使用 基础模型（Foundation Models） 对齐模态来改进，重点是视觉和语言。机器学习社区中有大量研究多模态的文献，感兴趣的读者可以参考调查论文 [161]，其中提出了多模态学习的分类法

。我们专注于多模态模型在机器人领域的应用。
A. 开放词汇目标检测和 3D 分类

1）物体检测：零样本物体检测使机器人能够识别和定位他们以前从未遇到过的物体

。扎根语言图像预训练（GLIP）[52]通过将对象检测重新定义为短语扎根来集成对象检测和扎根

术语保留：落地关联（Grounding）

。这种重新表述使得能够学习在对象级别上具有语言感知和语义丰富的视觉表示

。在此框架中，检测模型的输入不仅包括图像，还包括描述检测任务的所有潜在类别的文本提示

。为了训练 GLIP，编译了包含 2700 万个 落地关联（Grounding） 实例的数据集，

TABLE II
PRETRAINED MODELS FOR ROBOTICS

Paper	Backbone	Size (Parameters)	Pretrained Task	Inference Speed	Hardware *
RoboCat [153]	decoder-only transformer	1.18B	manipulation	10-20Hz	
Gato [154]	decoder-only transformer	1.2B	generalist agent	20Hz	4 days on 16x16 TPU v3 slice
PaLM-E-562B [6]	decoder-only transformer	562B	1Hz for Language subgoals + 5Hz low-level control policies	5-6Hz	runs on multi-TPU cloud service
ViNT [137]	EfficientNet+ decoder transformer	31M	visual navigation	4Hz	variety of GPU configurations is used including 2×4090, 3×Titan Xp, 4×P100, 8×1080Ti, 8×V100, and 8×A100
VPT [72]	a temporal convolution layer, a ResNet 62 image processing stack, and residual unmasked attention layers,	0.5B	embodied agent in Minecraft	20Hz	9 days on 720 V100 GPUs
RT-1 [44]	Conditioned EfficientNet + TokenLearner + decoder-only transformer	35M	real-world robotics tasks	3Hz	
RT-2 [45]	PaLI-X	55B	real-world robotics tasks	1-3Hz	runs on multi-TPU cloud service
RT-2-X [46]	ViT and Language model UL2 [159]	55B	real-world robotics tasks	1-3Hz	runs on multi-TPU cloud service
LIV [37]	CLIP		reward learning	15Hz	8 NVIDIA V100 GPUs
SMART [134]	decoder-only transformer	11M	bidirectional dynamics prediction and masked hindsight control	1 Hz	8 Nvidia V100 GPUs
COMPASS [160]	3D-Resnet encoder	20M	Contrastive loss	30 Hz	8 Nvidia V100 GPUs
PACT [47]	decoder-only transformer	12M	forward dynamics and next action prediction	10 Hz (edge) / 50 Hz	Nvidia Xavier NX (edge) / 8 Nvidia V100 GPUs

* Empty fields in the table denote no data is reported.

consisting of 3 million human-annotated pairs and 24 million image-text pairs obtained by web crawling. The results of the study demonstrate the remarkable zero-shot and few-shot transferability of GLIP to a wide range of object-level recognition tasks. Recently, PartSLIP [162] demonstrated that GLIP can be used for low-shot part segmentation on 3D objects. PartSLIP renders a 3D point cloud of an object from multiple views and combines 2D bounding boxes in these views to detect object parts. To deal with noisy 2D bounding boxes from different views, PartSLIP runs a voting and grouping method on super points from 3D, assigns multi-view 2D labels to super points, and finally groups super points to obtain a precise part segmentation. To enable few-shot learning of 3D part segmentation, prompt tuning, and multi-view feature aggregation are proposed to improve performance.

OWL-ViT [51] is an open-vocabulary object detector. OWL-ViT uses a vision transformer architecture with contrastive image-text pre-training and detection end-to-end fine-tuning. Unlike GLIP, which frames detection as a phrase grounding problem with a single text query and limits the number of possible object categories, OWL-ViT can handle multiple text-based or image-driven queries. OWL-ViT has been applied to robot learning for example in VoxPoser [34] as the open-vocabulary object detector to find "entities of interest" (e.g., vase or drawer handles) and ultimately define value maps for optimizing manipulation trajectories.

Grounding DINO [53] combines DINO [105] with grounded pre-training, extending the closed-set DINO model to open-set detection by fusing vision and language. Grounding DINO outperforms GLIP in open-set object detection. This superior performance is mainly due to the transformer architecture of Grounding DINO, which facilitates multi-modal feature fusion at multiple stages.

2) 3D Classification: Zero-shot 3D classifiers can enable robots to classify objects in their environments without explicit training data. Foundation models are strong candidates for performing 3D classification. PointCLIP [54] transfers CLIP's pre-trained knowledge of 2D images to 3D point cloud understanding by aligning point clouds with text. The authors propose to project each point onto a series of pre-defined image planes to generate depth maps. Then, the CLIP visual encoder is used to encode multi-view features of the point cloud and predict labels in natural language for each view. The final prediction for the point cloud is computed via weighted aggregation of the predictions for each view. PointBERT [55] uses a transformer-based architecture to extract features from point clouds, generalizing the concept of BERT into 3D point clouds.

Unlike PointCLIP which converts the task of matching point clouds and text to image-text alignment, ULIP [56], [57] is a Unified representation of Language, Images, and Point clouds for 3D understanding. It achieves this by pre-training with object triplets (image, text, point cloud). The model is trained using a small number of automatically synthesized triplets from ShapeNet55 [163], which is a large-scale 3D model repository. ULIP uses CLIP as the vision-language

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表 II 机器人预训练模型

|论文|主干|大小（参数）|预训练任务|推理
速度|硬件*|
|---|---|---|---|---|---|
|RoboCat [153]|仅解码器变压器|1.18B|操纵|10-20Hz||
|Gato [154]|仅解码器变压器|1.2B|通才智能体|20Hz|16x16 TPU v3 切片上的 4 天|
|PaLM-E-562B [6]|仅解码器变压器|562B|1Hz 用于语言
子目标 + 5Hz
低级控制
策略|5-6Hz|在多 TPU 云服务上运行|
|ViNT [137]|EfficientNet+解码变压器|31M|视觉导航|4Hz|使用多种GPU配置
包括2×4090、3×Titan
Xp、4×P100、8×1080Ti、8×V100、
和8×A100|
|VPT [72]|时间卷积层、
ResNet 62 图像处理
堆栈和残余未屏蔽
注意力层，|0.5B|
Minecraft|20Hz|720 V100 GPU 上的 9 天体现智能体|
|RT-1 [44]|条件 EfficientNet +
TokenLearner + 仅解码器
变压器|35M|现实世界机器人
任务|3Hz||
|RT-2 [45]|PaLI-X|55B|现实世界的机器人
任务|1-3Hz|在多 TPU 云服务上运行|
|RT-2-X [46]|ViT 和语言模型 UL2
[159]|55B|现实世界的机器人
任务|1-3Hz|在多 TPU 云服务上运行|
|LIV [37]|CLIP||奖励学习|15Hz|8 个 NVIDIA V100 GPU|
|SMART [134]|仅解码器变压器|11M|双向
动态预测
和屏蔽后见之明
控制|1 Hz|8 Nvidia V100 GPU|
|COMPASS [160]|3D-Resnet 编码器|20M|对比损失|30 Hz|8 个 Nvidia V100 GPU|
|PACT [47]|仅解码器变压器|12M|前向动态
和下一步动作
预测|10 Hz
（边缘）/ 50
Hz|Nvidia Xavier NX（边缘）/8

Nvidia V100 GPU|

> * 表中的空字段表示没有报告数据。

由 300 万个人工注释对和通过网络爬行获得的 2400 万个图像文本对组成。研究结果证明了 GLIP 在广泛的物体级识别任务中具有显着的零样本和少样本可迁移性。最近，PartSLIP [162] 证明 GLIP 可用于 3D 对象的低镜头部分分割。 PartSLIP 从多个视图渲染对象的 3D 点云，并结合这些视图中的 2D 边界框来检测对象部分。为了处理来自不同视图的噪声 2D 边界框，PartSLIP 对 3D 超级点运行投票和分组方法，为超级点分配多视图 2D 标签，最后对超级点进行分组以获得精确的零件分割。为了实现 3D 零件分割的少样本学习，提出了即时调整和多视图特征聚合来提高性能。

OWL-ViT [51] 是一种开放词汇对象检测器。 OWLViT 使用视觉转换器架构，具有对比图像文本预训练和检测端到端微调。与 GLIP 不同，GLIP 将检测框架为单个文本查询的短语 落地关联（Grounding） 问题并限制可能的对象类别的数量，OWL-ViT 可以处理多个基于文本或图像驱动的查询。 OWL-ViT 已应用于机器人学习，例如在 VoxPoser [34] 中作为开放词汇对象检测器来查找“感兴趣的实体”（例如花瓶或抽屉把手）并最终定义用于优化操作轨迹的值图。

落地关联（Grounding） DINO [53] 将 DINO [105] 与接地预训练相结合，将闭集 DINO 模型扩展到开放式

通过融合视觉和语言来设置检测。 落地关联（Grounding） DINO 在开放集目标检测中优于 GLIP。这种优越的性能主要归功于落地关联（Grounding） DINO的变压器架构，它有利于多个阶段的多模态特征融合。

2）3D分类：零样本3D分类器可以使机器人在没有明确训练数据的情况下对环境中的物体进行分类。 基础模型（Foundation Models） 是执行 3D 分类的有力候选者。 PointCLIP [54] 通过将点云与文本对齐，将 CLIP 预先训练的 2D 图像知识转移到 3D 点云理解。作者建议将每个点投影到一系列预定义的图像平面上以生成深度图。然后，使用CLIP视觉编码器对点云的多视图特征进行编码，并预测每个视图的自然语言标签。点云的最终预测是通过每个视图的预测的加权聚合来计算的。 PointBERT [55] 使用基于转换器的架构从点云中提取特征，将 BERT 的概念推广到 3D 点云中。

与将点云和文本匹配任务转换为图像文本对齐的 PointCLIP 不同，ULIP [56]、[57] 是用于 3D 理解的语言、图像和点云的统一表示。它通过使用对象三元组（图像、文本、点云）进行预训练来实现这一点。该模型使用来自 ShapeNet55 [163] 的少量自动合成的三元组进行训练，ShapeNet55 是一个大型 3D 模型存储库。 ULIP 使用 CLIP 作为视觉语言

model. During pretraining, the CLIP model is kept frozen and a 3D encoder is trained by aligning the 3D features of an object with its associated textual and visual features from CLIP using contrastive learning. The pretraining process allows ULIP to learn a joint embedding space where the three modalities are aligned. One of the major advantages of ULIP is that it can substantially improve the recognition ability of 3D backbone models. This is because the pretraining process allows ULIP to learn more robust and discriminative features for each modality, which can then be used to improve the performance of 3D models. Another advantage of ULIP is that it is agnostic to the 3D model architecture, and thus can be easily integrated into the pretraining process of existing 3D pipelines. ULIP adopts masked language modeling from BERT to 3D by tokenizing 3D patches randomly masking out 3D tokens and predicting them back during pretraining. ULIP [56], [57] has shown that the performance of recognition capability of models such as PointBERT can be improved by using a unified multimodal representation of ULIP.

B. Open-Vocabulary Semantic Segmentation

Semantic segmentation classifies each pixel in an image into semantic classes. This provides fine-grained information about object boundaries and locations within an image and enables embodied agents to understand and interact with the environment at a more granular level. Several works explore how foundation models such as CLIP can enhance the generalizability and flexibility of semantic segmentation tasks.

LSeg is a language-driven semantic segmentation model [58] that associates semantically similar labels to similar regions in an embedding space. LSeg uses a text encoder based on the CLIP architecture to compute text embeddings and an image encoder with the underlying architecture of Dense Prediction Transformer (DPT) [164]. Similar to CLIP, LSeg creates a joint embedding space using text and image embeddings. LSeg freezes the text encoder at training time and trains the image encoder to maximize the correlation between the text embedding and the image pixel embedding of the ground-truth pixel class. It allows users to arbitrarily shrink, expand, or rearrange the label set (with unseen categories) for any image at test time.

Segment Anything Model (SAM) [59] introduces a framework for promptable segmentation consisting of the task definition for promptable segmentation, a segmentation foundation model (the Segment Anything Model, or SAM), and a data engine. SAM adapts a pretrained Vision Transformer from Masked Auto-Encoder (MAE) [113] as an image encoder while using a text encoder from CLIP [114] for sparse prompts (points, boxes, and text) and a separate dense prompt encoder for masks. In contrast to other foundation models that are trained in an unsupervised manner on web-scale data, SAM is trained using supervised learning with data engines that help scale the number of available annotations. Along with the model, the authors released the Segment Anything 1 Billion (SA-1B) dataset. It consists of 11M images and 1.1B segmentation masks. In this work, the authors conducted experiments on five zero-shot transfer tasks, including point-valid mask evaluation, edge detection, object proposal, instance segmentation, and text-to-mask. The system's composable design, facilitated by prompt engineering techniques, enables a broader range of applications compared to systems trained specifically for fixed task sets. However, one limitation of this work that is particularly relevant to robotic applications is that SAM cannot run in real-time.

FastSAM [60] and MobileSAM [61] achieve comparable performance to SAM at faster inference speeds. The Track Anything Model (TAM) [62] combines SAM and XMem [165], an advanced video object segmentation (VOS) model, to achieve interactive video object tracking and segmentation. Anything-3D [166] employs a collection of visual-language models and SAMs to elevate objects into the realm of 3D. It uses BLIP [108] to generate textual descriptions while using SAM to extract objects of interest from visual input. Then, Anything-3D lifts the extracted objects into a Neural Radiance Field (NeRF) [167] representation using a text-to-image diffusion model, enabling their integration into 3D scenes.

Amidst these remarkable advancements, achieving fine-grained detection with real-time performance still remains challenging. For example, LSeg [58] reports failure cases related to misclassification, when the test time input labels do not include the true label for the pixel, and the model thus assigns the highest probability to the closest label. Another failure case occurs when multiple labels can be correct for a particular pixel, and the model must classify it as just one of the categories. For example "window" and "house" may both be defined as labels, but during inference, a pixel representing a "window" may be labeled instead as "house". SAM also does not provide precise segmentation for fine structures and often fails to produce crisp boundaries. All models that use SAM as a sub-component may encounter similar limitations. In the future, fine-grained semantic segmentation models that can assign multiple labels to a pixel when there are multiple correct descriptions should be considered. Additionally, developing models that can run in real-time will be critical for robotics applications.

C. Open-Vocabulary 3D Scene and Object Representations

Scene representations allow robots to understand their surroundings, facilitate spatial reasoning, and provide contextual awareness. Language-driven scene representations align textual descriptions with visual scenes, enabling robots to associate words with objects, locations, and relationships. In this section, we study recent works that use foundation models to enhance scene representations.

1) Language Grounding in 3D Scene: Language grounding refers to combining geometric and semantic representations of an environment. One type of representation that can provide an agent with a strong geometric prior is an implicit representation. One example of an implicit representation is a Neural Radiance Field (NeRF) [167]–[169]. NeRF creates high-quality 3D reconstructions of scenes and objects from a set of 2D images captured from different viewpoints (without the need for explicit depth information). The NeRF neural

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模型。在预训练期间，CLIP 模型保持冻结状态，并通过使用对比学习将对象的 3D 特征与其来自 CLIP 的相关文本和视觉特征对齐来训练 3D 编码器。预训练过程允许 ULIP 学习三种模式对齐的联合嵌入空间。ULIP的主要优点之一是它可以大幅提高3D骨干模型的识别能力。这是因为预训练过程允许 ULIP 为每种模态学习更鲁棒和更具辨别力的特征，然后可以将其用于提高 3D 模型的性能。 ULIP 的另一个优点是它与 3D 模型架构无关，因此可以轻松集成到现有 3D 管道的预训练过程中。 ULIP 采用从 BERT 到 3D 的屏蔽语言建模，通过对 3D 补丁进行标记，随机屏蔽 3D 标记并在预训练期间将其预测回来。 ULIP[56]、[57]表明，通过使用 ULIP 的统一多模态表示，可以提高 PointBERT 等模型的识别能力性能。
B. 开放词汇语义分割

语义分割将图像中的每个像素分类为语义类别。这提供了有关图像中对象边界和位置的细粒度信息，并使具身智能体能够在更细粒度的级别上理解环境并与环境交互。一些工作探讨了基础模型（Foundation Models）（例如CLIP）如何增强语义分割任务的通用性和灵活性。

LSeg 是一种语言驱动的语义分割模型 [58]，它将语义相似的标签与嵌入空间中的相似区域相关联。LSeg 使用基于 CLIP 架构的文本编码器来计算文本嵌入，并使用具有密集预测变换器（DPT）[164]底层架构的图像编码器。与 CLIP 类似，LSeg 使用文本和图像嵌入创建联合嵌入空间。 LSeg 在训练时冻结文本编码器并训练图像编码器以最大化文本嵌入和地面实况像素类的图像像素嵌入之间的相关性。它允许用户在测试时任意缩小、扩展或重新排列任何图像的标签集（具有看不见的类别）。

分段任意模型（SAM）[59]引入了一个可提示分段的框架，由可提示分段的任务定义、分段 基础模型（Foundation Model）（分段任意模型，或 SAM）和数据引擎组成。 SAM 采用来自 Masked Auto-Encoder (MAE) [113] 的预训练视觉变换器作为图像编码器，同时使用来自 CLIP [114] 的文本编码器用于稀疏提示（点、框和文本），并使用单独的密集提示编码器用于掩模。与其他在网络规模数据上以无监督方式进行训练的 基础模型（Foundation Models） 相比，SAM 使用数据引擎的监督学习进行训练，有助于扩展可用注释的数量。除了模型之外，作者还发布了 Segment Anything 10 Billion (SA-1B) 数据集。它由 11M 图像和 1.1B 分割掩模组成。在这项工作中，作者对五个零样本迁移任务进行了实验，包括点有效

掩模评估、边缘检测、对象提议、实例分割和文本到掩模。与专门针对固定任务集训练的系统相比，该系统的可组合设计在快速工程技术的推动下能够实现更广泛的应用。然而，这项工作的一个与机器人应用特别相关的限制是 SAM 无法实时运行。

FastSAM [60] 和 MobileSAM [61] 以更快的推理速度实现了与 SAM 相当的性能。 Track Anything Model (TAM) [62] 结合了 SAM 和 XMem [165]，一种先进的视频对象分割 (VOS) 模型，以实现交互式视频对象跟踪和分割。 Anything-3D [166] 采用视觉语言模型和 SAM 的集合将对象提升到 3D 领域。它使用 BLIP [108] 生成文本描述，同时使用 SAM 从视觉输入中提取感兴趣的对象。然后，Anything-3D 使用文本到图像扩散模型将提取的对象提升为神经辐射场 (NeRF) [167] 表示形式，从而使其能够集成到 3D 场景中。

尽管取得了这些显着的进步，实现具有实时性能的细粒度检测仍然具有挑战性。例如，当测试时输入标签不包括像素的真实标签时，LSeg [58] 报告与错误分类相关的失败案例，因此模型将最高概率分配给最接近的标签。当多个标签对于特定像素来说是正确的，并且模型必须将其分类为其中一个类别时，会发生另一种失败情况。例如，“窗户”和“房子”都可以被定义为标签，但是在推理过程中，代表“窗户”的像素可以被标记为“房子”。 SAM 也不能为精细结构提供精确分割，并且通常无法产生清晰的边界。所有使用 SAM 作

为子组件的模型都可能遇到类似的限制。未来，应该考虑细粒度的语义分割模型，当存在多个正确的描述时，可以为一个像素分配多个标签。此外，开发可以实时运行的模型对于机器人应用至关重要。

C. 开放词汇 3D 场景和对象表示

场景表示使机器人能够理解周围环境，促进空间推理，并提供上下文感知。语言驱动的场景表示将文本描述与视觉场景结合起来，使机器人能够将单词与物体、位置和关系联系起来。在本节中，我们研究最近使用 基础模型（Foundation Models） 来增强场景表示的作品。

1）3D场景中的语言落地关联（Grounding）：语言落地关联（Grounding）是指组合环境的几何和语义表示。可以为智能体提供强几何先验的一种表示类型是隐式表示。隐式表示的一个例子是神经辐射场（NeRF）[167]-[169]。 NeRF 根据从不同视点捕获的一组 2D 图像创建场景和对象的高质量 3D 重建（无需显式深度信息）。 NeRF 神经网络

network takes camera poses as input and predicts the 3D geometry of the scene as well as color and intensity. Most NeRF-based models memorize the light field in a single environment and are not pre-trained on a large data set, hence they are not foundation models. However, foundation models such as CLIP can be combined with NeRFs to extract semantic information from an agent's environment.

Kerr et al. [64] propose language-embedded radiance fields (LERFs) that ground CLIP embeddings into a dense multi-scale 3D field. This results in a 3D representation of the environment that can be queried to produce semantic relevancy maps. The LERF model takes 3D position (x, y, z), viewing direction (ϕ, θ), and a scaling factor as input and outputs an RGB value, density (σ), as well as DINO [105] and CLIP features. The LERF is optimized in two stages: initially, a multi-scale feature pyramid of CLIP embeddings over training views is computed; then, the pyramid is interpolated using the image scale and pixel location to obtain the CLIP embedding; and finally, the CLIP embeddings are supervised through cosine similarity and the RGB and density are supervised using the standard mean squared-error.

Models such as LERF inherit the shortcomings of CLIP and NeRF. For example, CLIP exhibits difficulty in capturing spatial relationships between objects. In addition, language queries from CLIP can highlight a significant issue similar to the bag-of-words model, which struggles to distinguish terms with opposite sentiments. Also, NeRF relies on known camera poses associated with pre-captured multi-view images.

In CLIP-Fields [170], an implicit scene representation $g(x, y, z) : \mathcal{R}^3 \rightarrow \mathcal{R}^d$ is trained by decoding a d-dimensional latent vector to different modality-specific outputs. The model distills information from pretrained image models by back-projecting the pixel labels to 3D space and training the output heads to predict semantic labels from an open-vocab object detector called Detic, the CLIP visual representation, and one-hot instance labels using a contrastive loss. The scene representation can then be used as a spatial database for segmentation, instance identification, semantic search over space, and 3D view localization from images.

Another related work is VLMaps [171], which projects pixel embeddings from LSeg to grid cells in a top-down grid map. This method does not require training and instead directly backprojects pixel embeddings to grid cells and averages the values in overlapping regions. By combining a VLMap with a code-writing LLM, the authors demonstrate spatial goal navigation using landmarks (e.g., move to the plant) or spatial references with respect to landmarks (between the keyboard and the bowl). Semantic Abstraction (SemAbs) [172] presents another approach for 3D scene understanding by decoupling visual-semantic reasoning and 3D reasoning. In SemAbs, given an RGB-D image of a scene, a semantic-aware 2D VLM extracts 2D relevancy maps for each queried object while semantic-abstracted 3D modules predict the 3D occupancy of each object using the relevancy maps. Because the 3D modules are trained irrespective of the specific object labels, the system demonstrates strong generalization capabilities, including generalization to new object categories and from simulation to the real world.

Current VLMs can reason about 2D images, however, they are not grounded in the 3D world. The main challenge for building 3D VLM foundation models is the scarcity of 3D data. Particularly, 3D data paired with language description is scarce. One strategy to circumvent this issue is to take advantage of 2D models trained on large-scale data to supervise 3D models. For instance, the authors of FeatureNeRF [173] propose to learn 3D semantic representations by distilling 2D vision foundation models (i.e., DINO or Latent Diffusion) into 3D space via neural rendering. FeatureNeRF predicts a continuous 3D semantic feature volume from a single or few images which can be used for downstream tasks such as keypoint transfer or object part co-segmentation.

In 3D-LLM [11], the authors propose to use 2D VLMs as backbones to train a 3D-LLM that can take 3D representations (i.e., 3D point clouds with their features) as inputs and accomplish a series of diverse 3D-related tasks. The 3D features are extracted from 2D multi-view images and mapped to the feature space of 2D pretrained VLMs. To overcome 3D data scarcity, the authors propose an efficient prompting procedure for ChatGPT to generate 3D-language data encompassing a diverse set of tasks. These tasks include 3D captioning, dense captioning, 3D question answering, 3D task decomposition, 3D grounding, 3D-assisted dialog, and navigation. Also, to capture 3D spatial information, the authors propose a 3D localization mechanism by 1) augmenting 3D features with position embedding and 2) augmenting LLM vocabularies with 3D location tokens. In the first part, the position embeddings of the three dimensions are generated and concatenated with 3D features. In the second part, the coordinates of the bounding box representing the grounded region are discretized to voxel integers as location tokens $< x_{min}, y_{min}, z_{min}, x_{max}, y_{max}, z_{max} >$. It is important to highlight that, typically, creating 3D representations necessitates the use of 2D multi-view images and camera matrices. These resources are not as readily available as the vast amounts of internet-scale text and image data that current foundation models are trained on.

2) Scene Editing: When an embodied agent relies on an implicit representation of the world, the capability to edit and update this representation enhances the robot's adaptability. For instance, consider a scenario where a robot utilizes a pretrained NeRF model of an environment for navigation and manipulation. If a portion of the environment changes, being able to adjust the NeRF without retraining the model from scratch saves time and resources.

In the case of NeRFs, Wang et al. [63] propose a text and image-driven method for manipulating NeRFs called CLIP-NeRF. This approach uses CLIP to disentangle the dependence between shape and appearance in conditional neural radiance fields. CLIP-NeRF facilitates the editing of the shape and appearance of NeRFs using either image or text prompts. It is composed of two modules: the disentangled conditional NeRF and CLIP-driven manipulation. The former takes the positional encoding $\gamma(x, y, z)$, a shape code z_s, viewing direction $v(\phi, \theta)$, and appearance code z_a as an input and outputs color and density. The disentanglement is achieved using a deformation network that is appended as input to the traditional NeRF MLP

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网络将相机姿势作为输入并预测场景的 3D 几何形状以及颜色和强度。大多数基于 NeRF 的模型都会记住单一环境中的光场，并且没有在大型数据集上进行预训练，因此它们不是 基础模型（Foundation Models）。然而，基础模型（Foundation Models）（例如 CLIP）可以与 NeRF 结合，从智能体环境中提取语义信息。

克尔等人。 [64]提出了语言嵌入辐射场（LERF），将 CLIP 嵌入嵌入到密集的多尺度 3D 场中。这会产生环境的 3D 表示，可以查询该表示以生成语义相关性图。 LERF 模型将 3D 位置 (x, y, z)、观察方向 (φ, θ) 和缩放因子作为输入，并输出 RGB 值、密度 (σ) 以及 DINO [105] 和 CLIP 特征。 LERF 分两个阶段进行优化：首先，计算训练视图上 CLIP 嵌入的多尺度特征金字塔；然后，使用图像尺度和像素位置对金字塔进行插值以获得CLIP嵌入；最后，通过余弦相似度来监督 CLIP 嵌入，并使用标准均方误差来监督 RGB 和密度。

LERF等模型继承了CLIP和NeRF的缺点。例如，CLIP 在捕捉对象之间的空间关系方面表现出困难。此外，来自 CLIP 的语言查询可以突出类似于词袋模型的一个重要问题，该模型很难区分具有相反情绪的术语。此外，NeRF 依赖于与预先捕获的多视图图像相关的已知相机姿势。

在 CLIP-Fields [170] 中，隐式场景表示 g(x, y, z) : R3 →Rd 通过将 d 维潜在向量解码为不同模态特定输出来训练。该模型通过将像素标签反投影到 3D 空间并训练输出头来预测来自名为 Detic 的开放词汇对象检测器、CLIP 视觉表示和使用对比损失的单热实例标签的语义标签，从而从预训练的图像模型中提取信息。然后，场景表示可以用作空间数据库，用于分割、实例识别、空间语义搜索以及图像的 3D 视图定位。

另一项相关工作是 VLMaps [171]，它将像素嵌入从 LSeg 投影到自上而下的网格地图中的网格单元。该方法不需要训练，而是直接将像素嵌入反投影到网格单元并对重叠区域中的值进行平均。通过将 VLMap 与代码编写 LLM 相结合，作者演示了使用地标（例如，移动到工厂）或相对于地标的空间参考（键盘和碗之间）的空间目标导航。语义抽象 (SemAbs) [172] 通过解耦视觉语义推理和 3D 推理，提出了另一种 3D 场景理解的方法。在 SemAbs 中，给定场景的 RGB-D 图像，语义感知 2D VLM 为每个查询对象提取 2D 相关性图，而语义抽象 3D 模块则使用相关性图预测每个对象的 3D 占用情况

。由于 3D 模块的训练与特定对象标签无关，因此系统表现出强大的泛化能力，包括泛化到新的对象类别以及从模拟到现实世界。

当前的 VLM 可以推理 2D 图像，但是它们并不立足于 3D 世界。构建 3D VLM 基础模型（Foundation Models）的主要挑战是 3D 数据的稀缺。特别是，与语言描述配对的 3D 数据非常稀缺。规避此问题的一种策略是利用在大规模数据上训练的 2D 模型来监督 3D 模型。例如，FeatureNeRF [173] 的作者提出通过神经渲染将 2D 视觉 基础模型（Foundation Models）（即 DINO 或潜在扩散）提取到 3D 空间来学习 3D 语义表示。 FeatureNeRF 从单个或几个图像中预测连续的 3D 语义特征量，可用于下游任务，例如关键点传输或对象部分联合分割。

在 3D-LLM [11] 中，作者建议使用 2D VLM 作为骨干来训练 3D-LLM，该 3D-LLM 可以将 3D 表示（即具有其特征的 3D 点云）作为输入并完成一系列不同的 3D 相关任务。从 2D 多视图图像中提取 3D 特征，并将其映射到 2D 预训练 VLM 的特征空间。为了克服 3D 数据稀缺的问题，作者提出了一种有效的 ChatGPT 提示程序，以生成涵盖多种任务的 3D 语言数据。这些任务包括 3D 字幕、密集字幕、3D 问答、3D 任务分解、3D 落地关联（Grounding）、3D 辅助对话和导航。此外，为了捕获 3D 空间信息，作者提

出了一种 3D 定位机制，方法是：1）通过位置嵌入增强 3D 特征，2）通过 3D 位置标记增强 LLM 词汇表。 在第一部分中，生成三个维度的位置嵌入并将其与 3D 特征连接起来。在第二部分中，表示接地区域的边界框的坐标被离散化为体素整数作为位置标记<xmin，ymin，zmin，xmax，ymax，zmax>。需要强调的是，创建 3D 表示通常需要使用 2D 多视图图像和相机矩阵。这些资源并不像当前 基础模型（Foundation Models） 所训练的大量互联网规模的文本和图像数据那么容易获得。

2）场景编辑：当具身智能体依赖于世界的隐式表示时，编辑和更新该表示的能力增强了机器人的适应性。例如，考虑这样一个场景：机器人利用环境的预训练 NeRF 模型进行导航和操作。如果环境的一部分发生变化，无需从头开始重新训练模型即可调整 NeRF，从而节省时间和资源。

就 NeRF 而言，Wang 等人。 [63]提出了一种文本和图像驱动的方法来操纵 NeRF，称为 CLIPNeRF。这种方法使用 CLIP 来理清条件神经辐射场中形状和外观之间的依赖性。 CLIP-NeRF 有助于使用图像或文本提示编辑 NeRF 的形状和外观

。它由两个模块组成：解开的条件 NeRF 和 CLIP 驱动的操作。前者将位置编码γ(x,y,z)、形状代码zs、观察方向v(φ,θ)和外观代码za作为输入并输出颜色和密度。使用变形网络实现解缠结，该网络作为传统 NeRF MLP 的输入附加

that produces density, and by taking the output from this MLP and concatenating it with an appearance code to attain the color value. The CLIP-driven manipulation module takes an image example or text prompt as an input and outputs a shape deformation Δz_s and an appearance deformation Δz_a from shape mapping and appearance mapping MLPs respectively. These deformation values aim to perturb the shape code and appearance code in the disentangled conditional NeRF module to produce the desired output.

A key limitation of the CLIP-NeRF approach is that prompting can impact the entire scene rather than a selected region. For example, prompting to change the color of a flower's petals might also impact the shape and color of its leaves. To address this limitation, Kobayashi et al. propose to train distilled feature fields (DFFs) [65] and then manipulate DFFs through query-based scene decomposition and editing. Pre-trained 2D VLMs (such as LSeg [58] and DINO [105]) are employed as teacher networks and distilled into 3D distilled feature fields via volume rendering. Editing is achieved by alpha compositing the density and color values of the two NeRF scenes. When combined with CLIP-NeRF, this method enables CLIP-NeRF to selectively edit specific regions of multi-object scenes. A similar approach was explored by Tschernezki et al. in [174] where the authors show that enforcing the 3D consistency of features in the NeRF embedding improved segmentation performance compared to using features from the original 2D images.

Another approach to more controlled 3D scene editing is to use structured 3D scene representations. Nerflets [175] represent a 3D scene as a combination of local neural radiance fields where each maintains its own spatial position, orientation, and dimension. Instead of employing a single large MLP to predict colors and densities as standard NeRF, individual Nerflets are combined to predict these values, modulated by their weights. After optimizing posed 2D images and segmentations, Nerflets reflect the decomposed scene and support more controlled editing.

One application of image editing in robotics is for data augmentation during policy learning. ROSIE [176] use the Imagen editor [177] to modify training images to add additional distractors and unseen objects and backgrounds to train robust imitation learning policies. GenAug [178] similarly generates images with in-category and cross-category object substitutions, visual distractors, and diverse backgrounds. The CACTI [14] pipeline includes a step in-painting different plausible objects via Stable-Diffusion [117] onto training images. These approaches generate photorealistic images for training robust policies; however, generating images with sufficient diversity while also maintaining physical realism, e.g. for object contacts, remains a challenge. Existing approaches use learned or provided masks to specify areas of the image to keep, or heuristics based on the particular robotic task.

Another direction is to use generative models to define goal images for planning. DALL-E-Bot [157] uses DALL-E 2 to define a goal image of human-like arrangements from observations.

3) Object Representations: Learning correspondences between objects can facilitate manipulation by enabling skill transfer from trained objects to novel object instances in known categories or novel object categories at test time. Traditionally, object correspondences have been learned using strong supervision such as keypoints and keyframes. Neural descriptor fields (NDFs) [179] remove the need for dense annotation by leveraging layer-wise activations from an occupancy network; however, this approach still requires many training shapes for each target object category. Additional works have started to build object representations directly from image features of pretrained vision models.

Feature Fields for Robotic Manipulation (F3RM) [180] builds on DFF to develop scene representations that support finding corresponding object regions. F3RM uses a similar feature representation for 6-DoF poses relative to objects (e.g., a grasp on the handle of the mug) to NDF. Besides allowing corresponding 6-DoF poses to be found from a few demonstrations, the pose embeddings can also be directly compared to text embeddings from CLIP to leverage language guidance (e.g., pick up the bowl). Correspondences between objects have also been directly extracted from DINO features [181] without training. This method first extracts dense ViT feature maps of two objects using multiple views. Similar regions on the two objects are found by computing the cyclical distance metric [182] on the feature maps. With the 2D patch correspondences, a 7-D rigid body transform (i.e., a SO(3) pose, a translation, and a scaling scalar) between the objects can be solved together with RANSAC and Umeyama's method [183].

D. Learned Affordances

Affordances refer to the potential of objects, environments, or entities to offer specific functions or interactions to an agent. They can include actions such as pushing, pulling, sitting, or grasping. Detecting affordances bridges the gap between perception and action.

Affordance Diffusion [66] synthesizes complex interactions of e.g. an articulated hand with a given object. Given an RGB image, Affordance Diffusion aims to generate images of human hands for hand-object interaction (HOI). The authors propose a two-step generative approach based on large-scale pretrained diffusion models based on where to interact (layout) and how to interact (content). The layout network generates a 2D spatial arrangement of hand and object. The content network then synthesizes images of a hand grasping the object conditioned on the given object and the sampled HOI layout. Affordance Diffusion outputs both the hand articulation and approach orientation.

Vision-Robotic Bridge (VRB) [67] trains a visual affordance model on internet videos of human behavior. Particularly, it estimates the likely location and manner in which a human interacts within a scene. This model captures the structural information of these behavioral affordances. The authors seamlessly integrate the affordance model with four different robot learning paradigms. Firstly, they apply offline imitation learning, where the robot learns by imitating the observed human interactions from the videos. Secondly, they use exploration techniques to enable the robot to actively

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产生密度，并通过从该 MLP 获取输出并将其与外观代码连接以获得颜色值。 CLIP驱动的操作模块以图像示例或文本提示作为输入，并分别从形状映射和外观映射MLP输出形状变形Δzs和外观变形Δza。这些变形值旨在扰乱解开的条件 NeRF 模块中的形状代码和外观代码，以产生所需的输出。

CLIP-NeRF 方法的一个关键限制是提示可能会影响整个场景而不是选定的区域。例如，提示改变花瓣的颜色也可能会影响其叶子的形状和颜色。为了解决这个限制，Kobayashi 等人。建议训练蒸馏特征场（DFF）[65]，然后通过基于查询的场景分解和编辑来操作 DFF。预训练的 2D VLM（例如 LSeg [58] 和 DINO [105]）被用作教师网络，并通过体渲染提炼成 3D 蒸馏特征场。编辑是通过 Alpha 合成两个 NeRF 场景的密度和颜色值来实现的。当与 CLIP-NeRF 结合使用时，该方法使 CLIP-NeRF 能够选择性地编辑多对象场景的特定区域。 Tschernezki 等人探索了类似的方法。在[174]中，作者表明，与使用原始 2D 图像的特征相比，在 NeRF 嵌入中强制特征的 3D 一致性提高了分割性能。

更受控制的 3D 场景编辑的另一种方法是使用结构化 3D 场景表示。 Nerflets [175] 将 3D 场景表示为局部神经辐射场的组合，其中每个神经辐射场都保持自己的空间位置、方向和维度。不像标准 NeRF 那样使用单个大型 MLP 来预测颜色和密度，而是组合各个 Nerflet 来预测这些值，并通过其权重进行调制。在优化 2D 图像和分割后，Nerflets 可以反映分解的场景并支持更受控制的编辑。

图像编辑在机器人技术中的一项应用是在策略学习期间进行数据增强。 ROSIE [176] 使用 Imagen 编辑器 [177] 修改训练图像以添加额外的干扰物和看不见的对象和背景，以训练强大的 模仿学习（Imitation Learning）策略。 GenAug [178] 类似地生成具有类别内和跨类别对象替换、视觉干扰物和不同背景的图像。 CACTI [14] 管道包括通过稳定扩散 [117] 将不同的合理对象绘制到训练图像上的步骤。这些方法生成逼真的图像，用于训练稳健的策略；然而，生成具有足够多样性的图像，同时保持物理真实感，例如对于物体接触来说，仍然是一个挑战。现有方法使用学习或提供的掩模来指定要保留的图像区域，或基于特定机器人任务的启发式方法。

另一个方向是使用生成模型来定义规划的目标图像。 DALL-E-Bot [157] 使用 DALL-E 2 根据观察定义类人排列的目标图像。

3）对象表示：学习对象之间的对应关系可以通过启用技能来促进操作

在测试时从训练过的对象转移到已知类别中的新对象实例或新对象类别。传统上，对象对应关系是使用关键点和关键帧等强监督来学习的。神经描述符字段（NDF）[179]通过利用占用网络的逐层激活消除了对密集注释的需要；然而，这种方法仍然需要针对每个目标对象类别进行许多训练形状。其他工作已经开始直接从预训练视觉模型的图像特征构建对象表示。

机器人操作特征场 (F3RM) [180] 基于 DFF 构建，以开发支持查找相应对象区域的场景表示。 F3RM 使用与 NDF 类似的特征表示来表示相对于对象的 6-DoF 姿势（例如，抓住杯子的手柄）。除了允许从一些演示中找到相应的 6-DoF 姿势之外，姿势嵌入还可以直接与 CLIP 中的文本嵌入进行比较，以利用语言指导（例如，拿起碗）。物体之间的对应关系也可以直接从 DINO 特征中提取[181]，无需训练。该方法首先使用多个视图提取两个对象的密集 ViT 特征图。通过计算特征图上的循环距离度量[182]来找到两个对象上的相似区域。通过 2D 块对应，对象之间的 7-D 刚体变换（即 SO(3) 位姿、平移和缩放标量）可以与 RANSAC 和 Umeyama 的方法一起求解 [183]。

D.了解到可供性（Affordances）

可供性（Affordances） 指对象、环境或实体向智能体提供特定功能或交互的潜力。它们可以包括推、拉、坐或抓等动作。检测 可供性（Affordances） 弥合了感知和行动之间的差距。

可供性（Affordance） Diffusion [66] 合成了复杂的相互作用，例如拿着给定物体的铰接手。给定 RGB 图像，可供性（Affordance） Diffusion 旨在生成用于手部物体交互 (HOI) 的人手图像。作者提出了一种基于大规模预训练扩散模型的两步生成方法，该模型基于交互位置（布局）和交互方式（内容）。布局网络生成手和物体的二维空间排列。然后，内容网络根据给定对象和采样的 HOI 布局合成抓握对象的手的图像。 可供性（Affordance） 扩散输出手部关节和接近方向。

视觉机器人桥（VRB）[67]在人类行为的互联网视频上训练视觉 可供性（Affordance） 模型。特别是，它估计人类在场景中交互的可能位置和方式。该模型捕获了这些行为 可供性（Affordances） 的结构信息。作者将 可供性（Affordance） 模型与四种不同的机器人学习范例无缝集成。首先，他们应用离线 模仿学习（Imitation Learning），机器人通过模仿视频中观察到的人类交互来学习。其次，他们利用探索技术使机器人能够主动地

discover and learn new affordances in its environment. Thirdly, the authors incorporate goal-conditioned learning, allowing the robot to learn how to achieve specific objectives by leveraging the estimated affordances. Finally, they integrate action parameterization for reinforcement learning, enabling the robot to learn complex behaviors by optimizing its actions based on the estimated affordances.

E. Predictive Models

Predictive dynamics models, or world models, predict how the state of the world changes given particular agent actions, that is, they attempt to model the state transition function of the world [184]. When applied to visual observations, dynamics modeling can be formulated as a video prediction problem [185], [186]. While video generation and prediction, particularly over long horizons, is a longstanding challenge with many prior efforts, recent models based on vision transformers and diffusion models have demonstrated improvements [187], [188]. For instance, the Phenaki model [189] generates variable length video up to minutes in length conditioned on text prompts.

Several approaches apply these models to robotics in the literature. Note that while learned dynamics or world models in robotics have been explored in constrained or smaller-data regimes, we focus in this section on works that train on a diversity or volume of data that is characteristic of foundation models. One strategy is to learn an action-conditioned model that may be used directly for downstream planning by optimizing an action sequence [190], i.e. performing model-predictive control, or for policy learning via training on simulated rollouts. One example is the GAIA-1 model which generates predictions of driving video conditioned on arbitrary combinations of video, action, and text [191]. It was trained on 4700 hours of proprietary driving data. Another approach is to use a video prediction model to generate a plan of future states, and then learn a separate goal-conditioned policy or inverse dynamics model to infer control actions based on the current and target state. One line of work instantiates this by combining text-conditioned video diffusion models with image-goal-conditioned policies to solve manipulation tasks in simulated and real tabletop settings [192]. This approach has been extended to longer-horizon object manipulation tasks by using the PaLM-E VLM to break down a high-level language goal into smaller substeps, leveraging feedback between the VLM and video generation models [193].

Another example is COMPASS [160], which first constructs a comprehensive multimodal graph to capture crucial relational information across diverse modalities. The graph is then used to construct a rich spatio-temporal and semantic representation. Pretrained on the TartanAir multimodal dataset, COMPASS was demonstrated to address multiple robotic tasks including drone navigation, vehicle racing, and visual odometry.

V. Embodied AI

Recently, researchers have shown that the the success of LLMs can be extended to embodied AI domains [32], [33], [42], [194], where "embodied" typically refers to a virtual embodiment in a world simulator, not a physical robot embodiment. Statler [69] is a framework that endows LLMs with an explicit representation of the world state as a form of "memory" that is maintained over time. Statler uses two instances of general LLMs: a world-model reader and a world-model writer, that interface with and maintain the world state. Statler improves the ability of existing LLMs to reason over longer time horizons without the constraint of context length.

Large Scale Language Models (LSLMs) have exhibited strong reasoning ability and the ability to adapt to new tasks through in-context learning. Dasgupta et al. [195] combine these complementary abilities in a single system consisting of three parts: a Planner, an Actor, and a Reporter. The Planner is a pretrained language model that can issue commands to a simple embodied agent (the Actor), while the Reporter communicates with the Planner to inform its next command. Mu et al. [70] build EgoCOT, a dataset consisting of carefully selected videos from the Ego4D dataset, along with corresponding high-quality language instructions. EmbodiedGPT [70] utilizes prefix adapters to augment the 7B language model's capacity to generate high-quality planning, training it on the EgoCOT dataset to avoid overly divergent language model responses. Comprehensive experiments were conducted, demonstrating that the model effectively enhances the performance of embodied tasks such as Embodied Planning, Embodied Control, Visual Captioning, and Visual Q&A. Embodied agents should autonomously and endlessly explore the environment. They should actively seek new experiences, acquire new skills, and improve themselves.

The game of Minecraft [196] provides a platform for designing intelligent agents capable of operating in the open world. MineDojo [71] is a framework for developing generalist agents in the game of Minecraft. MineDojo offers thousands of open-ended and language-prompted tasks, where the agent can navigate in a progressively generated 3D environment to mine, craft tools, and build structures. As part of this work, the authors introduce MiniCLIP, a video-language model that learns to capture the correlations between a video clip and its time-aligned text that describes the video. The MineCLIP model, trained on YouTube videos, can be used as a reward function to train the agent with reinforcement learning. By maximizing this reward function, it incentivizes the agent to make progress toward solving tasks specified in natural language.

Voyager [73] introduces an LLM-powered embodied lifelong learning agent in the realm of Minecraft. Voyager uses GPT-4 to continuously explore the environment. It interacts with GPT-4 through in-context prompting and does not require model parameter fine-tuning. Exploration is maximized by querying GPT-4 to provide a stream of new tasks and challenges based on the agent's history interactions and current situations. Also, the iterative prompting mechanism generates code as the action space to control the Minecraft agent. Iterative prompting incorporates environment feedback provided by Minecraft, execution errors, and a self-verification scheme. For self-verification, GPT-4 acts as a critic by checking task success and providing suggestions for task completion in

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在其环境中发现和学习新的 可供性（Affordances）。第三，作者结合了目标条件学习，使机器人能够学习如何利用估计的 可供性（Affordances） 来实现特定目标。最后，他们集成了 强化学习（Reinforcement Learning） 的动作参数化，使机器人能够根据估计的 可供性（Affordances） 优化其动作来学习复杂的行为。

E. 预测模型

预测动力学模型（世界模型（World Models））预测给定特定智能体动作时世界状态如何变化，也就是说，它们试图对世界的状态转换函数进行建模[184]。当应用于视觉观察时，动态建模可以表述为视频预测问题[185]、[186]。虽然视频生成和预测，尤其是长范围内的视频生成和预测，对于许多先前的努力来说是一个长期的挑战，但基于视觉变换器和扩散模型的最新模型已经证明了改进[187]、[188]。例如，Phenaki 模型 [189] 根据文本提示生成长度长达几分钟的可变长度视频。

文献中的几种方法将这些模型应用于机器人技术。请注意，虽然机器人技术中的学习动态或 世界模型（World Models） 已经在受限或较小数据范围内进行了探索，但我们在本节中重点关注对 基础模型（Foundation Models） 特征的数据多样性或数据量进行训练的工作。一种策略是学习一个动作条件模型，该模型可以通过优化动作序列[190]直接用于下游规划，即执行模型预测控制，或通过模拟 展开轨迹（Rollouts） 的训练进行策略学习。一个例子是 GAIA-1 模型，它根据视频、动作和文本的任意组合生成驾驶视频的预测 [191]。它接受了 4700 小时专有驾驶数据的训练。另一种方法是使用视频预测模型生成未来状态的计划，然后学习单独的目标条件策略或逆动态模型，以根据当前和目标状态推断控制动作。其中一项工作通过将文本条件视频扩散模型与图像目标条件策略相结合来实例化这一点，以解决模拟和真实桌面设置中的操作任务[192]。通过使用 PaLM-E VLM 将高级语言目标分解为更小的子步骤，利用 VLM 和视频生成模型之间的反馈，这种方法已扩展到较长范围的对象操作任务[193]。

另一个例子是 COMPASS [160]，它首先构建一个全面的多模态图来捕获跨不同模态的关键关系信息。然后使用该图构建丰富的时空和语义表示。 COMPASS 在 TartanAir 多模式数据集上进行了预训练，被证明可以解决多种机器人任务，包括无人机导航、赛车和视觉里程计。

五、具身智能（Embodied AI）

最近，研究人员表明，法学硕士的成功可以扩展到 具身智能（Embodied AI） 领域 [32]、[33]、

[42]、[194]，其中“体现”通常是指世界模拟器中的虚拟实施例，而不是物理机器人实施例

。 Statler [69] 是一个框架，它赋予法学硕士以一种随着时间的推移而维持的“记忆”形式的世界状态的明确表示。 Statler 使用通用法学硕士的两个实例：世界模型读取器和世界模型编写器，它们与世界状态交互并维护世界状态。 Statler 提高了现有法学硕士在较长时间范围内进行推理的能力，而不受上下文长度的限制。

大规模语言模型（LSLM）表现出强大的推理能力和通过上下文学习适应新任务的能力。达斯古普塔等人。[195]将这些互补的能力结合在一个由三部分组成的系统中：规划者、行动者和报告者。 Planner 是一种预训练的语言模型，可以向简单的具身智能体（Actor）发出命令，而 Reporter 则与 Planner 通信以通知其下一个命令。穆等人。 [70] 构建 EgoCOT，一个由 Ego4D 数据集中精心挑选的视频以及相应的高质量语言指令组成的数据集。 EmbodiedGPT [70] 利用前缀适配器来增强 7B 语言模型生成高质量规划的能力，并在 EgoCOT 数据集上对其进行训练，以避免语言模型响应过于分歧。综合实验表明，该模型有效提

升了具体化规划、具体化控制、视觉字幕、视觉问答等具体化任务的性能。具身主体应该自主地、不断地探索环境。他们应该积极寻求新经验、掌握新技能、提高自我。

Minecraft [196] 游戏提供了一个设计能够在开放世界中运行的智能智能体的平台。 MineDojo [71] 是一个用于在 Minecraft 游戏中开发通才智能体的框架。 MineDojo 提供数千个开放式和语言提示的任务，智能体可以在逐步生成的 3D 环境中导航来挖掘、制作工具和构建结构。作为这项工作的一部分，作者介绍了 MiniCLIP，这是一种视频语言模型，可以学习捕获视频剪辑与其描述视频的时间对齐文本之间的相关性。在 YouTube 视频上训练的 MineCLIP 模型可用作奖励函数，以使用 强化学习（Reinforcement Learning） 训练智能体。通过最大化该奖励函数，它可以激励智能体在解决自然语言指定的任务方面取得进展。

Voyager [73] 在 Minecraft 领域引入了一个由 LLM 驱动的终身学习智能体。 Voyager 使用 GPT-4 不断探索环境。它通过上下文提示与 GPT-4 交互，不需要模型参数微调。通过查询 GPT-4 以根据智能体的历史交互和当前情况提供一系列新任务和挑战，可以最大限度地探索。此外，迭代提示机制生成代码作为控制 Minecraft 智能体的操作空间。迭代提示结合了 Minecraft 提供的环境反馈、执行错误和自我验证方案。对于自我验证，GPT-4 通过检查任务成功并提供任务完成建议来充当批评者

the case of failure. The GPT-4 critic can be replaced by a human critic to provide on-the-fly human feedback during task execution. Ghost in the Minecraft (GITM) [197] leverages LLM to break down goals into sub-goals and map them to structured actions for generating control signals. GITM consists of three components: an LLM Decomposer, an LLM Planner, and an LLM Interface. The LLM Decomposer is responsible for dividing the given Minecraft goal into a sub-goal tree. The LLM Planner then plans an action sequence for each sub-goal. Finally, the LLM Interface executes each action in the environment using keyboard and mouse operations.

Reinforcement learning in embodied AI virtual environments has the potential to improve the capabilities of real-world robotics by providing efficient training and optimizing control policies in a safe and controlled setting. Reward design is a crucial aspect of RL that influences the robot's learning process. Rewards should be aligned with the task's objective and guide the robot to achieve the desired task. Foundation models can be leveraged to design rewards. Kwon et al. [16] investigate the simplification of reward design by utilizing a large language model (LLM), such as GPT-3, as a proxy reward function. In this approach, users provide a textual prompt that contains a few examples (few-shots) or a description (zero-shot) of the desired behavior. The proposed method incorporates this proxy reward function within a reinforcement learning framework. Users specify a prompt at the start of the training process. During training, the RL agent's behavior is evaluated by the LLM against the desired behavior outlined in the prompt, resulting in a corresponding reward signal generated by the LLM. Subsequently, the RL agent employs this reward to update its behavior through the learning process.

In [74], the authors propose a method called Exploring with LLMs (ELLM) that rewards an agent for achieving goals suggested by a language model. The language model is prompted with a description of the agent's current state. Therefore, without having a human in the loop, ELMM guides agents toward meaningful behavior.

Zhang et al. [198] explore the potential relationship between offline reinforcement learning and language modeling. They hypothesize that RL and LM share similarities in predicting future states based on current and past states, considering both local and long-range dependencies across states. To validate this assumption, the authors pre-train Transformer models on different offline RL tasks and assess their performance on various language-related tasks. Tarasov et al. [199] present an approach to harness pretrained language models in deep offline reinforcement learning scenarios that are not inherently compatible with textual representations. The authors suggest a method that involves transforming the RL states into human-readable text and performing fine-tuning of the pretrained language model during training with deep offline RL algorithms.

Advances in model architecture (e.g. transformer) for foundation models allow the model to effectively model and predict sequences. To harness the power of these models, some recent studies investigate exploiting these architectures for sequence modeling in RL problems. Reid et al. [200] explore the potential of leveraging the sequence modeling formulation of reinforcement learning and examine the transferability of pretrained sequence models across different domains, such as vision and language. They specifically focus on the effectiveness of fine-tuning these pretrained models on offline RL tasks, including control and games. In addition to investigating the transferability of pretrained sequence models, the authors propose techniques to enhance the transfer of knowledge between these domains. These techniques aim to improve the adaptability and performance of the pretrained models when applied to new tasks or domains.

High-level task planning using LLMs is demonstrated in embodied AI environments. Huang et al. [68] propose employing pretrained Language Models (LMs) as zero-shot planners. The approach is evaluated in the VirtualHome [129] environment. In this work, first, an autoregressive LLM such as GPT-3 [2] or Codex [201] is quarried to generate action plans for high-level tasks. Some of these action plans might not be executable by the agent due to ambiguity in language or referring to objects that are not present or grounded in the environment. So, to select the admissible action plans, admissible environment actions, and generated actions by the causal LLM are embedded using a BERT-style LM. Then for each admissible environment action, its semantic distance to the generated action is computed using cousin similarity.

Chain of thought reasoning and action generation are proposed for embodied agents as well. ReAct [202] combines reasoning (e.g. chain of thought) and acting (e.g. sequence of action generation) within LLM. Reasoning traces enhance the model's ability to deduce, monitor, and revise action plans, along with managing exceptions effectively. Actions facilitate interaction with external resources, like knowledge bases or environments, enabling it to acquire supplementary information. ReAct showcases its proficiency across a wide array of language and decision-making tasks, including question-answering and fact verification. It enhances interpretability and trust for users by transparently illustrating the process through which it searches for evidence and formulates conclusions. Unlike prior methods that depend on a singular chain-of-thought, ReAct engages with a Wikipedia API for pertinent information retrieval and belief updating. This strategy effectively mitigates the issues commonly associated with chain-of-thought reasoning, such as hallucination and error propagation.

VPT [72] presents video pretraining in which the agent learns to act by watching unlabeled online videos. It is shown that an inverse dynamic model can be trained with a small labeled dataset and the model can be used to label a huge unlabeled data of the internet. Videos of people who have played Minecraft are used to train an embodied AI agent to play Minecraft. The model exhibits zero-shot performance and can be fine-tuned for more complex skills using imitation learning or reinforcement learning. The VPT model is trained with a standard behavioral cloning loss (9) (negative log-likelihood) while the actions are drawn from the inverse dynamic model.

A. Generalist AI

A long-standing challenge in robotics research is deploying robots or embodied AI agents in a variety of non-factory real-world applications, performing a range of tasks. To make

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失败的情况。 GPT-4 批评家可以被人类批评家取代，以在任务执行期间提供即时的人类反馈。 Ghost in the Minecraft (GITM) [197] 利用 LLM 将目标分解为子目标，并将它们映射到结构化动作以生成控制信号。 GITM 由三个组件组成：LLM 分解器、LLM 规划器和 LLM 接口。 LLM 分解器负责将给定的 Minecraft 目标划分为子目标树。然后，法学硕士规划师为每个子目标规划一个行动顺序。最后，LLM 接口使用键盘和鼠标操作执行环境中的每个操作。

具身智能（Embodied AI） 虚拟环境中的 强化学习（Reinforcement Learning） 有潜力通过在安全受控的环境中提供有效的训练和优化控制策略来提高现实世界机器人的能力。奖励设计是强化学习的一个重要方面，影响机器人的学习过程。奖励应与任务目标相一致，并引导机器人完成所需的任务。 基础模型（Foundation Models） 可用于设计奖励。权等人。 [16]通过利用大型语言模型（LLM）（例如 GPT-3）作为智能体奖励函数来研究奖励设计的简化。在这种方法中，用户提供文本提示，其中包含所需行为的一些示例（少数镜头）或描述（零镜头）。所提出的方法将此智能体奖励函数合并到 强化学习（Reinforcement Learning） 框架内。用户在训练过程开始时指定提示。在训练期间，LLM 会根据提示中概述的期望行为来评估 RL 智能体的行为，从而导致 LLM 生成相应的奖励信号。随后，强化学习智能体在学习过程中利用此奖励来更新其行为。

在[74]中，作者提出了一种名为 Exploring with LLMs (ELLM) 的方法，该方法奖励智能体实现语言模型建议的目标。语言模型会提示智能体当前状态的描述。因此，在没有人参与的情况下，ELMM 可以引导智能体采取有意义的行为。

张等人。 [198]探索离线强化学习（Reinforcement Learning）与语言建模之间的潜在关系。他们假设 RL 和 LM 在根据当前和过去的状态预测未来状态方面有相似之处，同时考虑到各状态之间的本地和远程依赖性。为了验证这一假设，作者在不同的离线 RL 任务上预训练 Transformer 模型，并评估它们在各种语言相关任务上的表现。塔拉索夫等人。 [199]提出了一种在深度离线 强化学习（Reinforcement Learning） 场景中预训练 运行支撑框架（Harness） 语言模型的方法，这些场景本质上与文本表示不兼容。作者提出了一种方法，包括将 RL 状态转换为人类可读的文本，并在使用深度离线 RL 算法进行训练期间对预训练语言模型进行微调。

基础模型（Foundation Models） 模型架构（例如变压器）的进步使模型能够有效地建模和预测序列

。为了利用这些模型的力量，最近的一些研究调查了利用这些架构在强化学习问题中进行序列建模。里德等人

术语保留：运行支撑框架（Harness）

。 [200]探索利用 强化学习（Reinforcement Learning） 序列建模公式的潜力并检查

跨不同领域（例如视觉和语言）的预训练序列模型。他们特别关注在离线 RL 任务（包括控制和游戏）上微调这些预训练模型的有效性

。除了研究预训练序列模型的可转移性之外，作者还提出了增强这些领域之间知识转移的技术

。这些技术旨在提高预训练模型应用于新任务或领域时的适应性和性能。

使用 LLM 进行高级任务规划在 具身智能（Embodied AI） 环境中进行了演示。黄等人。 [68]建议采用预训练语言模型（LM）作为零样本规划器。该方法在 VirtualHome [129] 环境中进行评估。在这项工作中，

首先，使用 GPT-3 [2] 或 Codex [201] 等自回归 LLM 来生成高级任务的行动计划。由于语言含糊或引用不存在或不存在于环境中的物体，智能体可能无法执行其中一些行动计划

。因此，为了选择可接受的行动计划、可接受的环境行动和因果 LLM 生成的行动，使用 BERT 式 LM 嵌入

。然后，对于每个可接受的环境动作，使用表兄弟相似度计算其与生成动作的语义距离。

还为具体智能体提出了思想推理链和行动生成。 ReAct [202] 在法学硕士中结合了推理（例如思想链）和行动（例如行动生成序列）。推理跟踪增强了模型推断、监控和修改行动计划以及有效管理异常的能力。操作促进与外部资源（如知识库或环境）的交互，使其能够获取补充信息。 ReAct 展示了其在多种语言和决策任务中的熟练程度，包括问答和事实验证。它通过透明地说明搜索证据和得出结论的过程来增强用户的可解释性和信任度。与依赖于单一思维链的先前方法不同，ReAct 与维基百科 API 相结合以进行相关信息检索和信念更新。这种策略有效地缓解了通常与思维链推理相关的问题，例如幻觉和错误传播。

VPT [72] 提出了视频预训练，其中智能体通过观看未标记的在线视频来学习行为。结果表明，可以使用小型标记数据集来训练逆动态模型，并且该模型可以用于标记互联网上的大量未标记数据。使用玩过 Minecraft 的人的视频来训练 具身智能（Embodied AI） 智能体玩 Minecraft

。该模型表现出零样本性能，并且可以使用 模仿学习（Imitation Learning） 或 强化学习（Reinforcement Learning） 进行微调以适应更复杂的技能

。 VPT 模型使用标准行为克隆损失 (9)（负对数似然）进行训练，而动作则来自逆动态模型。

A. 多面手人工智能

机器人研究中的一个长期挑战是在各种非工厂现实世界应用中部署机器人或 具身智能（Embodied AI） 智能体，执行一系列任务。使

generalist robots that can operate in diverse environments with diverse tasks, some researchers have proposed generative simulators for robot learning. For example, Generative Agents [203] discusses how generative agents can produce realistic imitations of human behavior for interactive applications, creating a miniature community of agents similar to those found in games like The Sims. The authors connect their architecture with the ChatGPT large language model to create a game environment with 25 agents. The study includes two evaluations, a controlled evaluation and an end-to-end evaluation, which demonstrate the causal effects of the various components of their architecture. Xian et al. [204], authors propose a fully automated generative pipeline, known as a generative simulation for robot learning, which utilizes models to generate diverse tasks, scenes, and training guidance on a large scale. This approach can facilitate the scaling up of low-level skill learning, ultimately leading to a foundational model for robotics that empowers generalist robots.

An alternative method for developing generalist AI involves using generalizable multi-modal representations. Gato [154] is a generalist agent that works as a multi-modal, multi-task, multi-embodiment generalist policy. Using the same neural network with the same set of weights, Gato can sense and act with different embodiments in various environments across different tasks. Gato can play Atari, chat, caption images, stack blocks with a real robot arm, navigate in a 3D simulated environment, and more. Gato is trained on 604 different tasks with various modalities, observations, and actions. In this setting, language acts as a common grounding across different embodiments. Gato has 1.2B parameters and is trained offline in a supervised way. Positioned at the confluence of representation learning and reinforcement learning (RL), RRL [205] learns behaviors directly from proprioceptive inputs. By harnessing pre-trained visual representations, RRL is able to learn from visual inputs, which typically pose challenges in conventional RL settings.

B. Simulators

High-quality simulators or benchmarks are crucial for robotics development. Hence, we put the "simulator" section here to highlight its essential role. To facilitate generalization from simulation to the real world, Gibson [206] emphasizes real-world perception for embodied agents. To bridge the gap between simulation and real-world, iGibson [146] and BEHAVIOR-1K [207] further support the simulation of a more diverse set of household tasks and reach high levels of simulation realism. As a simulation platform for research in Embodied AI, Habitat [208] consists of Habitat-Sim and Habitat-API. Habitat-Sim can achieve several thousand frames per second (fps) running single-threaded. Rather than modeling into low-level physics, Habitat-Lab [147], is a high-level library for embodied AI, giving a modular framework for end-to-end development. It facilitates the definition of embodied AI tasks, such as navigation, interaction, instruction following, and question answering. Additionally, it enables the configuration of embodied agents, encompassing their physical form, sensors, and capabilities. The library supports various training methodologies for these agents, including imitation learning, reinforcement learning, and traditional non-learning approaches like the SensePlanAct pipelines. Furthermore, it provides standard metrics for evaluating agent performance across these tasks. In line with this, the recent release of Habitat 3.0 [209] further expands these capabilities.

Similarly, RoboTHOR [210] serves as a platform for the development and evaluation of embodied AI agents, offering environments in both simulated and physical settings. Currently, RoboTHOR includes a training and validation set comprising 75 simulated scenes. Additionally, there are 14 scenes each for test-dev and test-standard in the simulation, with corresponding physical counterparts. Key features of RoboTHOR include its reconfigurability and benchmarking capabilities. The physical environments are constructed using modular, movable components, enabling the creation of diverse scene layouts and furniture configurations in a single physical area. Another simulator, VirtualHome [129], models complex activities that occur in a typical household. It supports program descriptions for a variety of activities that happen in people's homes. Huang et al. [33] use VirtualHome to evaluate the robot planning ability with language models. These simulators have the potential to be applied for evaluating LLMs on robotics tasks.

VI. Challenges and Future Directions

In this section, we examine challenges related to integrating foundation models into robotics settings. We also explore potential future avenues to address some of these challenges.

A. Overcoming Data Scarcity in Training Foundation Models for Robotics

One main challenge is that compared to the internet-scale text and image data that large models are trained on, robotic-specific data is scarce. We discuss various techniques to overcome data scarcity. For example, to scale up robot learning, some recent works suggest the use of play data instead of expert data for imitation learning. Another technique is data augmentation using in-painting techniques.

1) Scaling Robot Learning Using Unstructured Play Data and Unlabeled Videos of Humans: Language-conditioned learning such as language-conditioned behavioral cloning, or language-conditioned affordance learning requires having access to large annotated datasets. To scale up learning, in Play-LMP [26], the authors suggest using teleoperated human-provided play data instead of fully annotated expert demonstrations. Play data is unstructured, unlabeled, cheap to collect, but rich. Collecting play data does not require scene staging, task segmenting, or resetting to an initial state. Also, in MimicPlay [118] a goal-conditioned trajectory generation model is trained based on human-play data. The play data includes unlabeled video sequences of humans interacting with the environment with their hands. Recently works such as [125] have shown a very small percentage (as little as 1%) of language-annotated data is needed to train a visuo-lingual affordance model for robot manipulation tasks.

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由于通用机器人可以在不同的环境中执行不同的任务，一些研究人员提出了用于机器人学习的生成模拟器。例如，生成智能体[203]讨论了生成智能体如何为交互式应用程序生成对人类行为的真实模仿，从而创建类似于《模拟人生》等游戏中的微型智能体社区。作者将他们的架构与 ChatGPT 大语言模型连接起来，创建了一个包含 25 个智能体的游戏环境。该研究包括两项评估，一项受控评估和一项端到端评估，展示了其架构的各个组成部分的因果效应。西安等人。 [204]，作者提出了一种完全自动化的生成管道，称为机器人学习的生成模拟，它利用模型大规模生成不同的任务、场景和训练指导。这种方法可以促进低级技能学习的扩展，最终形成一个增强多面手机器人能力的机器人基础模型。

开发通用人工智能的另一种方法是使用可泛化的多模态表示。 Gato [154] 是一个多面手智能体，作为多模式、多任务、多实施方式的多面手策略。使用具有相同权重集的相同神经网络，Gato 可以在不同任务的不同环境中以不同的实施例进行感知和行动。 Gato 可以玩 Atari、聊天、为图像添加字幕、使用真实的机器人手臂堆叠积木、在 3D 模拟环境中导航等等。 Gato 接受了 604 项不同任务的训练，这些任务具有不同的模式、观察结果和行动。在此设置中，语言充当跨不同实施例的公共落地关联（Grounding）。 Gato 有 1.2B 参数，并以监督方式离线训练。 RRL [205] 位于表征学习和 强化学习（Reinforcement Learning）(RL) 的交汇处，直接从本体感受输入中学习行为。通过利用预先训练的视觉表示，RRL 能够从视觉输入中学习，这通常会给传统 RL 设置带来挑战。

B. 模拟器

高质量的模拟器或基准测试对于机器人开发至关重要。因此，我们将“模拟器”部分放在这里以强调其重要作用。为了促进从模拟到现实世界的泛化，Gibson [206] 强调对实体主体的现实世界感知。为了弥合模拟与现实世界之间的差距，iGibson [146] 和 BEHAVIOR-1K [207] 进一步支持模拟更多样化的家庭任务，并达到高水平的模拟真实感。 Habitat[208]作为具身智能（Embodied AI）研究的仿真平台，由Habitat-Sim和Habitat-API组成。 Habitat-Sim 单线程运行时可以达到每秒数千帧 (fps)。 Habitat-Lab [147] 不是对低级物理进行建模，而是 具身智能（Embodied AI） 的高级库，为端到端开发提供了模块化框架。它有助于定义 具身智能（Embodied AI） 任务，例如导航、交互、遵循指令和回答问题。此外，它还可以配置具体智能体，包括其物理形式、传感器和功能。该库支持各种

这些智能体的训练方法，包括 模仿学习（Imitation Learning）、强化学习（Reinforcement Learning） 和传统的非学习方法，如 SensePlanAct 管道。此外，它还提供了用于评估智能体在这些任务中的表现的标准指标。与此相一致的是，最近发布的 Habitat 3.0 [209] 进一步扩展了这些功能。

同样，RoboTHOR [210] 作为 具身智能（Embodied AI） 智能体的开发和评估平台，提供模拟和物理设置环境。目前，RoboTHOR 包括一个由 75 个模拟场景组成的训练和验证集。此外，模拟中的测试开发和测试标准各有 14 个场景，并具有相应的物理对应物。 RoboTHOR 的主要功能包括其可重构性和基准测试功能。物理环境是使用模块化、可移动组件构建的，从而能够在单个物理区域中创建不同的场景布局和家具配置。另一个模拟器 VirtualHome [129] 对典型家庭中发生的复杂活动进行建模。它支持对人们家中发生的各种活动的程序描述。黄等人。 [33]使用VirtualHome通过语言模型评估机器人规划能力。这些模拟器有潜力用于评估机器人任务的法学硕士。

六．挑战和未来方向

在本节中，我们将研究与将 基础模型（Foundation Models） 集成到机器人设置中相关的挑战。我们还探索解决其中一些挑战的潜在未来途径。

A. 克服机器人训练中的数据稀缺问题 基础模型（Foundation Models）

一个主要挑战是，与训练大型模型所依据的互联网规模的文本和图像数据相比，机器人特定的数据很少。我们讨论克服数据稀缺的各种技术。例如，为了扩大机器人学习的规模，最近的一些工作建议使用游戏数据而不是 模仿学习（Imitation Learning） 的专家数据。另一种技术是使用修复技术进行数据增强。

1) 使用非结构化游戏数据和未标记的人类视频扩展机器人学习：语言条件学习（例如语言条件行为克隆）或语言条件 可供性（Affordance） 学习需要访问大型带注释的数据集。为了扩大学习规模，在 Play-LMP [26] 中，作者建议使用远程操作的人类提供的游戏数据，而不是完全注释的专家演示。游戏数据是非结构化、无标签的，收集成本低，但内容丰富。收集播放数据不需要场景分段、任务分段或重置到初始状态。此外，在 MimicPlay [118] 中，目标条件轨迹生成模型是基于人类游戏数据进行训练的。游戏数据包括人类用手与环境交互的未标记视频序列。最近的研究（例如[125]）表明，训练用于机器人操作任务的视觉语言 可供性（Affordance） 模型需要非常小的百分比（低至 1%）的语言注释数据。

2) Data Augmentation using Inpainting: Collecting robotics data requires the robot to interact with the real physical world. This data collection process can be associated with significant costs and potential safety concerns. One way to tackle this challenge is to use generative AI such as text-to-image diffusion models for data augmentation. For example, ROSIE (Scaling Robot Learning with Semantically Imagined Experience) [176] presents a diffusion-based data augmentation. Given a robot manipulation dataset, they use inpainting to create various unseen objects, backgrounds, and distractors with textual guidance. One important challenge for these methods is developing inpainting strategies that can generate sufficient semantically and visually diverse data, while at the same time ensuring that this data is physically feasible and accurate. For instance, using inpainting to modify an image of an object within a robot's gripper may result in an image with a physically unrealistic grasp, leading to poor downstream training performance. Additional investigation into generative foundation models that are evaluated not only for visual quality but also for physical realism may improve the generality of these methods.

3) Overcoming 3D Data Scarcity for Training 3D Foundation Models: Currently, multi-modal Vision-and-Language Models (VLMs) can analyze 2D images, but they lack a connection to the 3D world, which encompasses 3D spatial relationships, 3D planning, 3D affordances, and more. The primary obstacle in developing foundational 3D VLM models lies in the scarcity of 3D data, especially data that is paired with language descriptions. As discussed, language-driven perception tasks such as language-driven 3D scene representation, language-driven 3D scene editing, language-driven 3D scene or shape generation, language-driven 3D classification, and affordance prediction require access to 3D data or multi-view images with camera matrices which are not readily available data types. New datasets or data generation methods need to be created in the future to overcome data scarcity in the 3D domain.

4) Synthetic Data Generation via High-Fidelity Simulation: High-fidelity simulation via gaming engines can provide an efficient means to collect data, especially to solve multimodal and 3D perception tasks on robots. For example, TartanAir [211], a dataset for robot navigation tasks, was collected in [212] with the presence of moving objects, changing light, and various weather conditions. By collecting data in simulations, it was possible to obtain multi-modal sensor data and precise ground truth labels such as the stereo RGB image, depth image, segmentation, optical flow, camera poses, and LiDAR point cloud. A large number of environments were set up with various styles and scenes, covering challenging viewpoints and diverse motion patterns that are difficult to achieve by using physical data collection platforms. An extension TartanAir-V2 (https://tartanair.org) furthers the dataset by incorporating additional environments and modalities, such as fisheye, panoramas, and pinholes, with arbitrary camera intrinsic and rotations.

5) Data Augmentation using VLMs: Data augmentation can be provided using Visual-Language Models (VLMs). In DIAL [213], Data-driven Instruction Augmentation for Language-conditioned control is introduced. DIAL uses VLM to label offline datasets for language-conditioned policy learning. DIAL performs instruction augmentation using VLMs to weakly relabel offline control datasets. DIAL consists of three steps 1) Contrastive fine-tuning of a VLM such as CLIP [4] on a small robot manipulation dataset of trajectories with crowd-sourced annotation, 2) producing new instruction labels by using the fine-tuned VLM to score relevancy of crowd-sourced annotations against a larger dataset of trajectories, 3) training a language-conditioned policy using behavior cloning on both, the original and re-annotated dataset.

6) Robot Physical Skills are Limited to Distribution of Skills: One key limitation of the existing robot transformers and other related works in robotics is that robot physical skills are limited to the distribution of skills observed within the robot data. Using these transformers, the robot lacks the capability to generate new movements. To address this constraint, an approach involves using motion data from videos that humans performing various tasks. The inherent motion information within these videos can then be employed to facilitate the acquisition of physical skills in robotics.

B. Real Time Performance (High Inference Time of Foundation Models)

Another bottleneck for deploying foundation models on robots is the inference time of these models. In Table II, the inference time for some of these models is reported. As seen, the inference time for some of the models still needs to be improved for reliable real-time deployment of the robotic systems. As real-time capability is an essential requirement for any robotic system, more research needs to be performed to improve the computational efficiency of foundation models.

Furthermore, foundation models are most often stored and run in remote data centers, and accessed through APIs that require network connectivity. Many foundation models (e.g., the GPT models, the Dall-E models) can only be accessed this way, while others are usually accessed this way, but can also be downloaded and run locally with sufficient local computing power (such as SAM [59], LLaMA [214], and DINOv2 [107]). Given this cloud-service paradigm, the latencies and service times in response to an API call for a foundation model depend on the underlying network over which the data is routed and the data center where the computation takes place—factors that are beyond the control of a robot. So network reliability should be taken into account before integrating a foundation model into a robot's autonomy stack.

For some robotics domains reliance on the network and 3rd party computing may not be a safe or realistic operating paradigm. In autonomous driving, autonomous aircraft, search and rescue or emergency response applications, and defense applications the robot cannot rely on network connectivity for time-critical perception or control computations. One option is to have a safe fall-back mode that relies on classical autonomy tools using only local computation, that can take over if access to the cloud is interrupted for some reason. Another potential longer-term solution for network-free autonomy is the distillation of large foundation models into smaller-sized

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2）使用修复进行数据增强：收集机器人数据需要机器人与真实的物理世界进行交互。该数据收集过程可能会带来巨大的成本和潜在的安全问题。应对这一挑战的一种方法是使用生成式人工智能，例如用于数据增强的文本到图像扩散模型。例如，ROSIE（Scaling Robot Learning with Semantically Imagined Experience）[176]提出了一种基于扩散的数据增强。给定机器人操作数据集，他们使用修复来创建各种看不见的物体、背景和带有文本指导的干扰物。这些方法面临的一个重要挑战是开发修复策略，该策略可以生成足够的语义和视觉上多样化的数据，同时确保这些数据在物理上可行且准确。例如，使用修复来修改机器人抓手内的物体图像可能会导致图像的抓取在物理上不真实，从而导致下游训练性能不佳。对生成 基础模型（Foundation Models） 的进一步研究不仅评估视觉质量，而且评估物理真实性，可能会提高这些方法的通用性。

3）克服训练3D 基础模型（Foundation Models）的3D数据稀缺性：目前，多模态视觉和语言模型（VLM）可以分析2D图像，但它们缺乏与3D世界的连接，其中包括3D空间关系、3D规划、3D 可供性（Affordances）等。开发基础 3D VLM 模型的主要障碍在于 3D 数据的稀缺，尤其是与语言描述配对的数据。如前所述，语言驱动的感知任务（例如语言驱动的 3D 场景表示、语言驱动的 3D 场景编辑、语言驱动的 3D 场景或形状生成、语言驱动的 3D 分类和 可供性（Affordance） 预测）需要访问具有相机矩阵的 3D 数据或多视图图像，而这些数据类型不是现成的。未来需要创建新的数据集或数据生成方法来克服 3D 领域的数据稀缺问题。

4）通过高保真模拟生成合成数据：通过游戏引擎进行高保真模拟可以提供一种有效的数据收集手段，特别是解决机器人的多模态和3D感知任务。例如，TartanAir [211]，一个机器人导航任务的数据集，在 [212] 中收集，存在移动物体、变化的光线和各种天气条件。通过在模拟中收集数据，可以获得多模态传感器数据和精确的地面实况标签，例如立体 RGB 图像、深度图像、分割、光流、相机姿态和 LiDAR 点云。设置了大量不同风格和场景的环境，涵盖了使用物理数据收集平台难以实现的具有挑战性的视角和多样化的运动模式。扩展 TartanAirV2 (https://tartanair.org) 通过合并额外的环境和模式（例如鱼眼、全景图和针孔）以及任意相机内在和旋转来进一步完善数据集。

5) 使用 VLM 进行数据增强：可以使用视觉语言模型 (VLM) 提供数据增强

。在 DIAL [213] 中，数据驱动指令增强

引入了语言条件控制。 DIAL 使用 VLM 来标记离线数据集，以进行语言条件策略学习。 DIAL 使用 VLM 执行指令增强，以弱重新标记离线控制数据集。 DIAL 包含三个步骤：1）在具有众包注释的小型机器人操作轨迹数据集上对 VLM（例如 CLIP [4]）进行对比微调，2）通过使用微调的 VLM 来对众包注释与更大的轨迹数据集的相关性进行评分，生成新的指令标签，3）使用 行为克隆（Behavior Cloning） 在原始和重新注释的数据集上训练语言条件策略数据集。

6）机器人物理技能仅限于技能分布：现有机器人变压器和机器人技术中其他相关工作的一个关键限制是机器人物理技能仅限于在机器人数据中观察到的技能分布。使用这些变压器，机器人缺乏产生新动作的能力。为了解决这个限制，一种方法涉及使用人类执行各种任务的视频中的运动数据。然后可以利用这些视频中的固有运动信息来促进机器人物理技能的获取。

B. 实时性能（基础模型（Foundation Models） 推理时间高）

在机器人上部署 基础模型（Foundation Models） 的另一个瓶颈是这些模型的推理时间。表 II 报告了其中一些模型的推理时间。正如所见，一些模型的推理时间仍需要改进，以实现机器人系统的可靠实时部署。由于实时能力是任何机器人系统的基本要求，因此需要进行更多的研究来提高 基础模型（Foundation Models） 的计算效率。

此外，基础模型（Foundation Models） 通常在远程数据中心存储和运行，并通过需要网络连接的 API 进行访问。许多基础模型（Foundation Models）（例如GPT模型、Dall-E模型）只能通过这种方式访问，而其他模型通常通过这种方式访问，但也可以在具有足够本地计算能力的情况下下载并在本地运行（例如SAM[59]、LLaMA[214]和DINOv2[107]）。考虑到这种云服务范例，响应 基础模型（Foundation Model）的 API 调用的延迟和服务时间取决于路由数据的底层网络和进行计算的数据中心，这些因素超出了机器人的控制范围。因此，在将 基础模型（Foundation Model） 集成到机器人的自主堆栈中之前，应考虑网络可靠性。

对于某些机器人领域来说，对网络和第三方计算的依赖可能不是安全或现实的操作范例。在自动驾驶、自动飞机、搜索和救援或紧急响应应用以及国防应用中，机器人不能依赖网络连接来进行时间关键的感知或控制计算。一种选择是采用安全的后备模式，该模式依赖于仅使用本地计算的经典自治工具，如果对云的访问由于某种原因中断，该模式可以接管

。无网络自治的另一个潜在的长期解决方案是将大型 基础模型（Foundation Models） 升华为较小尺寸

specialized models that run on onboard robot hardware. Some recent work has attempted this approach (though without an explicit link to robotics) [215]. Such distilled models would likely give up some aspect of the full model, e.g. restricting operation to a certain limited context, in exchange for smaller size and faster compute. This could be an interesting future direction for bringing the power of foundation models to safety-critical robotics systems.

C. Limitations in Multimodal Representation

Multimodal interaction implicitly assumes that the modality is tokenizable and can be standardized into input sequences without losing information. The Multimodal models provide information sharing between multiple modalities and are some variation of multimodal transformers with cross-modal attention between every pair of inputs. In multimodal representation learning, it is assumed that cross-modal interactions and the dimension of heterogeneity between different modalities can all be captured by simple embeddings. In other words, a simple embedding is assumed to be sufficient to identify the modality or for example, how different language is from vision. In the realm of multimodal representation learning, the question of whether a single multimodal model can accommodate all modalities remains an open challenge.

Additionally, when paired data between a modality and text is available one can embed that modality into text directly. In robotics applications there are some modalities for which sufficient data is not available and to be able to align them with other modalities, they need to be first converted to other modalities and then be used. For example, 3D point cloud data has various applications in robotics but training a foundation model using this type of data is challenging since data is scarce and is not aligned with text. So, one way to overcome this challenge is first converting this 3D point cloud data to other modalities such as images and subsequently images to text as the secondary step of alignment. Then they can be used in foundation model training. As another example, in Socratic models [194], each modality, whether visual or auditory, is initially translated into language, after which language models attempt to respond to these modalities.

D. Uncertainty Quantification

How can we provide assurances on the reliability of foundation models when they are deployed in potentially safety-critical robotics applications [188]? Current foundation models such as LLMs often *hallucinate*, i.e., produce outputs that are factually incorrect, logically inconsistent, or physically infeasible. While such failures may be acceptable in applications where the outputs from the model can be checked by a human in real-time (e.g., as is often the case for LLM-based conversational agents), they are not acceptable when deploying autonomous robots that use the outputs of foundation models in order to act in human-centered environments. Rigorous uncertainty quantification is a key step toward addressing this challenge and safely integrating foundation models into robotic systems. Below, we highlight challenges and recent progress in uncertainty quantification for foundation models in robotics.

1) Instance-Level Uncertainty Quantification: How can we quantify the uncertainty in the output of a foundation model for a *particular* input? As an example, consider the problem of image classification; given a particular image, one may quantify uncertainty in the output by producing a *set* of object labels that the model is uncertain among or a *distribution* over object labels. Instance-level uncertainty quantification can inform the robot's decisions at runtime. For example, if an image classification model running on an autonomous vehicle produces a prediction set $\{\texttt{Pedestrian, Bicyclist}\}$ representing that it is uncertain whether a particular agent is a pedestrian or a bicyclist, the autonomous vehicle can take actions that consider both possibilities.

2) Distribution-Level Uncertainty Quantification: How can we quantify the uncertainty in the correctness of a foundation model that will be deployed on a *distribution* of possible future inputs? For the problem of image classification, one may want to compute or bound the probability of errors over the distribution of inputs that a robot may encounter when deployed. Distribution-level uncertainty quantification allows us to decide whether a given model is sufficiently reliable to deploy in our target distribution of scenarios. For example, we may want to collect additional data or fine-tune the model if the computed probability of error is too high.

3) Calibration: In order to be useful, estimates of uncertainty (both at the instance-level and distribution level) should be *calibrated*. If we perform instance-level uncertainty quantification using prediction sets, calibration asks for the prediction set to contain the true label with a user-specified probability (e.g., 95%) over future inputs. If instance-level uncertainty is quantified using a distribution over outputs, it should be the case that outputs that are assigned confidence p are in fact correct with probability p over future inputs. Similarly, distribution-level uncertainty estimates should bound the true probability of errors when encountering inputs from the target distribution.

We highlight a subtle, but important, point that is often overlooked when performing uncertainty quantification in robotics: it can be crucial to pay attention to the distinction between Frequentist and Bayesian interpretations of probabilities. In many robotics contexts — particularly safety-critical ones — the desired interpretation is often Frequentist in nature. For example, if we produce a bound ϵ for the probability of collision of an autonomous vehicle, this should bound the actual observed rate of collisions when the vehicle is deployed. Bayesian techniques (e.g., Gaussian processes or Bayesian ensembles) do not necessarily produce estimates of uncertainty that are calibrated in this Frequentist sense (since the estimates depend on the specific prior that is used to produce the estimates). Trusting the resulting uncertainty estimates may lead one astray if the goal is to provide statistical guarantees on the safety or performance of the robotic system when it is deployed.

4) Distribution Shift: An important challenge in performing calibrated uncertainty quantification is *distribution shift*. A foundation model trained on a particular distribution of inputs may not produce calibrated estimates of uncertainty when deployed on a different distribution for a downstream task. A

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在机载机器人硬件上运行的专用模型。最近的一些工作尝试了这种方法（尽管没有与机器人技术的明确联系）[215]。这种蒸馏模型可能会放弃完整模型的某些方面，例如将操作限制在某个有限的上下文中，以换取更小的尺寸和更快的计算。将 基础模型（Foundation Models） 的功能引入安全关键型机器人系统可能是一个有趣的未来方向。

C. 多模态表示的局限性

多模态交互隐含地假设模态是可标记的，并且可以标准化为输入序列而不会丢失信息。多模态模型提供多种模态之间的信息共享，是多模态转换器的一些变体，在每对输入之间具有跨模态关注。在多模态表示学习中，假设跨模态交互和不同模态之间的异质性维度都可以通过简单的嵌入来捕获。换句话说，假设简单的嵌入足以识别模态，或者例如语言与视觉的不同程度。在多模态表示学习领域，单个多模态模型是否可以容纳所有模态的问题仍然是一个悬而未决的挑战。

此外，当模态和文本之间的配对数据可用时，可以将该模态直接嵌入到文本中。在机器人应用中，有些模式没有足够的数据可用，为了能够将它们与其他模式对齐，需要首先将它们转换为其他模式，然后再使用。例如，3D 点云数据在机器人领域有多种应用，但使用此类数据训练 基础模型（Foundation Model） 具有挑战性，因为数据稀缺且与文本不对齐。因此，克服这一挑战的一种方法是首先将此 3D 点云数据转换为其他模式（例如图像），然后将图像转换为文本，作为对齐的第二步。然后它们就可以用于 基础模型（Foundation Model） 训练。另一个例子，在苏格拉底模型[194]中，每种模态，无论是视觉还是听觉，最初都被翻译成语言，之后语言模型尝试对这些模态做出反应。

D. 不确定性量化

当 基础模型（Foundation Models） 部署在潜在的安全关键机器人应用中时，我们如何保证其可靠性[188]？当前的 基础模型（Foundation Models）（例如法学硕士）经常产生幻觉，即产生实际上不正确、逻辑上不一致或物理上不可行的输出。虽然在人类可以实时检查模型输出的应用中（例如，基于 LLM 的对话智能体通常会出现这种情况），此类故障可能是可以接受的，但在部署使用 基础模型（Foundation Models） 的输出以便在以人为中心的环境中行动的自主机器人时，它们是不可接受的。严格的不确定性量化是应对这一挑战并将 基础模型（Foundation Models） 安全集成到机器人系统中的关键一步。下面，我们重点介绍机器人 基础模型（Foundation Models） 不确定性量化方面的挑战和最新进展。

1) 实例级不确定性量化：我们如何量化特定输入的 基础模型（Foundation Model） 输出的不确定性？举个例子，考虑图像分类问题；给定特定图像，可以通过生成一组对象标签来量化输出中的不确定性，模型在对象标签之间或对象标签上的分布是不确定的。实例级不确定性量化可以为机器人在运行时的决策提供信息。例如，如果在自动驾驶车辆上运行的图像分类模型生成预测集{行人，骑自行车者}，表示不确定特定主体是行人还是骑自行车者，则自动驾驶车辆可以采取考虑这两种可能性的行动。

2）分布级不确定性量化：我们如何量化将部署在未来可能输入的分布上的 基础模型（Foundation Model）的正确性的不确定性？对于图像分类问题，人们可能想要计算或限制机器人在部署时可能遇到的输入分布的错误概率。分布级不确定性量化使我们能够确定给定模型是否足够可靠，可以部署在我们的目标场景分布中。例如，如果计算出的错误概率太高，我们可能想要收集额外的数据或微调模型。

3）校准：为了有用，应该校准不确定性的估计（实例级别和分布级别）。如果我们使用预测集执行实例级不确定性量化，则校准要求预测集包含真实标签，并且对未来的输入具有用户指定的概率（例如 95%）。如果使用输出分布来量化实例级不确定性，则分配置信度 p 的输出实际上对于未来输入的概率 p 是正确的。

同样，当遇到来自目标分布的输入时，分布级别的不确定性估计应该限制真实的错误概率。

我们强调一个微妙但重要的点，在机器人技术中进行不确定性量化时经常被忽视：注意概率论和贝叶斯论解释之间的区别至关重要。在许多机器人技术环境中，尤其是安全关键的环境中，所需的解释本质上通常是频率论。例如，如果我们为自动驾驶车辆的碰撞概率生成一个界限 ε，那么这应该会限制车辆部署时实际观察到的碰撞率。贝叶斯技术（例如，高斯过程或贝叶斯集合）不一定会产生在频率论意义上校准的不确定性估计（因为估计取决于用于生成估计的特定先验）。如果目标是为机器人系统部署时的安全或性能提供统计保证，那么相信由此产生的不确定性估计可能会导致误入歧途。

4) 分布偏移：执行校准不确定性量化的一个重要挑战是分布偏移

。当部署在下游任务的不同分布上时，在特定输入分布上训练的 基础模型（Foundation Model） 可能不会产生不确定性的校准估计。一个

more subtle cause of distribution shift in robotics arises from *closed-loop* deployment of a model. For example, imagine an autonomous vehicle that chooses actions using the output of a perception system that relies on a pretrained foundation model; since the robot's actions influence future states and observations, the distribution of inputs the perception system receives can be potentially very different from the one it was trained on.

5) Case Study: Uncertainty Quantification for Language-Instructed Robots: Recently, there has been exciting progress in performing rigorous uncertainty quantification for language-instructed robots [216]. This work proposes an approach called KNOWNO for endowing language-instructed robots with the ability to *know when they don't know* and to ask for help or clarification from humans in order to resolve uncertainty. KNOWNO performs both instance-level and distribution-level uncertainty quantification in a calibrated manner using the theory of *conformal prediction*. In particular, given a language instruction (and a description of the robot's environment generated using its sensors), conformal prediction is used to generate a prediction set of candidate actions. If this set is a singleton, the robot executes the corresponding action; otherwise, the robot seeks help from a human by asking them to choose an action from the generated set. Using conformal prediction, KNOWNO ensures that asking for help in this manner results in a statistically guaranteed level of task success (i.e., distribution-level uncertainty quantification). KNOWNO tackles potential challenges with distribution shift by collecting a small amount of calibration data from the target distribution of environments, tasks, and language instructions, and using this as part of the conformal prediction calibration procedure. While KNOWNO serves as an example of calibrated instance-level and distribution-level uncertainty quantification for LLMs, future research should also explore assessing and ensuring the reliability of various other foundation models, such as vision-language models, vision-navigation models, and vision-language-action models, commonly employed in robotics. In addition, exploring how Bayesian uncertainty quantification techniques (e.g., ensembling [217], [218]) can be combined with approaches such as conformal prediction to produce calibrated estimates of instance-level and distribution-level uncertainty is a promising direction.

E. Safety Evaluation

The problem of *safety evaluation* is closely related to uncertainty quantification. How can we rigorously *test* for the safety of a foundation model-based robotic system (i) before deployment, (ii) as the model is updated during its lifecycle, and (iii) as the robot operates in its target environments? We highlight challenges and research opportunities related to these problems below.

1) Pre-deployment safety tests: Rigorous pre-deployment testing is crucial for ensuring the safety of any robotic system. However, this can be particularly challenging for robots that incorporate foundation models. First, foundation models are trained on vast amounts of data; thus, a rigorous testing procedure should ensure that test scenarios were not seen by the model during training. Second, foundation models often commit errors in ways that are hard to predict *a priori*; thus, tests need to cover a diverse enough range of scenarios to uncover flaws. Third, foundation models such as LLMs are often used to produce open-ended outputs (e.g., a plan for a robot described in natural language). The correctness of such outputs can be challenging to evaluate in an automated manner if these outputs are evaluated in isolation from the entire system.

The deployment cycle of current foundation models (in non-robotics applications) involves thorough *red-teaming* by human evaluators [3], [219]. Recent work has also considered partially automating this process by using foundation models themselves to perform red-teaming [220], [221]. Developing ways to perform red-teaming (both by humans and in a partially automated way) for foundation models in robotics is an exciting direction for future research.

In addition to evaluating the foundation model in isolation, it is also critical to assess the safety of the end-to-end robotic system. Simulation can play a critical role here, and already does so for current field-deployed systems such as autonomous vehicles [222], [223]. The primary challenges are to ensure that (i) the simulator has high enough fidelity for results to meaningfully transfer to the real world, and (ii) test scenarios (manually specified, replicated from real-world scenarios, or automatically generated via adversarial methods [224]) are representative of real-world scenarios and are diverse enough to expose flaws in the underlying foundation models. In addition, finding ways to augment large-scale simulation-based testing with smaller-scale real-world testing is an important direction for future work. We emphasize the need for performing such testing throughout the lifecycle of a field-deployed robotic system, especially as updates are made to different components (which may interact in unpredictable ways with foundation models).

2) Runtime monitoring and out-of-distribution detection: In addition to performing rigorous testing *offline*, robots with foundation model-based components should also perform *runtime monitoring*. This can take the form of *failure prediction* in a given scenario, which can allow the robot to deploy a safety-preserving fallback policy [225]–[229]. Alternately, the robot can perform *out-of-distribution (OOD) detection* using experiences collected from a small batch of scenarios in a novel distribution [230]–[233]; this can potentially trigger the robot to cease its operations and collect additional training data in the novel distribution in order to re-train its policy. Developing techniques that perform runtime monitoring and OOD detection with statistical guarantees on false positive/negative error rates in a data-efficient manner remains an important research direction.

F. Using Existing Foundation Models as Plug-and-Play or Building New Foundation Models for Robotics

To incorporate foundation models into robotics, either the existing pretrained large models can be employed as plug-and-play or new foundation models can be built using robotics data. Using foundation models as plug-and-play refers to integrating foundation models into various applications without

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机器人技术中分布变化的更微妙原因源于模型的闭环部署。例如，想象一辆自动驾驶汽车使用依赖于预训练 基础模型（Foundation Model） 的感知系统的输出来选择动作；由于机器人的行为会影响未来的状态和观察结果，感知系统接收到的输入分布可能与其训练时的分布有很大不同。

5）案例研究：语言指令机器人的不确定性量化：最近，在对语言指令机器人进行严格的不确定性量化方面取得了令人兴奋的进展[216]。这项工作提出了一种名为“KNOWNO”的方法，让语言指导的机器人能够知道自己何时不知道，并向人类寻求帮助或澄清，以解决不确定性。 KNOWNO 使用共形预测理论以校准的方式执行实例级和分布级不确定性量化。特别是，给定语言指令（以及使用其传感器生成的机器人环境的描述），共形预测用于生成候选动作的预测集。如果这个集合是单例，则机器人执行相应的动作；否则，机器人会通过要求人类从生成的集合中选择一个动作来寻求人类的帮助。使用保形预测，KNOWNO 确保以这种方式寻求帮助会导致任务成功的统计保证水平（即分布级别的不确定性量化）。 KNOWNO 通过从环境、任务和语言指令的目标分布中收集少量校准数据，并将其用作共形预测校准过程的一部分，解决了分布偏移的潜在挑战。虽然 KNOWNO 作为法学硕士校准实例级和分布级不确定性量化的示例，但未来的研究还应该探索评估和确保各种其他 基础模型（Foundation Models） 的可靠性，例如机器人技术中常用的视觉语言模型、视觉导航模型和 视觉语言动作（Vision-Language-Action） 模型。此外，探索如何将贝叶斯不确定性量化技术（例如，集成[217]、[218]）与保形预测等方法相结合，以生成实例级和分布级不确定性的校准估计，是一个有前途的方向。

E.安全性评价

安全评价问题与不确定性量化密切相关。我们如何严格测试基于 基础模型（Foundation Model） 的机器人系统的安全性（i）在部署之前，（ii）模型在其生命周期内更新，以及（iii）机器人在其目标环境中运行时？我们在下面重点介绍与这些问题相关的挑战和研究机会。

1）部署前安全测试：严格的部署前测试对于确保任何机器人系统的安全至关重要。然而，这对于采用 基础模型（Foundation Models） 的机器人来说尤其具有挑战性。首先，基础模型（Foundation Models）经过大量数据的训练；因此，严格的测试程序应确保测试场景不会被其他人看到

训练期间的模型

。其次，基础模型（Foundation Models） 经常会以难以先验预测的方式犯下错误；因此，测试需要涵盖足够多的场景才能发现缺陷。第三，基础模型（Foundation Models）（例如法学硕士）通常用于产生开放式输出（例如，用自然语言描述的机器人计划）。如果这些输出是独立于整个系统进行评估的，那么以自动化方式评估这些输出的正确性可能会很困难。

当前 基础模型（Foundation Models）（在非机器人应用中）的部署周期涉及人类评估人员的彻底红队[3]、[219]。最近的工作还考虑通过使用 基础模型（Foundation Models） 本身来执行红队 [220]、[221] 来部分自动化此过程。开发在机器人技术中为 基础模型（Foundation Models） 执行红队（由人类和部分自动化方式）的方法是未来研究的一个令人兴奋的方向。

除了单独评估 基础模型（Foundation Model） 之外，评估端到端机器人系统的安全性也至关重要。仿真可以在这里发挥关键作用，并且对于自动驾驶汽车等当前现场部署的系统来说已经发挥了重要作用[222]、[223]。主要挑战是确保（i）模拟器具有足够高的保真度，以便将结果有意义地转移到现实世界，以及（ii）测试场景（手动指定、从现实世界场景复制或通过对抗方法自动生成[224]）能够代表现实世界场景，并且

足够多样化以暴露底层 基础模型（Foundation Models） 中的缺陷。此外，寻找通过小规模真实世界测试来增强大规模基于模拟的测试的方法是未来工作的一个重要方向。我们强调需要在现场部署的机器人系统的整个生命周期中执行此类测试，特别是在对不同组件进行更新时（这些组件可能以不可预测的方式与 基础模型（Foundation Models） 进行交互）。

2）运行时监控和分布外检测：除了离线执行严格的测试外，带有基于基础模型（Foundation Model）的组件的机器人还应该执行运行时监控。这可以采取给定场景中故障预测的形式，这可以允许机器人部署安全保护后备策略[225]-[229]。或者，机器人可以使用从新分布中的一小批场景中收集的经验来执行分布外（OOD）检测[230]-[233]；这可能会触发机器人停止运行并收集新分布中的额外训练数据，以便重新训练其策略。开发执行运行时监控和 OOD 检测的技术，并以数据有效的方式对误报/阴性错误率进行统计保证，仍然是一个重要的研究方向。

F. 使用现有的 基础模型（Foundation Models） 作为即插即用或为机器人构建新的 基础模型（Foundation Models）

为了将 基础模型（Foundation Models） 融入机器人技术中，现有的预训练大型模型可以即插即用，也可以使用机器人数据构建新的 基础模型（Foundation Models）。使用基础模型（Foundation Models）即插即用是指将基础模型（Foundation Models）集成到各种应用程序中，而无需

the need for extensive customization. A large body of the existing literature on foundation models in robotics is centered around the use of foundation models from other domains such as language or vision as plug-and-play. The plug-and-play approach simplifies and facilitates the integration of recent AI advances into the robotics domain. While employing these models as plug-and-play offers a convenient way to harness the power of AI and provide rapid implementation, versatility, and scalability, they are not always customized to specific applications. When specific domain expertise is needed, it is necessary to build a foundation model from scratch or fine-tune the existing models. Building a foundation model from scratch is resource-intensive and demands significant computational power. However, it provides fine-grained control over the architecture, training parameters, and overall behavior.

G. High Variability in Robotic Settings

Another challenge is the high variability in robotic settings. Robot platforms are inherently diverse with different physical characteristics, configurations, and capabilities. Real-world environments that robots operate in are also diverse and uncertain with a wide range of variations. Due to all these variabilities, robotic solutions are usually tailored to specific robot platforms with specific layouts, environments, and objects for specific tasks. These solutions are not generalizable across various embodiments, environments, or tasks. Hence, to build general-purpose pretrained robotic foundation models, a key factor is to pre-train large models that are task-agnostic, cross-embodiment, and open-ended and capture diverse robotic data. In ROSIE [176] a diverse dataset is generated for robot learning by performing inpainting of various unseen objects, backgrounds, and distractors with semantic textual guidance. To overcome variability in robotic settings and improve generalization, another solution as ViNT [137] presents is to train foundation models on diverse robotic data across various embodiments. RT-X [46] also investigates the possibility of training large cross-embodied robotic models in the domain of robotic manipulation. RT-X is trained using a multi-embodiment dataset which is created by collecting data from different robot platforms collected through a collaboration between 21 institutions, demonstrating 160266 tasks. RT-X demonstrates transfer across embodiment improves robot capabilities by employing experience from diverse robotic platforms.

H. Benchmarking and Reproducibility in Robotics Settings

Another significant obstacle in incorporating foundation models into robotics research is the necessary reliance on real-world hardware experiments. This creates challenges for reproducibility, as replicating results obtained from hardware experiments may necessitate access to the exact equipment employed. Conversely, many recent works have relied on non-physics-based simulators (e.g., ignoring or greatly simplifying contact physics in gasping) that instead focus on high-level, long-term tasks and visual environment models. Examples of this class of simulators are common and include many of the simulators described above in Sec. V. For example the Gibson family of simulators [146], [206], the Habitat family [147], [208], [209], RobotTHOR [210], and VirtualHome [129] all neglect low-level physics in favor of simulating higher level tasks with high visual fidelity. This leads to a large sim-to-real gap and introduces variability in real-world performance based on how low-level planning and control modules handle the true physics of the scenario. Even when physics-based simulators are used (e.g., PyBullet or MuJoCo), the absence of standardized simulation settings, computing environments, and a persistent sim-to-real gap impede efforts to benchmark and compare performance across various research endeavors. A combination of open hardware, benchmarking in physics-based simulators, and promoting transparency in experimental and simulation setups can significantly alleviate the challenges associated with reproducibility in the integration of foundation models into robotics research. These practices contribute to the development of a more robust and collaborative research ecosystem within the field.

VII. Conclusion

Through examination of the recent literature, we have surveyed the diverse and promising applications of foundation models in robotics. We have delved into how these models have enhanced the capabilities of robots in areas such as decision-making, planning and control, and perception. We also discussed the literature on embodied AI and generalist AI, with an eye toward opportunities for roboticists to extend the concepts in that research field to real-world robotic applications. Generalization, zero-shot capabilities, multimodal capabilities, and scalability of foundation models have the potential to transform robotics. However, as we navigate through this paradigm shift in incorporating foundation models in robotics applications, it is imperative to recognize the challenges and potential risks that must be addressed in future research. Data scarcity in robotics applications, high variability in robotics settings, uncertainty quantification, safety evaluation, and real-time performance remain significant concerns that demand future research. We have delved into some of these challenges and have discussed potential avenues for improvement.

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广泛定制的需求。机器人领域有关 基础模型（Foundation Models） 的大量现有文献都集中在使用来自其他领域（例如语言或视觉）的 基础模型（Foundation Models） 作为即插即用。即插即用方法简化并促进了最新人工智能进步与机器人领域的集成

。虽然使用这些模型作为即插即用提供了一种便利的方式来利用人工智能的力量并提供快速实施、多功能性和可扩展性，但它们并不总是针对特定应用程序进行定制

术语保留：运行支撑框架（Harness）

。当需要特定领域的专业知识时，需要从头开始构建 基础模型（Foundation Model） 或对现有模型进行微调

。从头开始构建 基础模型（Foundation Model） 需要大量资源，并且需要大量的计算能力。然而，它提供了对架构、训练参数和整体行为的细粒度控制。

G. 机器人设置的高度可变性

另一个挑战是机器人设置的高度可变性。机器人平台本质上是多样化的，具有不同的物理特征、配置和功能。机器人运行的现实世界环境也是多种多样且不确定的，并且变化范围很大。由于所有这些变化，机器人解决方案通常针对具有特定布局、环境和用于特定任务的对象的特定机器人平台进行定制

。这些解决方案不可推广到各种实施例、环境或任务

。因此，要构建 通用（General-Purpose） 预训练机器人 基础模型（Foundation Models），一个关键因素是预训练与任务无关、跨实施例、开放式的大型模型并捕获不同的机器人数据。在 ROSIE [176] 中，通过使用语义文本指导对各种看不见的物体、背景和干扰物进行修复，生成了用于机器人学习的多样化数据集。为了克服机器人设置的可变性并提高泛化能力，ViNT [137]提出的另一个解决方案是在不同实施例的不同机器人数据上训练 基础模型（Foundation Models）。 RT-X [46] 还研究了在机器人操作领域训练大型跨实体机器人模型的可能性。 RT-X 使用多实施数据集进行训练，该数据集是通过收集来自不同机器人平台的数据而创建的，这些数据是通过 21 个机构之间的合作收集的，演示了 160266 个任务。 RTX 展示了跨实施例的传输通过利用来自不同机器人平台的经验来提高机器人的能力。

H. 机器人设置中的基准测试和可重复性

将 基础模型（Foundation Models） 纳入机器人研究的另一个重大障碍是对现实世界硬件实验的必要依赖。这给再现性带来了挑战，因为复制从硬件实验中获得的结果可能需要使用所使用的确切设备。相反，最近的许多工作都依赖于基于非物理的模拟器（例如，忽略或大大简化了喘气中的接触物理），而是专注于高级、长期任务和视觉环境模型

。此类模拟器的示例很常见，包括上面第 2 节中描述的许多模拟器。五、例如吉布森

模拟器系列[146]、[206]、Habitat系列[147]、[208]、[209]、RobotTHOR[210]和VirtualHome[129]都忽略了低级物理，有利于以高视觉保真度模拟更高级别的任务。这导致了巨大的模拟与真实差距，并根据低级规划和控制模块处理场景真实物理的方式引入了现实世界性能的变化。即使使用基于物理的模拟器（例如 PyBullet 或 MuJoCo），标准化模拟设置、计算环境的缺乏以及持续存在的 仿真到现实（Sim-to-Real） 差距也会阻碍对各种研究工作进行基准测试和比较性能的努力。开放硬件、基于物理的模拟器的基准测试以及提高实验和模拟设置的透明度的结合可以显着缓解与 基础模型（Foundation Models） 集成到机器人

研究中的可重复性相关的挑战。这些实践有助于在该领域内发展更加强大和协作的研究生态系统。

七．结论

通过查阅最近的文献，我们调查了 基础模型（Foundation Models） 在机器人领域的多样化且有前景的应用。我们深入研究了这些模型如何增强机器人在决策、规划和控制以及感知等领域的能力。我们还讨论了有关 具身智能（Embodied AI） 和通用人工智能的文献，着眼于机器人专家将该研究领域的概念扩展到现实世界的机器人应用的机会。 基础模型（Foundation Models） 的泛化、零射击功能、多模式功能和可扩展性具有改变机器人技术的潜力。然而，当我们经历这种将 基础模型（Foundation Models） 纳入机器人应用的范式转变时，必须认识到未来研究中必须解决的挑战和潜在风险。机器人应用中的数据稀缺、机器人设置的高度可变性、不确定性量化、安全评估和实时性能仍然是需要未来研究的重要问题。我们深入研究了其中一些挑战，并讨论了潜在的改进途径。

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